

ORDINANCE NO. _____

ENTITLED: “AN ORDINANCE AMENDING SECTION 13.20.050, GENERAL PROHIBITIONS, AND SECTION 13.20.070, LOCAL LIMITS, OF CHAPTER 13.20, WASTEWATER TREATMENT AND DISPOSAL, OF THE CODE OF THE CITY OF CHEYENNE, WYOMING, TO REVISE LOCAL DISCHARGE LIMITS AND RELATED PROVISIONS IN COMPLIANCE WITH NPDES AND WPDES PERMIT REQUIREMENTS.”

BE IT ORDAINED BY THE GOVERNING BODY OF THE CITY OF CHEYENNE, WYOMING:

Section 1. The Governing Body of the City of Cheyenne finds and declares that:

A. State and federal wastewater discharge permitting regulations require that publicly owned treatment works (POTWs) provide a written technical evaluation of the need to revise local limits following the issuance or reissuance of a National Pollutant Discharge Elimination System (NPDES) permit;

B. On February 20, 2024, the State of Wyoming issued permits authorizing the discharge of treated wastewater from the Dry Creek Water Reclamation Facility and the Crow Creek Water Reclamation Facility to receiving waters identified as Crow Creek, South Platte River Basin, in accordance with the requirements of the Wyoming Pollution Discharge Elimination System (WPDES) Permits WY0022934 and WY0022381, respectively;

C. In compliance with the NPDES permitting requirements, the City of Cheyenne Board of Public Utilities (BOPU) submitted a proposed Sampling Plan for review by the Environmental Protection Agency (EPA) on January 22, 2025, and the EPA approved the Sampling Plan on June 2, 2025;

D. In compliance with the NPDES permitting requirements, BOPU submitted a proposed Pollutants of Concern (POCs) report for EPA review on May 29, 2025, providing information pertaining to the screening and evaluation of pollutants and the proposed final determination of POCs to be included in the Local Limits Evaluation, and the EPA agreed with the POCs Evaluation on June 2, 2025;

E. In compliance with the NPDES permitting requirements, BOPU submitted a proposed technically based Local Limits Evaluation for EPA review on August 6, 2025, and the EPA deemed the proposed local limitations approvable on January 28, 2026; and

F. To remain in compliance with the NPDES permitting requirements, to preserve the quality of Crow Creek, to maintain the plant at the highest performance levels, to protect the plant’s operating staff, and to protect the overall environment of the South Platte River Basin, the Governing Body determines that it is necessary and appropriate to amend certain wastewater treatment and disposal provisions of the Cheyenne Municipal Code as provided in this Ordinance.

Section 2. That Sections 13.20.050, General Prohibitions, 13.20.070, Local Limits; of Chapter 13.20, Wastewater Treatment and Disposal, of Title 13, Public Services, of the code of the City of Cheyenne, Wyoming, are amended to read:

13.20.050 - General prohibitions.

- A. No industrial user shall contribute or cause to be contributed, directly or indirectly, to the POTW any pollutant or wastewater which will pass through or cause interference with the operation or performance of the POTW. These general prohibitions and the specific prohibitions in subsection B of this section shall apply to all the industrial users of the POTW, whether or not they are subject to pretreatment standards or requirements.
- B. Specific Prohibitions. No industrial user shall contribute the following substances to the POTW:
 - 1. Any substances which by reason of their nature or quantity will create a fire or explosion hazard or be injurious to the POTW or to the operation of the POTW, including, but not limited to, waste streams with a closed cup flashpoint of less than one hundred forty (140) degrees Fahrenheit or sixty (60) degrees Celsius using the test methods in 40 CFR Section 261.21;
 - 2. Any solid or viscous substances which will cause obstruction to sewer flow or other interference with the operation of the POTW such as, but not limited to, fats, oils and grease, garbage with particles greater than one-half inch in any dimension, or any material which can be disposed of as trash;
 - 3. Any biological, chemical, or enzyme product, used on its own or in conjunction with other substances, that causes the liquefaction or emulsification or the act of liquifying or emulsifying fats, oil or grease;
 - 4. Petroleum oil, nonbiodegradable cutting oil, or products of mineral oil origin in amounts that will cause pass through or interference;
 - 5. Any wastewater having a pH less than five or having any other corrosive property capable of causing damage or hazard to structures, equipment or personnel of the POTW;
 - 6. Any substance which may cause a public nuisance, cause hazard to life or prevent entry into the sewers for maintenance and repair;
 - 7. Heated substances in amounts which will prevent entry into the sewer system of authorized personnel or will inhibit biological activity in the POTW resulting in interference, but in no case heat in such quantities that the temperature at the POTW treatment plant exceeds one hundred four (104) degrees Fahrenheit or forty (40) degrees Celsius. A limit may be required in particular situations where the temperature has caused or has potential to cause pass through of an interceptor, pass through or interference in the POTW;

8. Any pollutant which results in the presence of toxic gases, vapors or fumes within the POTW in a quantity that may cause acute worker health and safety problems;
 9. Any trucked or hauled pollutants, except at discharge points designated by the BOPU;
 10. Any pollutant, including oxygen demanding pollutants, released in a discharge at a flow rate and/or concentration (including any slug load) which will cause interference to the POTW;
 11. Wastewater which imparts color which cannot be removed by the treatment process, such as, but not limited to, dye wastes and vegetable tanning solutions, which consequently imparts color to the treatment plant's effluent, thereby violating the BOPU's WYPDES permit;
 12. Wastewater containing any radioactive wastes or isotopes except in compliance with applicable state or federal regulations;
 13. Stormwater, surface water, ground water, artesian well water, roof runoff, subsurface drainage, swimming pool drainage, condensate, deionized water, noncontact cooling water, and unpolluted wastewater, unless specifically authorized by the BOPU;
 14. Sludges, screenings, or other residues from the pretreatment of industrial wastes;
 15. Wastewater causing, alone or in conjunction with other sources, the treatment plant's effluent to fail toxicity test;
 16. Detergents, surface-active agents, or other substances that might cause excessive foaming in the POTW;
 17. Bulk, expired, outdated, or concentrated prescription and non-prescription drugs, that may cause or contribute to pass through or interference.
- C. No chemicals, or substances, including, but not limited to, paints, solvents, boiler or water treatment chemicals, sludges or wastes shall be stored in proximity to a floor drain or other sewer openings unless secondary containment is provided or there are physical barriers to entry to the wastewater collection system. The storage of any chemicals, materials, substances, or wastes that leak or have potential to leak or discharge into the POTW which may create an explosion hazard or in any way have a deleterious effect to the POTW or constitute a nuisance or a hazard to POTW personnel, the general public, the environment, or the receiving stream shall be prohibited.

13.20.070 - Local limits.

- A. The BOPU is authorized to establish local limits pursuant to 40 CFR Section 403.5(c).
- B. It shall be unlawful for any significant industrial user or other designated and permitted industrial user to discharge or cause to be discharged any waste or wastewater to the Crow Creek Water Reclamation Facility or the Dry Creek Water Reclamation Facility that

exceeds the following daily maximum limit or pollutant allocations as specified in the individual wastewater discharge permit established in Section 13.20.210 of this chapter and issued by the BOPU.

Dry Creek Water Reclamation Facility		
Pollutant	Daily Discharge mg/L	Maximum Limits Total Allowable Load (MAIL) (*) lbs/day
Arsenic	0.106	0.4250
Cadmium	0.009	0.0376
Chromium	3.653	14.5843
Copper	1.773	7.0791
Lead	0.490	1.9565
Mercury	0.001	0.0040
Molybdenum	0.186	0.7417
Nickel	0.89	3.5409
Selenium	0.179	0.7137
Silver	0.285	1.1391
Zinc	4.933	19.6951
Biochemical Oxygen Demand (BOD ₅)	579	2312.51
Total Suspended Solid (TSS)	448	1786.94

Crow Creek Water Reclamation Facility		
Pollutant	Daily Discharge mg/L	Maximum Limits Total Allowable Load (MAIL) (*) lbs/day
Arsenic	0.102	0.4737
Cadmium	0.007	0.0321
Chromium	2.437	11.2876
Copper	0.108	0.4996
Lead	1.476	6.8349
Mercury	0.004	0.0192

Nickel	1.22	5.6478
Selenium	0.120	0.5560
Silver	0.193	0.8936
Zinc	3.964	18.3618
Biochemical Oxygen Demand (BOD ₅)	745	3448.92
Total Suspended Solid (TSS)	365	1689.66

(*) The BOPU may allocate the MAIL for a pollutant on a case-by-case basis for the Crow Creek and Dry Creek facilities to significant industrial users or other permitted industrial user discharges. The total allocations included in individual wastewater discharge permits shall not exceed the lbs/day values listed. The BOPU will base its allocation decisions upon available loading and industrial user needs. Allocations are at the sole discretion of the BOPU.

C. Benzene and BTEX. The following limits shall apply to wastewaters that are discharged from:

1. Groundwater cleanup of petroleum or gasoline underground storage tanks or other remediation wastewaters containing these pollutants;
2. Discharges where one or more of these pollutants are present; or
3. Where these pollutants are appropriate surrogates.

It shall be unlawful for any industrial user to discharge or cause to be discharged any waste or wastewater to the POTW that exceeds the following limits:

Pollutant ⁽¹⁾⁽³⁾	Daily Maximum Limit (mg/L)
Benzene	0.050
BTEX ⁽²⁾	0.750

⁽¹⁾ All pollutants shown in the table are total.

⁽²⁾ BTEX shall be measured as the sum of Benzene, Ethylbenzene, Toluene and Xylenes.

⁽³⁾ These limits are based upon installation of air stripping technology as described in the EPA document: "Model NPDES Permit for Discharges Resulting from the Cleanup of Gasoline Released from Underground Storage Tanks. June 1989."

D. It shall be unlawful for any industrial user to discharge or cause to be discharged any waste or wastewater to the Crow Creek Water Reclamation Facility or the Dry Creek Water Reclamation Facility that exceeds the following limits.

Pollutant	Daily Maximum Limit (mg/L)
Total Petroleum Hydrocarbons	50
Total Fats, Oils and Grease	100

- E. The BOPU may establish more stringent pollutant limits, additional site-specific pollutant limits, BMPs, or additional pretreatment requirements, by ordinance or in individual wastewater discharge permits when, in the judgment of the BOPU, such limitations are necessary to implement the provisions of this chapter.

Section 3. Sections, paragraphs, sentences, clauses, and phrases of this ordinance are severable. If any section, paragraph, sentence, clause, or phrase of this ordinance is declared unconstitutional or otherwise unlawful, such declaration shall not affect any of the remaining sections, paragraphs, sentences, clauses, and phrases.

Section 4. This ordinance shall be in full force and effect upon approval and publication.

FIRST READING: _____

SECOND READING: _____

THIRD AND FINAL READING: _____

PATRICK COLLINS, MAYOR

(SEAL)

ATTEST:

KYLIE SODEN, CITY CLERK

Published: _____
Wyoming Tribune-Eagle

Local Limits Evaluation Package



**City of Cheyenne Board of Public Utilities
2416 Snyder Ave.
Cheyenne, WY 82001**

February 03, 2026

Table of Contents

EXECUTIVE SUMMARY	8
1.0 INTRODUCTION.....	11
2.0 FACILITY DESCRIPTIONS.....	13
2.1 Dry Creek Water Reclamation Facility – Aerial Photo	13
2.2 Dry Creek Water Reclamation Facility Plant Schematic	14
2.3 Dry Creek Water Reclamation Facility Process Descriptions	14
2.4 Crow Creek Water Reclamation Facility – Aerial Photo.....	18
2.5 Crow Creek Water Reclamation Facility Plant Schematic	19
2.6 Crow Creek Water Reclamation Facility Process Descriptions.....	19
3.0 SAMPLING PLAN	21
3.1 Influent and Effluent Sampling	22
3.2 Hauled Wastes	23
3.3 Biosolids Sampling	23
3.4 Significant Industrial User Sampling.....	23
3.5 Residential/Commercial Sampling	24
3.6 Receiving Stream Sampling.....	25
4.0 ANALYTICAL METHODS.....	26
5.0 DETERMINATION OF FLOW VOLUMES	28
5.1 Plant Influent Flows	28
5.2 Significant Industrial User Flow.....	28
5.3 Combined Residential/Commercial Flow	29
5.4 Sludge Flow to Disposal	29
6.0 POLLUTANTS OF CONCERN.....	30
7.0 MINIMUM DETECTION LEVELS AND BELOW DETECTION LEVELS	32
8.0 ENVIRONMENTAL CRITERIA.....	32
8.1 Hardness-Dependent Acute and Chronic Criteria	33
8.2 Metals Translator.....	34
8.3 Sludge (Biosolids) Land Application Criteria	36
8.4 Final Environmental Criteria Table.....	37
9.0 REMOVAL EFFICIENCY CALCULATIONS.....	38
9.1 ADRE Method.....	38

9.2	MRE Method.....	39
9.3	Decile Method.....	39
9.4	Literature Values.....	40
9.5	Determination of Final Removal Efficiencies.....	41
10.0	CALCULATION OF ALLOWABLE HEADWORKS LOADING.....	43
10.1	Allowable Headworks Loading Based on Acute and Chronic WPDES Permit Criteria.....	44
10.2	Allowable Headworks Loading Based on Acute Water Quality Criteria.....	45
10.3	Allowable Headworks Loading Based on Chronic Water Quality Criteria.....	47
10.4	Allowable Headworks Loading Based on Land Application Criteria.....	48
11.0	MAXIMUM OF ALLOWABLE HEADWORKS LOADING (MAHL).....	49
12.0	TECHNOLOGY BASED LIMITATION FOR BTEX.....	51
13.0	MAXIMUM ALLOWABLE INDUSTRIAL LOADING (MAIL).....	51
13.1	Maximum Allowable Headworks Loading Safety Factor.....	52
13.2	Loading From Uncontrolled Sources.....	54
13.3	Determination of the Maximum Allowable Loadings (MAL).....	56
13.4	Growth Allowance.....	58
13.5	Determination of Final Maximum Allowable Industrial Loadings.....	58
14.0	RESERVE LOADING.....	60
15.0	ALLOCATION OF MAILS AMONG CONTROLLED SOURCES.....	61
16.0	LIMIT DURATION.....	62
17.0	CALCULATED UNIFORM LOCAL LIMITS.....	62
18.0	LOCAL LIMITS ASSESSMENT.....	64
18.1	Achievability of Calculated Local Limits by Treatment.....	65
18.2	Compliance of Calculated Local Limits versus Sampling Method Detection Levels.....	65
18.3	Assessment of Calculated Local Limits versus Previously Calculated Limits Table.....	65
18.4	Assessment of Calculated Local Limits versus Previously Calculated Limits Narrative – Dry Creek Water Reclamation Facility.....	67
18.5	Assessment of Calculated Local Limits versus Previously Calculated Limits Narrative – Crow Creek Water Reclamation Facility.....	69
19.0	EPA REGION 8 LOCAL LIMITS DEVELOPMENT SPREADSHEET.....	70
	ATTACHMENT A.....	71
	Sampling Data - Dry Creek Water Reclamation Facility.....	71
	Sampling Data - Crow Creek Water Reclamation Facility.....	72
	ATTACHMENT B.....	75
	Biosolids Sampling Results.....	75

ATTACHMENT C	76
Significant Industrial Users Sampling Results.....	76
ATTACHMENT D	67
Residential/Commercial Sampling Results	67
ATTACHMENT E	71
EPA Local Limits Spreadsheet - Dry Creek Water Reclamation Facility	71
ATTACHMENT F	83
EPA Local Limits Spreadsheet - Crow Creek Water Reclamation Facility	83
ATTACHMENT G.....	95
Revised Legal Authority Language.....	95

List of Tables

Table 1 – Dry Creek Water Reclamation Facility Proposed Local Limits	10
Table 2 – Crow Creek Water Reclamation Facility Proposed Local Limits	11
Table 3 – Discharging Facility Monitoring Frequencies	24
Table 4 – Wastewater Analytical Methods (40 CFR Part 136).....	27
Table 5 – Biosolids Analytical Methods (40 CFR Part 503.8)	27
Table 6 – Influent Flow Volumes.....	28
Table 7 – SIU Discharge Volumes	29
Table 8 – Sludge Flow to Disposal Calculation	30
Table 9 – Pollutants of Concern.....	32
Table 10 – Hardness Dependent Metals.....	33
Table 11 – Metals Translator Dissolved to Total Recoverable.....	35
Table 12 – Calculated Final Aquatic Life Water Quality Standards.....	35
Table 13 – Biosolids Land Application Criteria	36
Table 14 – Environmental Criteria.....	37
Table 15 – Calculated Removal Efficiencies	41
Table 16 – Final Determination of Percent Removal Efficiencies	43
Table 17 – Allowable Headworks Loading Based on WPDES Criteria	44
Table 18 – Allowable Headworks Loading Based on Acute WQS Criteria.....	46
Table 19 – Allowable Headworks Loading Based on Chronic WQS Criteria.....	48
Table 20 – Allowable Headworks Loading Based on Biosolids Land Application Criteria	49
Table 21 – Maximum Allowable Headworks Loadings (MAHL)	50
Table 22 – MAHL Less Safety Factor	53
Table 23 – Residential/Commercial Headworks Loading	55
Table 24 – Maximum Allowable Loadings (MAL)	57
Table 25 – Maximum Allowable Industrial Loadings (MAIL).....	59
Table 26 – Reserved Loadings	61
Table 27 – Calculated Uniform Local Limits.....	64
Table 28 – Calculated Local Limits Versus Previously Established Limits.....	66

Acronyms/Abbreviations

ADRE	Average Daily Removal Efficiency
AHL	Allowable Headworks Loading
BCF	Bioconcentration Factor
BOD	Biochemical Oxygen Demand
BOPU	Board of Public Utilities
BTEX	Benzene, Toluene, Ethylbenzene and Xylene
CC WRF	Crow Creek Water Reclamation Facility
CFR	Code of Federal Regulations
CIU	Categorical Industrial User
COD	Chemical Oxygen Demand
DC WRF	Dry Creek Water Reclamation Facility
DMT	Dry Metric Ton
DO	Dissolved Oxygen
DRE	Daily Removal Efficiency
EPA	United States Environmental Protection Agency
HDPE	High-Density Polyethylene
GA	Growth and Expansion
GPD	Gallons per Day
lbs/Day	Pounds per Day
IFAS	Integrated Fixed-Film Activated Sludge
IPS	Influent Pump Station
MAL	Maximum Allowable Loading
MAHL	Maximum Allowable Headworks Loading
MAIL	Maximum Allowable Industrial Loading
MBBR	Moving Bed Biofilm Reactor
MGD	Million Gallons per Day
mg/L	Milligrams per Liter
MRE	Mean Removal Efficiency
NPDES	National Pollutant Discharge Elimination System
ORP	Oxidation Reduction Potential
PEFS	Primary Effluent Flow Splitter
POC	Pollutant of Concern
POTW	Publicly Owned Treatment Works
RAS	Return Activated Sludge
RL	Reporting Limit
RP	Reasonable Potential
SCADA	Supervisory Control And Data Acquisition
SIU	Significant Industrial User
TBLL	Technically Based Local Limits
TSS	Total Suspended Solids
USEPA	United States Environmental Protection Agency
UV	Ultraviolet
WAS	Waste Activated Sludge
WPDES	Wyoming Pollution Discharge Elimination System
WQS	Water Quality Standard

References

Code of Federal Regulations, 40 CFR Part 403.1 through 40 CFR Part 403.20

Code of Federal Regulations, 40 CFR Part 122 Appendix D, Tables II, III, IV, and V

Code of Federal Regulations, 40 CFR Part 503

USEPA Local Limits Development Guidance and Appendices, (July 2004)

USEPA - Region 8 Technically-Based Local Limits Development Strategy and Appendices, (April 11, 2003)

Authorization to Discharge under Wyoming Pollution Discharge Elimination System (WPDES) Permits WY0022934 and WY0022381, issued to the BOPU, Effective date February 20, 2024

Wyoming Department of Environmental Quality, Water Quality Rules and Regulations, Chapters 1 and 2, Wyoming Surface Water Quality Standards (as amended).

Wyoming Department of Environmental Quality, Wyoming Surface Water Classification List, Water Quality Division Surface Water Standards (April, 2024).

The Wyoming Department of Environmental Quality's Surface Water Quality Standards Program Calculator Spreadsheet

EPA National Recommended Water Quality Criteria - Aquatic Life Criteria Table

EPA National Recommended Water Quality Criteria - Human Health Criteria Table

USEPA Technical Support Document for Water Quality-Based Toxics Control (1991)

EPA ProUCL Version 5.2 User Guide (June, 2022)

EPA ProUCL Version 5.2 Technical Guide (June, 2022)

EPA Office of Wastewater Management -NPDES Permit Writers Manual (September 2010)
Code of Federal Regulations, 40 CFR Part 403.1 through 40 CFR Part 403.20

EPA "The Metals Translator: Guidance for Calculating a Total Recoverable Permit Limit from a Dissolved Criterion", (1996)

Local Limits Evaluation

**Prepared By:
City of Cheyenne Board of Public Utilities
2416 Snyder Ave.
Cheyenne, WY 82001**

EXECUTIVE SUMMARY

The City of Cheyenne Board of Public Utilities (BOPU) is authorized by the State of Wyoming to discharge from the Dry Creek Water Reclamation Facility and Crow Creek Water Reclamation Facility to receiving waters named Crow Creek, South Platte River Basin in accordance with the requirements of Wyoming Pollution Discharge Elimination System (WPDES) Permits WY0022934 and WY0022381 respectively. The issuance date for both permits is February 20, 2024. This Local Limits Evaluation is for both treatment facilities.

The General Pretreatment Regulations per 40 CFR Part 403 require that every POTW develop and implement local limits to protect their treatment plant, sewage system, biosolids and receiving water from adverse impacts from industrial and commercial wastewater dischargers. Each POTW with an approved pretreatment program shall continue to develop these limits as necessary and effectively enforce such limits. EPA regulations require that POTWs must provide a written technical evaluation of the need to revise local limits following NPDES permit issuance or reissuance. This technical evaluation has been conducted in accordance with EPA's "Local Limits Development Guidance" July 2004 and the USEPA - Region 8 "Technically-Based Local Limits Development Strategy" April 2003 including the associated appendices and the Region 8 Local Limits Spreadsheet.

All records of the monitoring data used in the development of these technically-based local limits are on-file and available for review at the Dry Creek Water Reclamation Facility. These records will be maintained on-file as long as the current local limits are in effect.

The development of this technically based local limits evaluation includes the following steps:

- Preliminary data collection
 - Plant characteristics including compliance history
 - Influent flows (domestic, non-domestic, SIUs, hauled wastes)
 - Sampling data (plant influent/effluent scans, domestic and commercial sampling, SIUs, and Biosolids)
- Compilation of information
 - WYDES Permit limitations
 - Water Quality Standards

- Sludge Quality Standards
- Preliminary identification of Pollutants of Concern (POC) from the preliminary data and compilation of information
- Calculation of plant removal efficiencies
- Calculation of Allowable Headworks Loading (AHL), Maximum Allowable Headworks Loadings (MAHL), and Maximum Allowable Industrial Loading (MAIL)
- Calculation of Uniform Local Limits
- Assessment of calculated Local Limits

In preparation for this technically-based study, the BOPU submitted a proposed Sampling Plan for EPA review on January 22, 2025. The plan, reviewed and approved on March 3, 2025, contained information regarding sampling practices conducted by BOPU and identified the monitoring data and information used in the preparation of the local limits evaluation. The BOPU also submitted a proposed Pollutants of Concern Evaluation for EPA Review on May 29, 2025; this submittal provided information pertaining to the screening and evaluation of pollutants and proposed final determination of POCs to be included in the Local Limits evaluation. The EPA reviewed and deemed the Pollutants of Concern Evaluation approvable as a component of the Local Limits Development Evaluation on June 2, 2025.

Based on EPA guidance, the following proposed local discharge limitations have been developed based on the Maximum Allowable Headworks Loading (MAHL), calculated in pounds per day, for each pollutant of concern. MAHLs estimate the maximum combined loadings that can be received at the POTW's headworks from all sources. A pollutants MAHL is determined by first calculating its Allowable Headworks Loading (AHL) for each environmental criterion; the most stringent AHL constitutes the MAHL. A safety factor is applied to protect the POTW from uncertainties in data. Region 8 requires a safety + growth factor of at least ten (10) percent. After the MAHL, adjusted with a safety factor, has been determined for each POC, the portion of the pollutant loading from non-controllable domestic sources and pollutant loading from commercial sources not operating under a discharge permit is subtracted leaving the Maximum Allowable Loading (MAL) available to Significant Industrial Users (SIUs). An additional growth allowance is applied to protect the POTW against growth and expansion of existing SIUs. The calculated MAL less the growth factor determines the Maximum Allowable Industrial Loading (MAIL). Reserve loading is subtracted from the MAIL to allow for loading from future SIUs. The MAIL less reserve loading is then pollutant loadings allocated to SIUs. The BOPU will allocate the loadings to the SIUs through uniform concentration-based discharge limitations.

The USEPA considers the adoption of, or revisions to the MAIL a Substantial Program Modification. As such, the Approval Authority (EPA Region 8 Pretreatment Administration) must issue a public notice of the POTWs request for approval and must provide an opportunity

for interested parties to provide comments (typically a 30-day period). The BOPU shall present to the City of Cheyenne City Council the newly developed discharge limitations for incorporation into the city's ordinance. After the first reading of the proposed ordinance revisions, the BOPU will submit a request for Substantial Program Modification approval package to the EPA for public notification, the package shall contain the Local Limits Evaluation Package, current and proposed revised Legal Authority Language for Local Limits (Attachment G), and a Statement of Basis for Pretreatment Program Modifications. The request for approval package shall also include an Attorney's Statement confirming that the modified legal authority:

- (1) allows the municipality to fully implement and enforce Pretreatment Standards and Requirements,
- (2) Be processed and adopted by administrative procedures established in local laws and regulations and will include an opportunity for the public to participate,
- (3) Assures the changes will comply with state laws and established standards and requirements in the municipality's WYDES discharge permits.

After the 30-day public notification period, the revisions to the Municipal Code will be presented to the City of Cheyenne City Council for a second reading and incorporation into the city's ordinance.

Tables 1 and 2 provide the proposed Daily Local Limits, MAHL, and MAIL applicable to SIUs discharging to the Dry Creek Water Reclamation Facility and Crow Creek Water Reclamation Facility.

Table 1 – Dry Creek Water Reclamation Facility Proposed Local Limits

Pollutant	Local Limit	Units	MAHL (lbs/day)	MAIL (lbs/day)
Arsenic (As) ⁽¹⁾	0.106	mg/L	0.6554	0.4250
Cadmium (Cd) ⁽¹⁾⁽²⁾	0.009	mg/L	0.0687	0.0376
Chromium (Cr) ⁽¹⁾⁽²⁾⁽³⁾	3.653	mg/L	18.1885	14.5843
Copper (Cu) ⁽¹⁾⁽²⁾	1.773	mg/L	11.4660	7.0791
Lead (Pb) ⁽¹⁾⁽²⁾	0.490	mg/L	2.4697	1.9565
Mercury (Hg) ⁽¹⁾⁽²⁾	0.001	mg/L	0.0063	0.0040
Molybdenum (Mo) ⁽¹⁾	0.186	mg/L	1.0882	0.7417
Nickel (Ni) ⁽¹⁾⁽²⁾	0.887	mg/L	4.5769	3.5409
Selenium (Se) ⁽¹⁾	0.179	mg/L	0.9242	0.7137
Silver (Ag) ⁽¹⁾⁽²⁾	0.285	mg/L	1.4388	1.1391
Zinc (Zn) ⁽¹⁾⁽²⁾	4.933	mg/L	29.5299	19.6951
BOD ⁽⁴⁾	579	mg/L	17200	2313
TSS ⁽⁴⁾	448	mg/L	19940	1787

Table 2 – Crow Creek Water Reclamation Facility Proposed Local Limits

Pollutant	Local Limit	Units	MAHL (lbs/day)	MAIL (lbs/day)
Arsenic (As) ⁽¹⁾	0.102	mg/L	0.6294	0.4737
Cadmium (Cd) ⁽¹⁾⁽²⁾	0.007	mg/L	0.0506	0.0321
Chromium (Cr) ⁽¹⁾⁽²⁾⁽³⁾	2.437	mg/L	14.0127	11.2876
Copper (Cu) ⁽¹⁾⁽²⁾	0.108	mg/L	2.2397	0.4996
Lead (Pb) ⁽¹⁾⁽²⁾	1.476	mg/L	8.5839	6.8349
Mercury (Hg) ⁽¹⁾⁽²⁾	0.0041	mg/L	0.0277	0.0192
Nickel (Ni) ⁽¹⁾⁽²⁾	1.219	mg/L	7.0720	5.6478
Selenium (Se) ⁽¹⁾	0.120	mg/L	0.7212	0.5560
Silver (Ag) ⁽¹⁾⁽²⁾	0.193	mg/L	1.1319	0.8936
Zinc (Zn) ⁽¹⁾⁽²⁾	3.964	mg/L	25.7965	18.3618
BOD ⁽⁴⁾	745	mg/L	9340	3449
TSS ⁽⁴⁾	365	mg/L	9340	1690

(1) Pollutant in which the Local Limit was developed for Total Recoverable metal concentration.

(2) WQS for this pollutant are expressed as dissolved. In the development of the MAIL and Local Limit, the WQS was converted to Total Recoverable based on approved guidance calculations.

(3) Local Limit developed based on Chromium III standards

(4) Local Limit developed based on POTW influent design capacity

1.0 INTRODUCTION

The City of Cheyenne governing body, operating under the Mayor/Council form of government, reviews and approves the Sewer Enterprise Fund as recommended by the BOPU. The BOPU has exclusive control of all sewer utility systems owned by the City of Cheyenne. The BOPU provides water and sewer services for the City of Cheyenne, Warren Air Force Base, and South Cheyenne Water and Sanitation District. Approximately 23,000 residential, 1630 commercial, and 20 industrial sewer taps have been authorized by the BOPU.

Treated wastewater produced at the Dry Creek and Crow Creek Water Reclamation Facilities discharge to receiving waters named Crow Creek, which is part of the South Platte River Basin. Wyoming Pollution Discharge Elimination System (WPDES) Permits WY0022934 and WY0022381 issued to the Dry Creek and Crow Creek treatment facilities respectively, establish end-of-pipe effluent limits that are protective of all receiving water designated uses as defined in the Wyoming Water Quality Rules and Regulations. Designated uses applicable to the receiving water include warm water aquatic life and fish consumption. Wyoming Water Quality Rules and Regulations, Chapter 1 establish the State Water Quality Standards for the protection of aquatic life and human health.

Sewer sludge produced at the Crow Creek facility is sent to the Dry Creek facility for further processing. The produced Biosolids are recycled as a fertilizer and soil amendment and land applied per EPA Biosolids requirements provided in 40 CFR Part 503.

The purpose and goals of an Industrial Pretreatment Program are:

- To prevent the introduction of pollutants into the Publicly-Owned Treatment Works (POTW) that will interfere with the operation of the system or contaminate the resulting sludge;
- To prevent the introduction of pollutants into the POTW which will pass through the system, inadequately treated, into receiving waters or the atmosphere or otherwise be incompatible with the system;
- To improve the opportunity to recycle and reclaim wastewaters and sludges from the system;
- To provide for equitable distribution among users of the cost of the POTW;
- To provide for and promote the general health, safety and welfare of the citizens residing within the City and connecting jurisdictions;
- To enable the City to comply with its WYDES discharge permit conditions, sludge use and disposal requirements, and any other Federal or State laws to which the POTW is subject; and
- To prevent adverse impacts to worker health and safety due to the discharge of pollutants from industrial users.

These goals are accomplished through administering:

- National Categorical Standards (industry-specific effluent limits)
- Prohibited Discharge Standards (general and specific prohibitions)
- Enforceable **Technically Based Local Limits (TBLLs)**

The development and adoption of TBLLs, based on site- specific conditions, protects the treatment plant, collection system, biosolids, and receiving stream from adverse impacts from industrial and commercial dischargers.

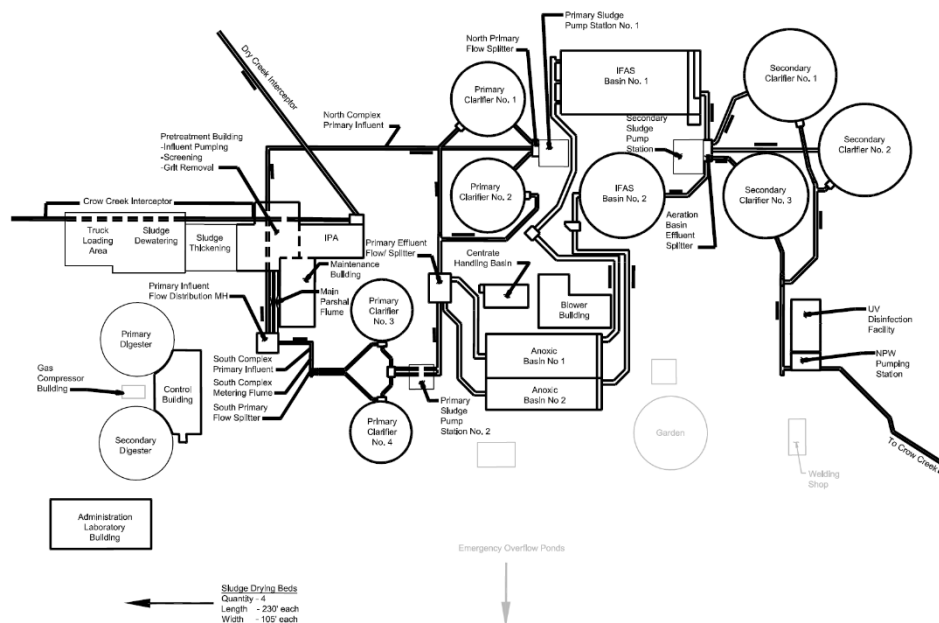
The development of the proposed TBLLs have been based on site conditions, sampling data collected over the previous five years, current regulations pertaining to water quality and biosolids standards, and the best professional judgement of BOPU.

2.0 FACILITY DESCRIPTIONS

2.1 Dry Creek Water Reclamation Facility – Aerial Photo



2.2 Dry Creek Water Reclamation Facility Plant Schematic



2.3 Dry Creek Water Reclamation Facility Process Descriptions

Wastewater from residential, commercial, and industrial sources is conveyed to the Dry Creek Water Reclamation Facility (DC WRF) via the Crow Creek and Dry Creek interceptors. The DC WRF site is located approximately one-half mile southwest of the Campstool Road interchange with Interstate 80 (8911 Campstool Road). The plant discharges into Crow Creek and is at an elevation of approximately 5,900 feet above mean sea level.

Liquid Stream Wastewater Treatment and Handling:

Raw wastewater from the Crow Creek and Dry Creek interceptors is directed to the wet well of Dry Creek's Influent Pumping Station (IPS). The IPS pumps the raw wastewater to the Headworks Building for preliminary treatment. There, the flow is divided into two parallel flow channels that pass through fine stair screens (1/4-inch opening) designed to remove rags and debris. The stair screens also include a washing/compacting system. The washed and compacted screenings are pressed into a pipe which discharges into a dumpster located in the North Side of the Headworks Building.

Following screening, the wastewater is conveyed to a vortex grit separation (Pista Grit) basin. The grit basin is located inside the Headworks Building. The grit slurry is pumped from the bottom of the grit basin to a classifier/dewatering unit located in the truck-loading bay. The dewatered grit is deposited in a second dumpster in the North Side of

the Headworks Building. The dumpsters are emptied daily and taken to landfills. Following the grit-removal process, the wastewater flows through the main Parshall flume located on the west side of the Maintenance Building. The flume is fitted with an ultrasonic level sensor for flow measurement. The level/flow signal from the Parshall flume is transmitted to the supervisory control and data acquisition (SCADA) system and is logged as the DC WRF's influent flow. The flow is split between the North Primary Complex and the South Primary Complex.

The North Primary Complex flows are directed to the North Primary flow splitter, where the flow is split evenly to the two North Complex primary clarifiers. Flow to the South Primary Complex is split evenly to the two South Complex primary clarifiers. The North and South primary clarifiers provide for removal of settleable solids (primary sludge). The settled sludge is removed from the clarifiers via pumps located in the Primary Sludge Pumping Stations. Floating fats, oils, and greases are also removed from the wastewater at the primary clarifiers to a solids pit that is pumped out weekly. Effluent from the north and south primary clarifiers recombine at the Primary Effluent Flow Splitter (PEFS) and then spill over a set of weir troughs to discharge boxes on the north and south sides of the structure.

Wastewater spilling into the north discharge box is directed to Anoxic Basin Train 1 (north). Similarly, wastewater spilling into the south discharge box is directed to Anoxic Basin Train 2 (south). Return activated sludge (RAS) from the Secondary Sludge Pumping Station is also directed to the center of the PEFS prior to conveyance to the anoxic basins. The weirs divide primary effluent and RAS flows proportionally between the two anoxic basin trains based on the volume of each train. Each anoxic basin train is divided into three reactors (in series) by submerged baffle walls. Each reactor is provided with a floating mixer to keep solids in suspension as the liquid flows through the train. The anoxic basins are not aerated. Anoxic Basin Train 1 discharges to Integrated Fixed-Film Activated Sludge (IFAS) Basin 1, and Anoxic Basin Train 2 discharges to IFAS Basin 2.

The anoxic (unaerated) process trains provide for the biological treatment of nitrates & nitrites, as well as some BOD treatment. The biomass (microorganisms) performing the treatment are supplied by the RAS and circulate freely within the wastewater flow to form mixed liquor. Each IFAS basin is divided into two reactors in series. The IFAS process uses high-density polyethylene (HDPE) plastic media suspended in the basins as biofilm carrier elements. The IFAS process trains provide for the biological treatment of BOD and ammonia. The biomass (microorganisms) performing the treatment are supported by the plastic media, as well as circulating freely within mixed liquor. The discharge side of each IFAS reactor is provided with submerged cylindrical stainless steel sieve screens that allow the mixed liquor to exit the reactor but retain the media. The floor of each IFAS reactor is provided with a coarse medium-bubble stainless steel aeration grid. The aeration grid provides oxygen to the biomass, as well as mixing energy for the reactor.

The effluent structure at the effluent end of each IFAS basin is provided with two submersible type recycle pumps. The recycle pumps direct a portion of the IFAS effluent flow back to the head of the anoxic basins. Each IFAS basin's effluent structure is provided with a fixed effluent control weir such that any IFAS effluent not recycled back to the anoxic trains, spills over the weirs and is directed to the Secondary Clarifiers. The Blower Building is located south of IFAS Basin 2. Five centrifugal blowers located in the Blower Building provide process air to the IFAS basins. Dissolved oxygen (DO) probes, oxidation reduction potential (ORP) probes located in the IFAS basins, provide process control data to the SCADA system. The SCADA system uses the process control signal from the DO probes to automatically adjust the airflow delivered to each IFAS basin.

The non-recycled portion of the IFAS effluent flows by gravity to the Secondary Clarifier Influent Splitter Box where it is proportionally distributed to Secondary Clarifiers No. 1, No. 2, and No. 3. The biological solids (secondary sludge) settle out from the treated wastewater in the secondary clarifiers. The base of each clarifier is provided with a hopper for collection of the settled sludge. The settled sludge is removed from the clarifiers via pumps located in the Secondary Sludge Pumping Station. Most of the secondary sludge is directed to the PEFS as RAS. A portion of the secondary sludge is directed to the sludge thickening facility as waste activated sludge (WAS). Floating scum is also removed at the secondary clarifiers via rotating skimmer arms. Clarifier overflow (secondary effluent) is piped to the Ultraviolet (UV) Disinfection Facility.

The UV Disinfection Facility consists of two parallel open channels fitted with UV-generating modules. The UV modules, cleaning tank, jib crane, and electrical support equipment are housed inside the UV Building. A common effluent channel fitted with weir troughs is provided downstream of the UV treatment channels. The disinfected final effluent exits the effluent chamber by spilling over the weir troughs and entering the non-potable water (NPW) Pumping Station wetwell. Any effluent not pumped out by the NPW pumps spills over a control weir and enters the final effluent pipe. The final effluent pipe is fitted with a flowmeter for flow measurement prior to exiting the plant site and discharging to Crow Creek. The flow signal from the effluent flowmeter is transmitted to the SCADA system and is logged as effluent flow.

Solids Stream Treatment and Handling:

The solids stream consists of the organic and inorganic solids (sludge) removed from the wastewater in the primary and secondary clarifiers. The sludge is pumped from the bottom of the clarifiers and stabilized in the anaerobic digesters. All of the BOPU's sludge stabilization and processing facilities are located at the DC WRF. Therefore, all of the solids collected at the Crow Creek Water Reclamation Facility (CC WRF) are directed to the DC WRF via the Crow Creek Interceptor.

When the CC WRF solids reach the DC WRF, they are settled out in the DC WRF primary clarifiers. Floating scum (grease, oil, fat, etc.) that has been removed from the surface of the primary and secondary clarifiers is not normally directed to the digesters.

Primary scum is normally stabilized with fly ash in the drying beds. Secondary scum is normally directed back to the head of the plant. Primary sludge is pumped directly into the Primary Digester for stabilization. Secondary sludge (WAS) must first be thickened before it is fed to the Primary Digester. WAS is pumped to the Sludge Thickening Building where it is thickened (5 percent solids) by one of the two rotary drum thickeners. The Sludge Thickening Building houses two rotary drum thickeners, a polymer feed system, thickened sludge holding tanks, thickened sludge pumps, and electrical support equipment. Polymer is added to the sludge ahead of the drum thickeners to enhance thickening performance. The thickened WAS is pumped from the thickened sludge holding tanks to the Primary Digester by the thickened sludge pumps.

The liquid separated from the WAS in the thickening process is directed to the head of the plant for treatment. The Primary and Secondary Digesters at the DC WRF stabilize the sludge prior to dewatering, air drying, and final land application. The sludge in the digesters is heated via a heat exchanger. The combination of heat and detention time provides the degree of stabilization needed to meet the regulations. The two tanks are connected with a gravity transfer line. The liquid level in the Primary Digester is held at a constant elevation by a fixed position effluent pipe connected to the gravity transfer line. As the liquid level in the Primary Digester rises, flow is transferred to the Secondary Digester by gravity by the fixed position effluent pipe.

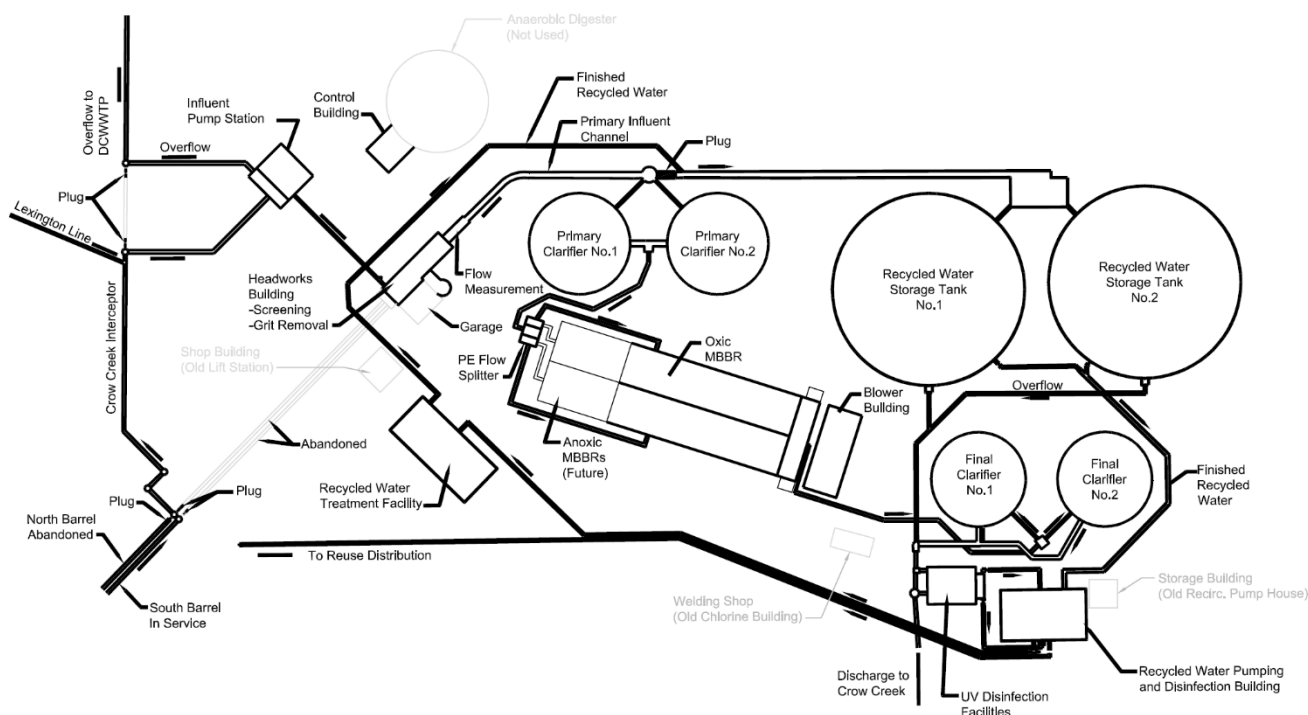
The digested sludge is pumped from the Secondary Digester to the Sludge Dewatering Building. The Secondary Digester provides additional detention time, as well as acting as a flow equalization tank so that flows to the Sludge Dewatering Building do not have to exactly match flows into the digester. The solids in the digester are kept in suspension with gas-mixing guns distributed inside the digesters. Methane gas is generated during the digestion process and is collected for use in powering the heat exchanger. Excess gas is burned at the digester's flare. Dewatered sludge pumps, located in the basement of the Control Building, convey the digested sludge to the centrifuges located in the Sludge Dewatering Building where it is dewatered by one of the two high solids centrifuges.

The Sludge Dewatering Building houses two high solids centrifuges, a polymer feed system, a dewatered sludge truck-loading bay, and electrical and mechanical support equipment. Polymer is added to the sludge ahead of the centrifuges to enhance dewatering performance. The dewatered digested sludge is deposited on a screw conveyor which moves the sludge into the truck loading bay. The dewatered sludge is then trucked to the drying beds for supplemental drying prior to being hauled off-site and land applied. The liquid separated from the sludge in the dewatering process is called centrate. The centrate is conveyed by gravity to the Centrate Handling Basin. Centrate has a high concentration of ammonia, so it should either receive some treatment prior to entering the main flow stream or it should be metered into the flow stream during periods of low influent ammonia concentration. The Centrate Handling Basin can serve as a pretreatment tank or as a fill and dump tank.

2.4 Crow Creek Water Reclamation Facility – Aerial Photo



2.5 Crow Creek Water Reclamation Facility Plant Schematic



2.6 Crow Creek Water Reclamation Facility Process Descriptions

Wastewater from residential, commercial, and industrial sources is conveyed to the CC WRF via the Crow Creek Interceptor and the Lexington Line. The CC WRF site is located approximately one-half mile southeast of the College Avenue interchange with Interstate 80 (4403 Livingston Avenue). The plant discharges into Crow Creek and is at an elevation of approximately 6,000 feet above mean sea level.

Liquid Stream Wastewater Treatment and Handling

Raw wastewater from the Crow Creek Interceptor and Lexington Line is directed to the wetwell of Crow Creek's IPS. The IPS is located upstream of the screening and degritting processes across from the Headworks and Control Buildings. The pumps are designed to handle raw wastewater, including rags and solids up to 6 inches in diameter. IPS pumps the raw wastewater to the Headworks Building for preliminary treatment. There, the flow passes through a fine screen (1/4-inch opening) designed to remove rags and debris. The Headworks Building contains a single flow channel in which a Parkson Bioguard screening/washing system is installed.

The Parkson Bioguard system consists of a fine screen attached to a screenings washing/compacting/dewatering system. The washed and compacted screenings are deposited onto a belt conveyor which discharges into a dumpster located in the attached garage. Following screening, the wastewater exits the Headworks Building superstructure and enters a grit separation basin. The grit basin is attached to the south side of the Headworks Building but is otherwise open to the atmosphere. The grit slurry is pumped from the bottom of the grit basin back inside the Headworks Building to a classifier/dewatering unit. The dewatered grit is deposited on the same conveyor as the washed screenings and conveyed to the same dumpster. The dumpster is emptied weekly and taken to landfill.

Following the grit-removal process, the wastewater flows through a Parshall flume located on the east side of the Headworks Building. The flow signal from the Parshall flume is transmitted to the SCADA system and is logged as the CC WRF's influent flow. From the Parshall flume, the screened and degritted wastewater flows via an open channel to the primary influent distribution manhole where the flow is split evenly to the two primary clarifiers. Effluent from the two primary clarifiers recombines and flows by gravity to the PEFS structure through a 30-inch pipe.

Primary effluent enters the PEFS from the north in the center of the structure and then spills over weirs on the east and west side of the structure. Wastewater spilling over the east weir is directed to the east moving bed biofilm reactor (MBBR) train via a dedicated 30-inch pipe. Similarly, wastewater spilling over the west weir is directed to the west MBBR train via its own 30-inch pipe. The weirs divide flow equally between the east and west MBBR trains.

Two parallel oxic MBBR (aerated) process trains provide for biological treatment of the primary effluent prior to discharge to the secondary clarifiers. Each process train consists of four oxic MBBR reactors in series. The MBBR process uses high-density polyethylene (HDPE) plastic media suspended in the basins as biofilm carrier elements. Cylindrical stainless steel submerged sieves retain the media within each reactor. The first two reactors in each train are designated for BOD removal, and the last two reactors in each train are designated for nitrification (ammonia removal). The north exterior wall of the basin and the influent structure and piping are designed to accommodate addition of the pre-denitrification Anoxic MBBR basins in the future.

The effluent structure at the effluent end of the MBBR Complex is sized and configured to provide for the future addition of recycle pumps as required for pre-denitrification. The effluent structure consists of a fixed effluent control weir and effluent chamber. The Blower Building is located immediately south of the MBBR Complex. Four centrifugal blowers located in the Blower Building provide process air to each oxic MBBR reactor. Dissolved oxygen (DO) probes and oxidation reduction potential (ORP) probes located in the MBBR basins, provide process control data to the SCADA system. The SCADA system uses the process control signal from the DO probes to automatically adjust the airflow delivered to each MBBR train. The MBBR effluent flows by gravity to the

Secondary Clarifier Influent Distribution Manhole where it is evenly distributed to Secondary Clarifiers No. 1 and No. 2. The biological solids sloughed from the media in the MBBR reactors settle out from the treated wastewater in the secondary clarifiers. Clarifier overflow (secondary effluent) is conveyed to the UV Disinfection Facility via a combination of flumes and piping for disinfection.

The UV Disinfection Facility consists of a single open channel fitted with UV-generating modules. The UV modules, cleaning tank, lifting hoist, and electrical support equipment are housed inside the UV Building. An effluent chamber fitted with control weirs is provided at the effluent end of the UV Building. The disinfected final effluent exits the effluent chamber by spilling over the control weirs and flowing into a 30-inch final effluent pipe. The final effluent pipe is fitted with a flowmeter and discharges to Crow Creek. The flow signal from the 30-inch flowmeter is transmitted to the SCADA system and is logged as the CC WRF's effluent flow. Two 24-inch pipes, located upstream of the control weirs, connect the UV effluent chamber to the wet pit for the Recycled Water Transfer Pumps. The wet pit and pumps are located in the Secondary Disinfection and Pumping Building. Since the transfer pump wet pit and UV Treatment Facility are hydraulically connected, they will operate at the same water surface elevation. Any disinfected UV effluent not pumped out by the reuse feed pumps spills over the effluent chamber control weirs and is discharged to Crow Creek. The SCADA system uses the effluent flow signal and level sensors in the effluent chamber to prevent the feed pumps from dropping the water surface at the UV Modules below the control weir level.

Solids Stream Treatment and Handling

The solids stream consists of the organic and inorganic solids (sludge) removed from the wastewater in the primary and secondary clarifiers. The settled solids are pumped from the bottom of the clarifiers and combined with the floating scum (grease, oil, fat, etc.) that has been removed from the surface of the clarifiers. All of the BOPU's sludge stabilization and processing facilities are located at the DC WRF. Therefore, all of the solids collected at the CC WRF are pumped into the IPS overflow line which connects to the interceptor feeding the DC WRF. A small bypass line is provided in the IPS discharge piping that provides a flushing flow of raw wastewater into the overflow line keep the sludge/solid in suspension until they reach the DC WRF. When the CC WRF solids reach the DC WRF, they are settled out in the primary clarifiers at that site. The sludge drying beds at the CC WRF are currently used for stabilization of grease collected through the BOPU's oil and grease mitigation program.

3.0 SAMPLING PLAN

A formal sampling plan was submitted to US EPA Region 8 for review on January 22, 2025. The EPA reviewed and approved the Sampling Plan on March 3, 2025.

3.1 Influent and Effluent Sampling

Per the State WPDES permits issued to the Cheyenne Board of Public Utilities, the Dry Creek and Crow Creek water reclamation facilities must analyze the treatment plant influent and effluent for the presence of the toxic pollutants listed in 40 CFR Part 122, Appendix D; Table II at least one (1) time per year; and pollutants listed in Table III at least once per quarter (4) times per year. If, based upon information available, there is reason to suspect the presence of any toxic or hazardous pollutant listed in Table V, or any other pollutant in quantity or concentration known or suspected to adversely affect POTW operation, receiving water quality, or solids disposal procedures, analysis for these pollutants shall be performed at least once per quarter (4) times per year on both the influent and effluent.

The BOPU monitors both treatment facilities influent and effluent for the pollutants listed in Table II and Table III four (4) times per year, and Table V pollutants Styrene and Xylene four (4) times per year. Per WPDES permit requirements, beginning in 2024, BOPU also monitors annually both facilities effluent for the specific volatile and semi-volatile pollutants listed in Table C of the permits for submittal to the Wyoming Department of Environmental Quality.

BOPU monitors the treatment facilities influent for BOD, TSS, Ammonia, Nitrate, and Total Dissolved Solids (TDS) twice weekly as plant data. Ammonia, BOD, Nitrate, Oil & Grease, Chloride, Sulfate, Total Cyanide, Total Phenolics, Phosphorus, TSS, TDS, Total Sulfide, and TPH (SGT-HEM). These pollutants are also monitored with the quarterly sampling.

The BOPU monitors the treatment facilities' effluent for BOD, TSS, TDS, Ammonia, Nitrate, Nitrite, and phosphorus twice weekly as plant data. Ammonia, BOD, Nitrate, Oil & Grease, Chloride, Sulfate, Total Cyanide, Total Phenolics, Phosphorus, TSS, TDS, Total Sulfide, and TPH (SGT-HEM). These pollutants are also monitored with the quarterly sampling.

The BOPU monitors the treatment facilities effluent for Copper (dissolved), Cadmium (dissolved), Selenium, Phosphorus, and Oil and Grease to meet daily or monthly per permit requirements.

A comprehensive listing of sampling results was provided in the Pollutants of Concern Evaluation and formed the basis for final determination of POCs to be included in the Local Limits evaluation.

A listing of the parameters in which analysis provided a result at or above the reporting limit in the plant's influent or effluent from January 1, 2020 through December 31, 2024 has been compiled and is provided in Attachment A.

3.2 Hauled Wastes

The BOPU receives Trucked and Hauled Septage waste only at the Dry Creek facility using a manifest system. The pollutant concentrations and loadings are included as part of influent monitoring at the Dry Creek facility.

3.3 Biosolids Sampling

All sludge generated at the Crow Creek facility is transferred to the Dry Creek facility for treatment.

The Dry Creek treatment system produces Class "B" Biosolids. The Biosolids are transferred to drying beds and achieve Class A designation for pathogen reduction prior to use in an agricultural land application program.

The Dry Creek facility monitors the Biosolids for the pollutants in which ceiling concentrations have been established per EPA 40 CFR Part 503.13. In addition, the Biosolids are monitored for Boron, Chromium, Iron, Potassium, Magnesium, Manganese, Silver, Phosphorus, Chloride, Ammonia, Nitrate/Nitrite, and Total Nitrogen (Kjeldahl). Biosolids monitoring is performed six (6) times per year.

A listing of the parameters and sampling analysis of the Biosolids from January 1, 2022 through December 31, 2024 has been compiled and is provided in Attachment B.

3.4 Significant Industrial User Sampling

Per 40 CFR Part 403 requirements; Significant Industrial Users (SIUs) must have their effluent sampled at least once per year.

The BOPU performs authority monitoring (sampling) at all discharging SIUs per the frequency provided in Table 3.

Per 40 CFR Part 403 requirements; SIUs subject to Categorical Pretreatment Standards and SIUs not subject to Categorical Pretreatment Standards must submit to the control authority, twice per year, a report indicating the nature and concentration of pollutants in their effluent.

The BOPU requires all discharging SIUs to perform self-monitoring (sampling) per the frequency provided in Table 3.

Attachment C provides the average and maximum pollutant concentrations from Significant Industrial users based on authority monitoring sampling data from January 1, 2020 through December 31, 2024.

Table 3 – Discharging Facility Monitoring Frequencies

Permitted SIU	Permit #	Description	Receiving Facility	Self-Monitoring Frequency	Authority Monitoring Frequency
Black Hills Energy - Cheyenne Prairie Generating Station	BHE-2022	CIU	Dry Creek	Quarterly	Biannual
Cheyenne Renewable Diesel Company	CRDC-2022-Mod	SIU	Dry Creek / Crow Creek	Monthly	Annual
Union Pacific Railroad	UPRR-2022	SIU	Crow Creek	Monthly	Annual
Walmart Distribution Center #7077	WALMART DISTRIBUTION CENTER#7077-2022	SIU	Crow Creek	Quarterly	Annual
SPF Store No #46 LAUST Facility ID #3337	WSP-SPF46-2022	SIU	Crow Creek	Quarterly	Annual
Flying J LAUST Facility ID #3476	WSP-FLYING J-2022	SIU	Crow Creek	Quarterly	Annual

3.5 Residential/Commercial Sampling

EPA Local Limits Development Guidance recommends that POTWs base their local discharge limitations on maximum allowable headworks loading (MAHL) for each pollutant of concern from all sources controllable and non-controllable.

To determine the allocation of the MAHL available to the controllable sources, the POTW must determine the pollutant loadings from non-controllable sources (commercial non-permitted facilities and residential domestic sources). In simplified terms, the MAHL less the non-controllable loadings results in the maximum allowable industrial loadings (MAIL) available to permitted industries.

Per City of Cheyenne Municipal Code 13.20.070, Local Limits are applicable to Significant Industrial Users (SIUs) and other designated and permitted Industries (controllable sources).

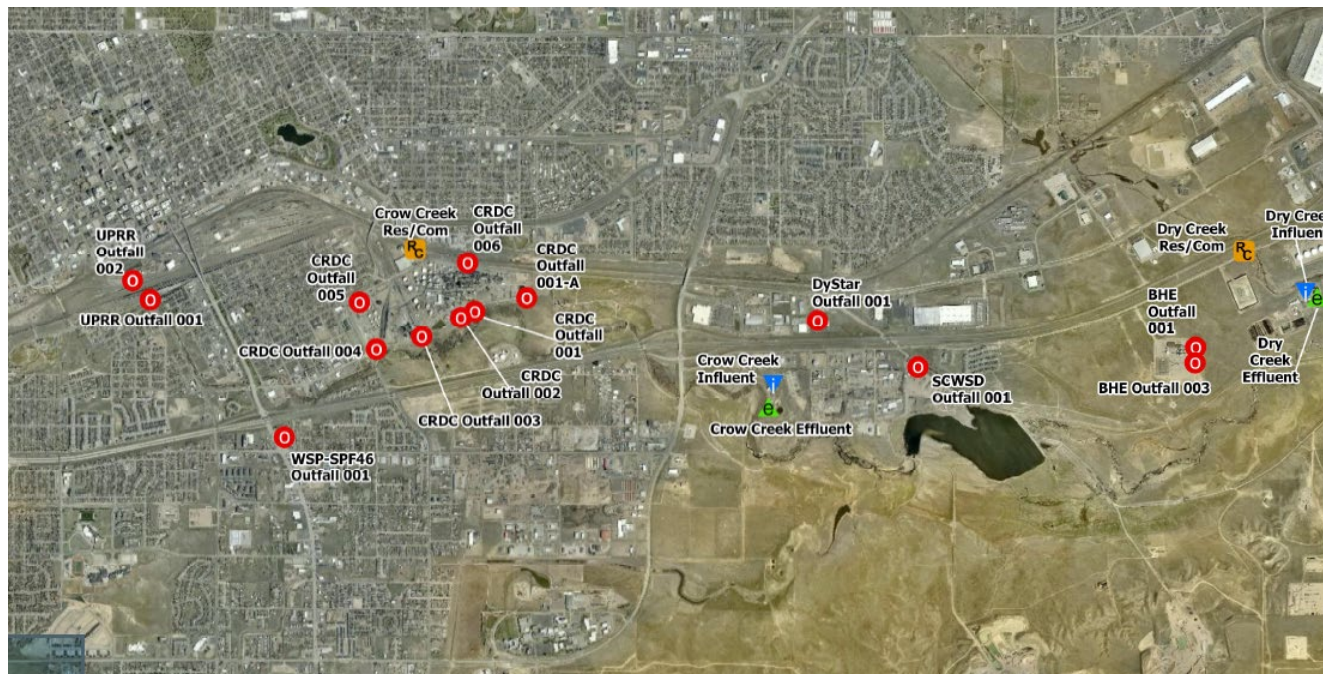
To determine the relative contributions of non-controllable sources (commercial non-permitted facilities and residential domestic sources); the BOPU has an established sampling program consisting of three (3) sampling locations in the service area collection system.

Two sampling locations, DC-Res/Com-1 and SCWSD-1 (South Cheyenne Water & Sanitation District), are representative of commercial non-permitted facilities and residential domestic sources pollutant loadings to the Dry Creek treatment system. One sampling location, CC-Res/Com-1, is representative of commercial non-permitted facilities and residential domestic sources pollutant loadings to the Crow Creek treatment system. The sampling locations are monitored twice per year.

Residential/Commercial sampling results from January 1, 2020 – December 31, 2024 are provided in Attachment D.

3.6 Receiving Stream Sampling

Per the Statement of Basis documents provided by the Wyoming Department of Environmental Quality Water Quality Division with the Dry Creek Water Reclamation Facility (WY0022934) and Crow Creek Water Reclamation Facility (WY0022381) discharge permits: “all water quality-quality based effluent limits established in the permits are equal to instream standards because zero receiving water dilution was considered in establishing the facilities water-quality based effluent limits.” As such, BOPU is not required to monitor for upstream receiving stream background pollutant concentrations, as they are not considered in the calculation of Allowable Headworks Loading Based on Acute and Chronic Water Quality Criteria.



Sampling Locations Map

4.0 ANALYTICAL METHODS

All sampling data collection and analysis including hold times and sample preservation have been performed in accordance with the procedures established in 40 CFR Part 136 and 40 CFR Part 503. Tables 4 and 5 provide the analytical testing and collection methods used during sampling data collection.

Table 4 – Wastewater Analytical Methods (40 CFR Part 136)

Parameter	Collection Method	Analytical Testing Method
Ammonia	Composite	SM 4500 NH3-D
Metals, Total	Composite/Grab*	EPA 200.8
Mercury	Composite/Grab*	EPA 245.1
Phosphorus, Total	Composite/Grab*	EPA 365.1
Cyanide, Total	Grab	SM 4500-CN E/EPA 335.4
Chloride	Composite	SM4500-C1-E
Oil & Grease/HEM-SGT	Composite	EPA 1664A
Hexavalent Chromium	Grab	SM3500-Cr B
Phenols, Total	Grab	EPA 420.4/SM 5530 D
Total Sulfide	Composite	EPA 376.2
Nitrate	Composite	SM4500 NO3-F
Sulfate	Composite	EPA 375.2
Total Dissolved Solids	Composite	SM 2520 B
Biological Oxygen Demand (BOD)	Composite/Grab*	SM 5210 B
Chemical Oxygen Demand (COD)	Composite/Grab*	SM 5220D/HACH 8000
Total Suspended Solids (TSS)	Composite/Grab*	SM2540 D
Table II Toxic Organics:		
Volatile Organics	Composite	EPA 624
Semi-volatile Organics	Composite	EPA 625
Pesticides/PCBs	Composite	EPA 608

* Composite or grab sample collection method per discharging permit requirements

Table 5 – Biosolids Analytical Methods (40 CFR Part 503.8)

Parameter	Collection Method	Analytical Testing Method
Metals, Total	Composite	SW 846 Method 6020
Mercury	Composite	SW 846 Method 7471
Phosphorus, Total	Composite	Standard Methods 4500 PC
Cyanide, Total	Composite	SW 846 Method 9010
Iron, Total	Composite	SW 846 Method 7000b
Potassium, Total	Composite	SW 846 Method 7000b
Phenols, Total	Composite	SW 846 Method 9065
Table II Toxic Organics:		
Volatile Organics	Composite	SW 846 Method 8260C
Semi-volatile Organics	Composite	SW 846 Method 8270C
Pesticides	Composite	SSW 846 Method 8081A
PCBs	Composite	SW 846 Method 8082

5.0 DETERMINATION OF FLOW VOLUMES

5.1 Plant Influent Flows

5.1.1 Dry Creek Water Reclamation Facility

Based on plant influent flow recorded data from 2020 through 2024, the average influent flow is 4.989 MGD, the highest daily flow observed during this time period was 8.669 MGD, and the maximum monthly average was 5.908 MGD.

5.1.2 Crow Creek Water Reclamation Facility

Based on plant influent flow recorded data from 2020 through 2024, the average influent flow is 2.756 MGD, the highest daily flow observed during this time period was 5.25 MGD, and the maximum monthly average was 4.088 MGD.

Table 6 provides the Influent flow volumes from 2020 – 2024 for the two Facilities:

Table 6 – Influent Flow Volumes

	2020	2021	2022	2023	2024	Avg or Maximum
Dry Creek Water Reclamation Facility						
Average Daily	5.058	5.091	4.657	4.788	5.349	4.989
Maximum Daily	6.233	8.427	6.224	8.669	6.947	8.669
Maximum Monthly Average	5.354	5.791	4.951	5.908	5.604	5.908
Crow Creek Water Reclamation Facility						
Average Daily	2.621	2.553	2.427	3.112	3.065	2.756
Maximum Daily	3.510	3.810	3.000	5.250	3.920	5.250
Maximum Monthly Average	2.865	3.216	2.660	4.088	3.265	4.088

5.2 Significant Industrial User Flow

Industrial Wastewater Discharge Permits issued by the BOPU to Significant Industrial Users (SIUs) include the requirement to monitor, record, and report discharge volumes. Discharge volumes have been calculated using data from January 1, 2020 through December 31, 2024. Table 7 provides the daily maximum discharge and the daily discharge average from SIUs.

Table 7 – SIU Discharge Volumes

Permitted SIU	Permit #	SIU Maximum Daily Discharge (gpd)	SIU Average Daily Discharge (gpd)	POTW	Total Maximum Daily Discharge (gpd)	Total Daily Avg Discharge (gpd)
Black Hills Energy - Cheyenne Prairie Generating Station	BHE-2022	59744	38220	Dry Creek		
Cheyenne Renewable Diesel Company Outfall 001A	CRDC-2022-Mod	323400	152860	Dry Creek	383144	191080
Walmart Distribution Center #7077	WALMART DC #7077 - 2022	12974	9481	Crow Creek		
WSP SPF Store No #46 LAUST Facility ID #3337	WSP-SPF46-2022	2766	461	Crow Creek		
Flying J LAUST Facility ID #3476	WSP-FLYING J- 2022	3709	573	Crow Creek		
Union Pacific Railroad	UPRR-2022	14543	5429	Crow Creek		
Cheyenne Renewable Diesel Company Outfall 001	CRDC-2022-Mod	410323	221475	Crow Creek	444315	237419

5.3 Combined Residential/Commercial Flow

Wastewater flow volumes from residential/commercial (non-controllable sources) were calculated by subtracting daily average SIU discharge volumes from the average daily influent flow volume. The average residential/commercial contribution for the Dry Creek Facility is 4.798 MGD and the average residential/commercial contribution for the Crow Creek Facility is 2.518 MGD.

5.4 Sludge Flow to Disposal

Sewer sludge produced at the Crow Creek facility is sent to the Dry Creek facility for further processing. The produced Biosolids are recycled as a fertilizer and soil amendment and land applied per EPA Biosolids requirements provided in 40 CFR Part 503. The BOPU records dry metric tons (DMT) of Biosolids produced. The average DMT for years 2022-2024 is 1310 DMT. Through sampling analysis (2022- 2024), the average percent solids of the produced Biosolids is 77.9%. The final calculated gallons of sludge flow sent to disposal at the Dry Creek Facility is 0.001103 (MGD). Table 8 provides the sludge flow to disposal calculations.

Table 8 – Sludge Flow to Disposal Calculation

	2022	2023	2024	Average for Final Calculation
Dry Metric Tons (DMT)	1547	1139	1243	1310
Percent Solids	0.814	0.782	0.742	0.779
Gallons (Total)	455865	349496	401998	402453
Gallons per Day	1249	958	1101	1103
Gallons to Disposal (MGD)	0.00125	0.00096	0.00110	0.001103

The standard formula for converting dry metric tons to gallons using percent solids is:

$$\text{Gallons} = (\text{dry tons} \times 2000 \text{ lbs./ton}) / (8.34 \text{ lbs./gal} \times \% \text{ solids (as decimal)})$$

Per EPA Guidance, determining gallons to disposal from DMT data eliminates the need for a specific gravity factor in the equations. Allowable Headworks calculations based on clean sludge standards (see section 10.4) will use a specific gravity factor of one.

6.0 POLLUTANTS OF CONCERN

The BOPU submitted a proposed Pollutants of Concern Evaluation for EPA Review on May 29, 2025. This submittal provided information pertaining to the screening and evaluation of pollutants and proposed final determination of POCs to be included in the Local Limits evaluation. The EPA reviewed and agreed with the Pollutants of Concern Evaluation on June 2, 2025.

Pollutants of Concern (POC) are pollutants that must be carried through the local limits evaluation. A POC is any pollutant that might reasonably be expected to be discharged to the POTW in sufficient amounts to cause pass through or interference, cause problems in its collection system, cause operational problems, or jeopardize its workers.

EPA Local Limits Development Guidance identifies POCs as any pollutants falling into the following categories:

1. National POCs - Arsenic, Cadmium, Copper, Chromium, Cyanide, Lead, Mercury, Molybdenum, Nickel, Selenium, Silver, Zinc, 5-day Biochemical Oxygen Demand, Total Suspended Solids, and Ammonia;
2. Any pollutant listed in your State Water Quality Standards for your receiving water or are listed in your NPDES permit;

3. Any pollutant specified in 40 CFR 122 Appendix D, Tables II or V that was detected in an influent or effluent sample and its concentration was equal to or greater than 0.05 mg/l;
4. Any pollutant that is present at 0.01 mg/l or greater (above the method detection limit if MDL is >0.01 mg/l) and has a Bioconcentration Factor (BCF) of 300 or greater;
5. Any other pollutant not covered by category 3 or 4, that is determined to be present at equal to, or greater than, 0.1 mg/l in the influent or effluent samples and that may be toxic to the POTW, receiving waters, worker health or biosolids if discharged;
6. Any pollutant with a current average influent (headworks) loading greater than the POTW design loading;
7. Any pollutant identified through WET characterization studies as being present and suspected or responsible for toxicity;
8. Any pollutant identified in biosolids analytical results which is not a pollutant that would be expected or is at a concentration that would be considered atypical;
9. Any pollutant that has the potential to interfere with POTW operations or are known to cause operational or maintenance problems;
10. Any other pollutant specifically designated by the POTW Pretreatment Program and/or Approval Authority as a pollutant of concern.

EPA recommends that any pollutant that has a “reasonable potential” to be discharged in amounts that could exceed water quality standards or other environmental criteria should be considered a POC and evaluated accordingly.

The BOPU evaluated pollutants per the guidance documents. To further identify potential pollutants of concern, the BOPU performed quantitative statistical analyses to determine if a Reasonable Potential (RP) exists to violate a state water quality standard or permit limit. The RP evaluation was based upon EPA’s guidance for establishing reasonable potential as found in Chapter 3 of the Technical Support Document for Water Quality-based Toxics Control (USEPA 1991). After the comprehensive evaluation, the Pollutants of Concern identified in Table 9 have been carried through the Local Limits Evaluation.

Table 9 – Pollutants of Concern

Pollutant	Comments
Arsenic (As)	Existing Local Limit / National POC
Cadmium (Cd)	Existing Local Limit / National POC
Chromium (Cr)	Existing Local Limit / National POC
Copper (Cu)	Existing Local Limit / National POC
Lead (Pb)	Existing Local Limit / National POC
Mercury (Hg)	Existing Local Limit / National POC
Molybdenum (Mo)	Existing Local Limit / National POC
Nickel (Ni)	Existing Local Limit / National POC
Selenium (Se)	Existing Local Limit / National POC
Silver (Ag)	Existing Local Limit / National POC
Zinc (Zn)	Existing Local Limit / National POC
BOD	Existing Local Limit / National POC
TSS	Existing Local Limit / National POC

7.0 MINIMUM DETECTION LEVELS AND BELOW DETECTION LEVELS

Environmental data sets frequently contain values that are below analytical detection methods. In the TBLL evaluation, the BOPU is following the EPA conservative position that a pollutant is present in some concentration even if the analysis provides a result below the reporting limit (non-detection). During the calculations of influent and effluent average concentrations used in maximum allowable loadings computations, the BOPU used a conservative approach of substituting $\frac{1}{2}$ the reporting limit for non-detections.

During the Pollutants of Concern evaluation, the pollutants warranting Reasonable Potential statistical analysis, the Robust Regression on Order Statistics (ROS) method to “censor” the non-detection results was used. The ROS method plots the detected values on a probability plot then calculates a linear regression line in order to approximate the parameters of the underlying (assumed) distribution. This fitted distribution is then utilized to generate imputed estimates for each of the censored measurements, which are then combined with the known (detected) values to the summary statistics of interest.

8.0 ENVIRONMENTAL CRITERIA

Discharge limitations have been established in the Wyoming Department of Environmental Quality, Water Quality Rules and Regulations, Chapter 1, EPA National Recommended Water Quality Criteria - Aquatic Life and Human Health Criteria Tables,

and Authorization to Discharge under Wyoming Pollution Discharge Elimination System (WPDES) Permits WY0022934 and WY0022381. Standards for the use or disposal of Sewage Sludge have been established in 40 CFR Part 503. The limitations for the Pollutants of Concern are provided in Table 14.

Acute and Chronic Water Quality Standards (WQS) are established for the protection of aquatic life and human health. These standards establish numeric limitations on the amount of pollutants and chemicals that can be present in surface water before it is likely to cause harm to human health or have effects on plants and animal life such as survival, growth, or reproduction abilities.

8.1 Hardness-Dependent Acute and Chronic Criteria

Water Quality Standards for some metals are established based on the measured Hardness of the receiving stream waters. The average water hardness for the Dry Creek facility is 210.4 mg/L as CaCO₃ and the average water hardness for the Crow Creek facility is 229.9 mg/L as CaCO₃. The average hardness measurements are provided in the Statement of Basis issued by the Water Quality Division of the Wyoming Department of Environmental Quality with each facilities discharge permits.

Per the Wyoming Water Quality Rules and Regulations, Chapter 1, the formulas provided in Table 10 are used to calculate numeric hardness-dependent metal limitations (Dissolved Criteria).

Table 10 – Hardness Dependent Metals

Parameter	Acute Value (µg/L)
Cadmium, Dissolved	$e(1.0166[\ln(\text{hardness})] - 3.924) * (1.136672 - [(\ln \text{hardness})(0.041838)])$
Chromium (III), Dissolved	$e(1.0166[\ln(\text{hardness})] + 3.7256) * (0.316)$
Copper, Dissolved	$e(0.9422[\ln(\text{hardness})] - 1.700) * (0.960)$
Lead, Dissolved	$e(1.273[\ln(\text{hardness})] - 1.460) * (1.46203 - [(\ln \text{hardness})(0.145712)])$
Nickel, Dissolved	$e(0.8460[\ln(\text{hardness})] + 2.255)(0.998)$
Silver, Dissolved	$e(1.72[\ln(\text{hardness})] - 6.52)(0.85)(0.5)$
Zinc, Dissolved	$e(0.8473[\ln(\text{hardness})] + 0.884)(0.978)$
Parameter	Chronic Value (µg/L)
Cadmium, Dissolved	$e(0.7409[\ln(\text{hardness})] - 4.719) * (1.101672 - [(\ln \text{hardness})(0.041838)])$
Chromium (III), Dissolved	$e(0.8190[\ln(\text{hardness})] + 0.6848) * (0.860)$
Copper, Dissolved	$e(0.8545[\ln(\text{hardness})] - 1.702) * (0.960)$
Lead, Dissolved	$e(1.273[\ln(\text{hardness})] - 4.705) * (1.46203 - [(\ln \text{hardness})(0.145712)])$
Nickel, Dissolved	$e(0.8460[\ln(\text{hardness})] + 0.0584)(0.997)$
Zinc, Dissolved	$e(0.8473[\ln(\text{hardness})] + 0.884)(0.986)$

Based on the formulas provided in Table 10, the BOPU has calculated hardness-dependent limitations (Dissolved Criteria) expressed in mg/L.

Hardness Dependent Parameters Aquatic Life Criteria	Dry Creek Hardness 210.4 mg/L as CaCO ₃		Crow Creek Hardness 229.9 mg/L as CaCO ₃	
	Acute	Chronic	Acute	Chronic
Cadmium (D) (mg/L)	0.0041	0.0004	0.0045	0.0004
Chromium III (D) (mg/L)	1.0478	0.1363	1.1267	0.1466
Copper (D) (mg/L)	0.0271	0.0169	0.0294	0.0182
Lead (D) (mg/L)	0.1437	0.0056	0.1578	0.0061
Nickel (D) (mg/L)	0.8785	0.0976	0.9469	0.1052
Silver (D) (mg/L)	0.0062	N/A	0.0072	N/A
Zinc (D) (mg/L)	0.2201	0.2219	0.2372	0.2392

8.2 Metals Translator

Water Quality Standards for some metals are expressed as dissolved criteria. Most metals measurements, however, are reported in the total or total recoverable criteria. Total and total recoverable metals concentrations are always at least as high as dissolved metals concentrations because a fraction of the metal has sorbed to particulate matter in the water. Per EPA Local limits Development Guidance (2004), POTWs should convert WQS for metals expressed as dissolved criteria into total recoverable criteria before using them to calculate water quality-based Allowable Headworks Loadings (AHLs). Based on this EPA recommendation, where a pollutant has been identified as being detected above the detection level through dissolved or total analysis, the POTW will evaluate the pollutant as total recoverable. Established conversion factors will be used when environmental criteria are expressed as limitations in the dissolved or partially dissolved form to convert the criteria to total recoverable in the development of a local limit.

The BOPU has converted WQS expressed as dissolved to total recoverable using the Conversion Tables provided in the EPA “The Metals Translator: Guidance for Calculating a Total Recoverable Permit Limit from a Dissolved Criterion”, (1996). The conversion factors for Cadmium and Lead are hardness-dependent. Tables 11 and 12 provide the dissolved to total recoverable conversion tables and final calculated aquatic life Water Quality Standards in total recoverable which will be used in the Local Limits Evaluation.

Table 11 – Metals Translator Dissolved to Total Recoverable

Conversion Factors Dissolved to Total	Dry Creek		Crow Creek	
	Acute	Chronic	Acute	Chronic
Cadmium * Hardness Dependent	0.91288	0.87788	0.90917	0.87417
Chromium III	0.316	0.86	0.316	0.86
Copper	0.96	0.96	0.96	0.96
Lead * Hardness Dependent	0.68261	0.68261	0.6697	0.6697
Mercury	0.85	N/A	0.85	N/A
Nickel	0.998	0.997	0.998	0.997
Silver	0.85	N/A	0.85	N/A
Zinc	0.978	0.986	0.978	0.986
Cadmium Hardness Dependent Conversion Factors	Acute: CF = 1.136672 - [ln (hardness) (0.041838)] Chronic: CF = 1.101672 - [ln (hardness) (0.041838)]			
Lead Hardness Dependent Conversion Factors	Acute and Chronic: CF = 1.46203 - [ln(hardness) (0.145712)]			

Table 12 – Calculated Final Aquatic Life Water Quality Standards

Calculated Final Aquatic Life WQS	Dry Creek		Crow Creek	
	Acute	Chronic	Acute	Chronic
Cadmium (T) (mg/L)	0.0045	0.00046	0.0049	0.00049
Chromium III (T) (mg/L)*	1.7645	0.1554	1.8973	0.1671
Copper (T) (mg/L)	0.0282	0.0176	0.0306	0.0189
Lead (T) (mg/L)	0.2105	0.0082	0.2098	0.0082
Mercury	0.0016	N/A	0.0016	N/A
Nickel (T) (mg/L)	0.8803	0.0979	0.9488	0.1055
Silver (T) (mg/L)	0.0073	N/A	0.0083	N/A
Zinc (T) (mg/L)	0.2250	0.2250	0.2425	0.2425

*State Water Quality Standards have been established by the Wyoming Department of Environmental Quality for Chromium III (aquatic life, hardness dependent). EPA Region 8 recommends setting local limits for Total Chromium based upon the Chromium (III) water quality standards.

Formula for calculation of final WQS:

$$EC_t = EC_d * [(1.0 - CF) + 1.0]$$

EC_t = Final Environmental Criteria (Total Recoverable) Table 12

EC_d = Hardness Dependent Environmental Criteria (expressed as Dissolved) per Table 10.

CF = Conversion Factor per Table 11.

8.3 Sludge (Biosolids) Land Application Criteria

The following table provides limitations or standards per 40 CFR Part 503.13 to protect public health and the environment from any reasonably anticipated adverse effects of certain pollutants and contaminants that may be present in biosolids.

Table 13 – Biosolids Land Application Criteria

	Biosolids (mg/kg) ⁽¹⁾		Biosolids (mg/kg) ⁽¹⁾
Arsenic (total)	41	Molybdenum (total)	75
Cadmium (total)	39	Nickel (total)	420
Copper (total)	1500	Selenium (total)	100
Lead (total)	300	Zinc (total)	2800
Mercury (total)	17		

(1) EPA 40 CFR Part 503.13 Allowable Pollutant Concentrations for Land Application

8.4 Final Environmental Criteria Table

Table 14 – Environmental Criteria

Dry Creek Water Reclamation Facility Environmental Criteria						
Parameter	Aquatic Life WQS Acute (1) mg/L	Aquatic Life WQS Chronic (1) mg/L	Human Health WQS (2) mg/L	WPDES Permit Chronic (3) mg/L	EPA Sewer Sludge (4) mg/kg	POTW Design Capacity lbs/Day
Arsenic	0.34	0.15	0.01		41	
Cadmium (5)(6)	0.0045	0.00046		0.00046	39	
Chromium (5)(6)(7)	1.7645	0.1554				
Copper (5)(6)	0.0282	0.0176		0.0176	1500	
Lead (5)(6)	0.2105	0.0082			300	
Mercury	0.0016 (6)	0.00077	0.000051		17	
Molybdenum					75	
Nickel (5)(6)	0.8803	0.0979	4.6		420	
Selenium	0.02	0.005	4.2	0.005	100	
Silver (5)(6)	0.0073					
Zinc (5)(6)	0.225	0.225	26		2800	
BOD						17200
TSS						19940

Crow Creek Water Reclamation Facility Environmental Criteria						
Parameter	Aquatic Life WQS Acute (1) mg/L	Aquatic Life WQS Chronic (1) mg/L	Human Health WQS (2) mg/L	WPDES Permit Chronic (3) mg/L	EPA Sewer Sludge (4) mg/kg	POTW Design Capacity lbs/Day
Arsenic	0.34	0.15	0.01		41	
Cadmium (5)(6)	0.0049	0.00049		0.00049	39	
Chromium (5)(6)(7)	1.8973	0.1671				
Copper (5)(6)	0.0306	0.0189		0.0189	1500	
Lead (5)(6)	0.2098	0.0082			300	
Mercury	0.0016 (6)	0.00077	0.000051		17	
Molybdenum						
Nickel (5)(6)	0.9488	0.1055	4.6		420	
Selenium	0.02	0.005	4.2	0.005	100	
Silver (5)(6)	0.0083					
Zinc (5)(6)	0.2425	0.2425	26		2800	
BOD						9340
TSS						9340

- (1) Aquatic Life Wyoming Numeric Water Quality Standards (Acute and Chronic) applicable to all surface waters of the State designated for aquatic life. Wyoming Water Quality Rules and Regulations Chapter 1.
- (2) Human Health Wyoming Numeric Water Quality Standards applicable to surface waters of the State designated as fish consumption classification. Wyoming Water Quality Rules and Regulations Chapter 1.
- (3) Numeric Limitation provided in WYPDES Permits WY0022934 (Dry Creek Water Reclamation Facility) and WY0022381 (Crow Creek Water Reclamation Facility)
- (4) EPA Standards for the Use or Disposal of Sewer Sludge per 40 CFR Part 403, Table 3 maximum monthly average pollutant concentration for land application except Molybdenum which has a ceiling concentration for land application per 40 CFR Part 403, Table 1.
- (5) Hardness Dependent, see Table 10
- (6) Aquatic Life WQS expressed as dissolved converted to total recoverable, see Table 11
- (7) Chromium Total Recoverable based on Chromium III standards

9.0 REMOVAL EFFICIENCY CALCULATIONS

Removal efficiency is the fraction or percentage of the influent pollutant loading that is removed from the waste stream across an entire wastewater treatment works or specific processes. Per EPA guidance, there are three removal efficiency calculation methodologies commonly used by POTWs: the Average Daily Removal Efficiency (ADRE) method, the Mean Removal Efficiency (MRE) method, and the Decile method. Each of these methods uses a data set consisting of influent and lagged effluent POTW sampling results.

Removal efficiency methodologies vary by degree of data quality and calculation method. A review of the data quality has been performed. Per EPA guidance, influent/effluent paired sampling events in which the influent level was below the detection limit were removed from the data set before calculating removal efficiencies. Because plant influent flows are relatively consistent, the BOPU calculated the removal efficiencies based on pollutant concentrations versus influent/effluent pollutant loadings. Data used in the removal efficiency calculations was obtained from influent/effluent pollutant scans (January 1, 2020 through December 31, 2024). A summary of the calculated removal efficiencies for the three methods is provided in Table 15.

9.1 ADRE Method

The ADRE is calculated using matched pairs of influent/effluent sampling results. The removal efficiency percentage for each matched pair is calculated using the following formula:

Daily Removal Efficiency (DRE) = $\frac{[\text{Influent Concentration} - \text{Effluent Concentration}]}{\text{Influent Concentration}} \times 100$

The individual daily removal results are then averaged to determine the final removal percentage (ADRE).

For the Dry Creek facility, removal efficiencies using the ADRE method were not calculated for Cadmium, Chromium, Lead, Mercury, and Silver due to the limited number of matched pairs of influent/effluent sampling results above the reporting limit.

For the Crow Creek facility, removal efficiencies using the ADRE method not calculated for Cadmium, Chromium, Lead, and Silver due to the limited number of matched pairs of influent/effluent sampling results above the reporting limit.

9.2 MRE Method

The MRE is calculated by using the same general formula as for the ADRE, but instead of averaging the individual matched pair daily removal percentages, the entire set of influent values is first averaged to determine the average influent value and the same is done for the entire set of effluent values. These average values are then used in the removal efficiency formula shown above. The MRE can be based on influent and effluent sample values that are not always paired. If the influent level was below the detection limit, the associated effluent result was removed from the data set. Censored data (1/2 RL) was substituted for effluent results reported as below the detection limit threshold when there was an associated influent result at or above the detection limit.

For the Dry Creek facility, the removal efficiency using the MRE method was not calculated for Silver due to the limited number of influent and effluent sampling results above the reporting limit.

For the Crow Creek facility, the removal efficiency using the MRE method was not calculated for Cadmium, and Silver due to the limited number of influent or effluent sampling results above the reporting limit.

9.3 Decile Method

The Decile Method, unlike the ADRE and MRE methods, considers how often the actual plant daily removal efficiency will be above or below a specified removal rate, thereby taking into account the variability of POTW removal efficiencies over time. Each decile is the value below which a certain percentage of the DREs fall. For

example, the first decile is the value below which 10% of the DREs fall. Similarly, the second decile is the value below which 20% of the DREs fall, on up to the ninth decile, which is the value below which 90% of the DREs fall. The fifth decile is the median and half of the DREs fall below this number. The main value of the decile approach is that it provides an estimate of how often a POTW is expected to exceed certain removal values, such as the ADRE and MRE. For the example, assuming the MRE for a particular pollutant is 35%; if the MRE falls in the third decile (25%ile), the plant's actual removal rate will exceed the MRE 75% of the time. If an Allowable Headworks Loading for that particular pollutant is based on sludge land application criteria and used the 35% MRE and the plant has an actual removal efficiency of 70%, more of the pollutant may end up in the sludge than allowed for, thus resulting in a potential exceedance. To apply the decile approach, a minimum of nine "matched pairs" of DREs are required. With this limitation, the BOPU was able to calculate Decile Removal Efficiencies for Arsenic, Copper, Molybdenum and Zinc for both facilities.

9.4 Literature Values

Removal efficiencies are based largely on site-specific conditions such as climate, POTW design, operation and maintenance, plant conditions, and sewage characteristics. Therefore, EPA strongly suggests that site-specific data be used to calculate removal efficiencies. However, after conducting site-specific sampling, there are some pollutants that do not have adequate data (observations above the reporting limit) to confidently calculate removals. In these instances, POTWs may selectively use removal efficiencies (Literature Values) reported by other POTWs. EPA urges POTWs to use performance data from plants employing the same treatment technology and similar contributing sources. The EPA guidance provides a listing of removal efficiency data for priority pollutants gathered from other POTWs. The EPA Literature Value removal efficiency based POTWs with similar treatment processes was used for Cadmium and Silver at the Crow Creek facility, and Cadmium, Mercury, and Silver at the Dry Creek facility.

Table 15 – Calculated Removal Efficiencies

Dry Creek Water Reclamation Facility										
	ADRE %	MRE %	Decile %							Literature Value
			10%ile	20%ile	25%ile	50%ile	75%ile	80%ile	90%ile	
Arsenic, Total	26.97	2.82	-8	0	0	44.83	50	50	50	45
Cadmium, Total	N/A	-39.18	N/A							67
Chromium, Total	N/A	57.90	N/A							82
Copper, Total	78.66	91.71	59.84	86.23	87.43	93.75	96.79	96.97	97.3	86
Lead, Total	N/A	87.05	N/A							61
Mercury, Total	N/A	97.12	N/A							60
Molybdenum, Total	34.24	44.92	0	1.6	7.688	49.39	53.68	61.96	70.14	N/A
Nickel, Total	60.10	65.76	N/A							42
Selenium, Total	81.20	77.54	N/A							50
Silver, Total	N/A	N/A	N/A							75
Zinc, Total	68.46	74.22	39.83	55.82	64.02	78.57	83.38	83.69	84.87	79
BOD	N/A	98.7	N/A							N/A
TSS	N/A	98.7	N/A							N/A

Crow Creek Water Reclamation Facility										
	ADRE %	MRE %	Decile %							Literature Value
			10%ile	20%ile	25%ile	50%ile	75%ile	80%ile	90%ile	
Arsenic, Total	38.06	21.66	25.26	31.72	31.82	45.83	50	50	50	45
Cadmium, Total	N/A	N/A	N/A							67
Chromium, Total	N/A	59.34	N/A							82
Copper, Total	75.55	83.08	58.61	65.49	67.86	71.23	81.98	86.59	93.45	86
Lead, Total	N/A	96.74	N/A							61
Mercury, Total	84.30	93.72	N/A							60
Molybdenum, Total	29.24	41.82	0	0	0	33.33	50	63.33	69.77	N/A
Nickel, Total	40.80	49.14	N/A							42
Selenium, Total	69.56	76.36	N/A							50
Silver, Total	N/A	N/A	N/A							75
Zinc, Total	71.48	76.56	55.25	60.79	61.95	67.95	76.5	80.82	88	79
BOD	N/A	98.7	N/A							N/A
TSS	N/A	98.7	N/A							N/A

9.5 Determination of Final Removal Efficiencies

Per EPA guidance, in developing local limits, appropriate removal efficiencies must be selected for calculation of AHLs for each pollutant. EPA recommends the MRE over the ADRE method if less than ten data pairs are available because the MRE method is generally less sensitive to variation in daily removal efficiencies. EPA also recommends that POTWs consider using the decile approach or the MRE method because they better account for variabilities in removal efficiencies over time. In the final

determination of removal efficiencies, which are used in the Allowable Headworks Calculations, the BOPU, using EPA recommendations and best professional judgement, will use the Decile method, when data sets allow, at the 50th percentile for pollutants that are subject to biosolids land application limitations per 40 CFR Part 503. The decile method will adequately protect against exceedances of sludge disposal standards. The Mean Removal Efficiency (MRE) will be used for pollutants with insufficient data sets for Decile Method calculation or have less than ten data pairs of influent/effluent sampling results. Literature median values provided by the EPA for similar POTWs will be used when data quality prohibits the use of the ADRE, MRE, and Decile methods. The Removal Efficiencies used in the Allowable Headworks Loading calculations are provided in Table 16.

Table 16 – Final Determination of Percent Removal Efficiencies

Dry Creek Water Reclamation Facility		
	Removal Efficiency (%)	Criteria
Arsenic, Total	44.83	Decile
Cadmium, Total	67.00	Literature
Chromium, Total	57.90	MRE
Copper, Total	93.75	Decile
Lead, Total	87.05	MRE
Mercury, Total	60	Literature
Molybdenum, Total	49.39	Decile
Nickel, Total	65.76	MRE
Selenium, Total	77.54	MRE
Silver, Total	75.00	Literature
Zinc, Total	67.95	Decile
BOD	98.7	MRE
TSS	98.7	MRE

Crow Creek Water Reclamation Facility		
	Removal Efficiency (%)	Criteria
Arsenic, Total	45.83	Decile
Cadmium, Total	67.00	Literature
Chromium, Total	59.34	MRE
Copper, Total	71.23	Decile
Lead, Total	96.74	MRE
Mercury, Total	93.72	MRE
Molybdenum, Total	33.33	Decile
Nickel, Total	49.14	MRE
Selenium, Total	76.36	MRE
Silver, Total	75.00	Literature
Zinc, Total	67.95	Decile
BOD	95.9	MRE
TSS	71.0	MRE

10.0 CALCULATION OF ALLOWABLE HEADWORKS LOADING

EPA guidance recommends that POTWs base their Technically-Based Local Limits (TBLLs) on Maximum Allowable Headworks Loading (MAHL) calculated for each pollutant of concern. A MAHL is the upper limit of pollutant loading at which a POTW will not violate any treatment plant or environmental criteria developed to prevent process inhibition, interference, effluent limitation, or sludge quality standards. MAHLs

are the basis for local limits. A pollutant's MAHL is determined by first calculating the Allowable Headworks Loading (AHL) for each limiting criterion for each pollutant; the most stringent AHL would be the MAHL. The AHL for each pollutant is a function of limiting criteria (environmental or process), plant flows, and plant removal efficiency.

10.1 Allowable Headworks Loading Based on Acute and Chronic WPDES Permit Criteria

As provided in Table 14, there are no numeric acute effluent limitations listed in either facilities' discharge permits for the pollutants being evaluated in this TBLL study. There are chronic (monthly) effluent limitations listed in both facilities discharge permits for Cadmium, Copper, and Selenium. The calculated AHLs for these pollutants are provided in Table 17.

Per EPA guidance, the standard equation used to determine AHLs based on NPDES Permit Criteria is:

$$\text{AHL} = (8.34 * \text{WPDES}_{\text{CRIT}} * Q_{\text{POTW}}) / (1 - (\text{RE}/100))$$

AHL = Allowable Headworks Loading in lbs/Day

$\text{WPDES}_{\text{CRIT}}$ = WYPDES Permit Chronic Limitation in mg/L (Table 14)

Q_{POTW} = Maximum Average Monthly Plant Influent Flow Rate, in MGD (Table 6)

RE = Removal Efficiency (Table 16)

Table 17 – Allowable Headworks Loading Based on WPDES Criteria

Dry Creek Water Reclamation Facility					
		WPDES Criteria (mg/L)	Maximum Monthly Influent Flow (MGD)	Removal Efficiency (%)	Allowable Headworks Loading (lbs/Day)
Cadmium	8.34	0.00046	5.908	67.00	0.0687
Copper	8.34	0.0176	5.908	93.75	13.8752
Selenium	8.34	0.005	5.908	77.54	1.0967
Crow Creek Water Reclamation Facility					
		WPDES Criteria (mg/L)	Maximum Monthly Influent Flow (MGD)	Removal Efficiency (%)	Allowable Headworks Loading (lbs/Day)
Cadmium	8.34	0.00049	4.088	67.00	0.0506
Copper	8.34	0.0189	4.088	71.23	2.2397
Selenium	8.34	0.005	4.088	76.36	0.7212

10.2 Allowable Headworks Loading Based on Acute Water Quality Criteria

The numeric acute Water Quality Criteria for both facilities is provided in Table 14. The basis for these standards is the Aquatic Life Wyoming Numeric Water Quality Standards applicable to all surface waters of the State designated for aquatic life. Wyoming Water Quality Rules and Regulations Chapter 1.

As discussed in Section 8.0, several of the parameters in which TBLLs are being developed are hardness dependent and the WQS criteria expressed as dissolved have been translated to total recoverable.

Per EPA guidance, the standard equation used to determine AHLs based on Acute Water Quality Standards is:

$$AHL = (8.34 [C_{AWQ} (Q_{STR} + Q_{POTW}) - (Q_{STR} * C_{STR})] / (1 - (RE/100)))$$

AHL = Allowable Headworks Loading in lbs/Day

C_{AWQ} = Acute Environmental Criteria, in mg/L (Table 14)

Q_{STR} = Acute Receiving Stream (upstream) Flow Rate, in mgd

Q_{POTW} = Maximum Average Monthly Plant Influent Flow Rate, in MGD (Table 6)

C_{STR} = Average Receiving Stream Background Concentration, in mg/L

RE = Removal Efficiency (Table 16)

As discussed in Section 3.6, per the Statement of Basis documents provided by the Wyoming Department of Environmental Quality Water Quality Division with both facilities discharge permits, “all water quality-quality based effluent limits established in the permits are equal to instream standards because zero receiving water dilution was considered in establishing the facilities water-quality based effluent limits.” As such, the receiving stream background concentrations (C_{STR}) and acute receiving stream flow rates (Q_{STR}) are set at zero in the calculations. The calculated AHLs based on Acute Water Quality Standards are provided in Table 18.

Table 18 – Allowable Headworks Loading Based on Acute WQS Criteria

Dry Creek Water Reclamation Facility					
		Acute WQS Criteria (mg/L)	Maximum Monthly Influent Flow (MGD)	Removal Efficiency (%)	Allowable Headworks Loading (lbs/Day)
Arsenic	8.34	0.34	5.908	44.83	30.3656
Cadmium	8.34	0.0045	5.908	67.00	0.6719
Chromium	8.34	1.7645	5.908	57.90	206.5227
Copper	8.34	0.0282	5.908	93.75	22.2319
Lead,	8.34	0.2105	5.908	87.05	80.0942
Mercury	8.34	0.0016	5.908	59.0	0.1923
Molybdenum	8.34	No Criteria			
Nickel	8.34	0.8803	5.908	65.76	126.6787
Selenium	8.34	0.02	5.908	77.54	4.3876
Silver	8.34	0.0073	5.908	75.00	1.4388
Zinc	8.34	0.225	5.908	67.95	34.5908
BOD		17200 (LBS/Day)			17200
TSS		19940 (LBS/Day)			19940
Crow Creek Water Reclamation Facility					
		Acute WQS Criteria (mg/L)	Maximum Monthly Influent Flow (MGD)	Removal Efficiency (%)	Allowable Headworks Loading (lbs/Day)
Arsenic	8.34	0.34	4.088	45.83	21.3992
Cadmium	8.34	0.0049	4.088	67.00	0.5062
Chromium	8.34	1.8973	4.088	59.34	159.0910
Copper	8.34	0.0306	4.088	71.23	3.6263
Lead,	8.34	0.2098	4.088	96.74	219.4142
Mercury	8.34	0.0016	4.088	93.72	0.8686
Molybdenum	8.34	No Criteria			
Nickel	8.34	0.9488	4.088	49.14	63.6027
Selenium	8.34	0.02	4.088	76.36	2.8844
Silver	8.34	0.0083	4.088	75.00	1.1319
Zinc	8.34	0.2425	4.088	67.95	25.7965
BOD		9340 (lbs/Day)			9340
TSS		9340 (lbs/Day)			9340

10.3 Allowable Headworks Loading Based on Chronic Water Quality Criteria

The numeric Chronic Water Quality Criteria for both facilities is provided in Table 14. The basis for these standards is the Chronic Aquatic Life criteria and Human Health (consumption of fish) standards established by the state of Wyoming that are applicable to all surface waters of the State as provided in the Wyoming Water Quality Rules and Regulations Chapter 1. As discussed in Section 8.0, several of the parameters in which TBLLs are being developed are hardness dependent and the WQS criteria expressed as dissolved have been translated to total recoverable.

Per EPA guidance, the standard equation used to determine AHLs based on Chronic Water Quality Standards is:

$$\text{AHL} = (8.34 [C_{\text{CWQ}} (Q_{\text{STR}} + Q_{\text{POTW}}) - (Q_{\text{STR}} * C_{\text{STR}})] / (1 - (\text{RE}/100))$$

AHL = Allowable Headworks Loading in lbs/Day

C_{CWQ} = Chronic Environmental Criteria, in mg/L (Table 14)

Q_{STR} = Chronic Receiving Stream (upstream) Flow Rate, in mgd

Q_{POTW} = Maximum Average Monthly Plant Influent Flow Rate, in MGD (Table 6)

C_{STR} = Average Receiving Stream Background Concentration, in mg/L

RE = Removal Efficiency (Table 16)

As discussed in Section 3.6, per the Statement of Basis documents provided by the Wyoming Department of Environmental Quality Water Quality Division with both facilities discharge permits, “all water quality-quality based effluent limits established in the permits are equal to instream standards because zero receiving water dilution was considered in establishing the facilities water-quality based effluent limits.” As such, the receiving stream background concentrations (C_{STR}) and acute receiving stream flow rates (Q_{STR}) are set at zero in the calculations. The calculated AHLs based on Chronic Water Quality Standards are provided in Table 19.

Table 19 – Allowable Headworks Loading Based on Chronic WQS Criteria

Dry Creek Water Reclamation Facility							
		Chronic WQS Criteria (mg/L)	Human Health WQS Criteria (mg/L)	Maximum Monthly Influent Flow (MGD)	Removal Efficiency (%)	Chronic WQS AHL Loading (lbs/Day)	Human Health WQS AHL (lbs/Day)
Arsenic	8.34	0.15	0.01	5.908	44.83	13.3958	0.8931
Cadmium	8.34	0.00046		5.908	67.00	0.0687	
Chromium	8.34	0.1554		5.908	57.90	18.1874	
Copper	8.34	0.0176		5.908	93.75	13.8744	
Lead,	8.34	0.0082		5.908	87.05	3.11987	
Mercury	8.34	0.00077	0.000051	5.908	60.00	0.0948	0.0063
Molybdenum	8.34	No Criteria					
Nickel	8.34	0.0979	4.6	5.908	65.76	14.0879	661.9440
Selenium	8.34	0.005	4.2	5.908	77.54	1.0966	921.1856
Silver	8.34			5.908	75.00		
Zinc	8.34	0.225	2.6	5.908	67.95	34.5888	3996.9229
Crow Creek Water Reclamation Facility							
		Chronic WQS Criteria (mg/L)	Human Health WQS Criteria (mg/L)	Maximum Monthly Influent Flow (MGD)	Removal Efficiency (%)	Chronic WQS AHL Loading (lbs/Day)	Human Health WQS AHL (lbs/Day)
Arsenic	8.34	0.15	0.01	4.088	45.83	9.4417	0.6294
Cadmium	8.34	0.00049		4.088	67.00	0.0506	
Chromium	8.34	0.1671		4.088	59.34	14.0141	
Copper	8.34	0.0189		4.088	71.23	2.2400	
Lead,	8.34	0.0082		4.088	96.74	8.5848	
Mercury	8.34	0.00077	0.000051	4.088	93.72	0.4183	0.0277
Molybdenum	8.34	No Criteria					
Nickel	8.34	0.1055	4.6	4.088	49.14	7.0727	308.3838
Selenium	8.34	0.005	4.2	4.088	76.36	0.7213	605.8801
Silver	8.34						
Zinc	8.34	0.2425	2.6	4.088	67.95	25.7989	2766.0713

10.4 Allowable Headworks Loading Based on Land Application Criteria

Per EPA guidance, the standard equation used to determine AHLs based on biosolids Land Application and Surface Disposal is:

$$\text{AHL} = [(8.34) (C_{\text{SLDSTD}}) (PS/100) (Q_{\text{SLDG}}) (G_{\text{SLDG}})] / (RE/100)$$

AHL = Allowable Headworks Loading in lbs/Day

C_{SLDSTD} = Sludge Standard, in mg/kg dry sludge (Table 14)

PS = Percent Solids of Sludge to Disposal = 77.9% (Section 5.4)

Q_{SLDG} = Total Sludge Flow Rate to Disposal = 0.001103 MGD (Table 8)

G_{SLDG} = Specific Gravity of Sludge, in kg/L=1 (Section 5.4)

RE = Removal Efficiency from Headworks to Effluent, in decimal (Table 16)

The calculated AHLs based on Sewage Sludge Criteria are provided in Table 20. All sludge generated at the Crow Creek facility is transferred to the Dry Creek facility for treatment.

Table 20 – Allowable Headworks Loading Based on Biosolids Land Application Criteria

Dry Creek Water Reclamation Facility							
		Sludge Standard (mg/kg)	% Solids of Sludge to Disposal	Sludge to Disposal (MGD)	Specific Gravity of Sludge	Removal Efficiency (%)	Allowable Headworks Loading (lbs/Day)
Arsenic	8.34	41	77.9	0.001103	1	44.83	0.6554
Cadmium	8.34	39	77.9	0.001103	1	67.00	0.4171
Chromium	8.34	No Criteria					
Copper	8.34	1500	77.9	0.001103	1	93.75	11.4660
Lead	8.34	300	77.9	0.001103	1	87.05	2.4697
Mercury	8.34	17	77.9	0.001103	1	60.00	0.2030
Molybdenum	8.34	75 ⁽¹⁾	77.9	0.001103	1	49.39	1.0882
Nickel	8.34	420	77.9	0.001103	1	65.76	4.5769
Selenium	8.34	100	77.9	0.001103	1	77.54	0.9242
Silver	8.34	No Criteria					
Zinc	8.34	2800	77.9	0.001103	1	67.95	29.5299

(1) Molybdenum Standard Based on Table 1 Maximum Ceiling Concentration Per 40 CFR Part 503.

11.0 MAXIMUM OF ALLOWABLE HEADWORKS LOADING (MAHL)

The Maximum Allowable Headworks Loading (MAHL) for each pollutant of concern is defined as the most stringent AHL. The Table 21 provides the MAHL based on the AHL calculations provided in Section 10.0.

Table 21 – Maximum Allowable Headworks Loadings (MAHL)

Dry Creek Water Reclamation Facility						
	WPDES Criteria (lbs/Day)	Acute WQS AHL (lbs/Day)	Chronic WQS AHL (lbs/Day)	Human Health AHL (lbs/Day)	Biosolids AHL (lbs/Day)	Maximum Allowable Headworks Loading (lbs/Day)
Arsenic		30.3656	13.3966	0.8931	0.6554	0.6554
Cadmium	0.0687	0.6719	0.06867		0.4171	0.0687
Chromium		206.5227	18.1885			18.1885
Copper	13.8752	22.2319	13.8752		11.4660	11.4660
Lead,		80.0942	3.1201		2.4697	2.4697
Mercury		0.1971	0.0949	0.0063	0.2030	0.0063
Molybdenum					1.0882	1.0882
Nickel		126.6835	14.0887	661.9837	4.5769	4.5769
Selenium	1.0969	4.3869	1.0967	921.2409	0.9242	0.9242
Silver		1.4388				1.4388
Zinc		34.5908	34.5908	3997.1629	29.5299	29.5299
BOD ⁽¹⁾	17200					17200
TSS ⁽¹⁾	19940					19940
Crow Creek Water Reclamation Facility						
	WPDES Criteria (lbs/Day)	Acute WQS AHL (lbs/Day)	Chronic WQS AHL (lbs/Day)	Human Health AHL (lbs/Day)	Biosolids AHL (lbs/Day)	Maximum Allowable Headworks Loading (lbs/Day)
Arsenic		21.3992	9.4408	0.6294		0.6294
Cadmium	0.0506	0.5062	0.0506			0.0506
Chromium		159.1044	14.0127			14.0127
Copper	2.2397	3.6263	2.2397			2.2397
Lead,		219.6234	8.5839			8.5839
Mercury		0.8691	0.4183	0.0277		0.0277
Molybdenum	No Criteria					
Nickel		63.6015	7.0720	308.3546		7.0720
Selenium	0.7212	2.8849	0.7212	605.8227		0.7212
Silver		1.1319				1.1319
Zinc		25.7965	25.7965	2765.8094		25.7965
BOD ⁽¹⁾	9340					9340
TSS ⁽¹⁾	9340					9340

(1) MAHL for BOD and TSS based on POTW Influent Design Capacity

12.0 TECHNOLOGY BASED LIMITATION FOR BTEX

BTEX is the sum of measured concentrations for Benzene, Toluene, Ethylbenzene, and Xylene. Per USEPA-Region 8 Technically-Based Local Limits Development Strategy, The POTW may establish local limits for BTEX on a variety of criteria including fume toxicity, aquatic life protection, etc. An additional alternative exists for the POTW. This includes establishing technically-based limits for BTEX based on treatment technology. Information on this may be found in the EPA publication "Model NPDES Permit for Discharges Resulting from the Cleanup of Gasoline Released from Underground Storage Tanks", June 1989. The technology-based limits for POTWs to accept ground water cleanup wastes are 0.05 mg/L for Benzene and 0.75 mg/L for BTEX.

Per Cheyenne Municipal Code 13.20.070 (C), the BOPU has established technology-based limits for Benzene and BTEX applicable to all industrial users.

Based on EPA guidance, the BOPU will maintain currently established wastewater discharge limitations for Benzene: 0.050 mg/L and BTEX: 0.750 mg/L.

13.0 MAXIMUM ALLOWABLE INDUSTRIAL LOADING (MAIL)

The MAHLs provided in Table 21 estimate the maximum combined loadings that can be received (domestic, commercial, and industrial) at the POTW's headworks from all sources and represent the upper limit of influent pollutant loadings at which a POTW will not violate any treatment plant or environmental criteria developed to prevent process inhibition or interference, or violation of effluent, or sludge quality standards.

The Maximum Allowable Industrial Loading (MAIL) developed for each pollutant of concern represents the amount of the calculated MAHLs the POTW is able to allocate to Industrial permitted users. The available loadings are typically allocated through concentration-based discharge limitations. Consistent with the City's Municipal Codes, the limitations developed by the BOPU are enforced upon Significant Industrial Users (SIUs) or Industrial Users issued a discharge permit.

The equation for calculating the MAIL for each pollutant is as follows:

$$\text{MAL} = (\text{MAHL Less Safety Factor} - L_{\text{UNC}})$$

$$\text{MAIL} = (\text{MAL Less GA})$$

MAL = Maximum Allowable Loading

MAIL = Maximum Allowable Industrial Loading in lbs/Day

SF = Safety Factor (percent)

L_{UNC} = Loading from uncontrolled sources (domestic + commercial) in lbs/Day
calculated as $8.34 \times \text{residential/commercial concentration in mg/L} \times \text{Average}$
residential/commercial influent flow in MGD

GA = Growth and Expansion Allowance in lbs/day, (portion of the loading held in
reserve to protect the POTW against expansion of current permitted industries)

The first step in calculating the final MAIL is to calculate the Maximum Allowable Loadings (MAL). The MAL is calculated by applying a safety factor to the Maximum Allowable Headworks Loadings. After the safety factor has been applied, loadings from residential/commercial sources are subtracted. The final step in calculating the MAIL is applying a Growth and Expansion factor to the MAL.

13.1 Maximum Allowable Headworks Loading Safety Factor

EPA Region 8 requires that a safety + growth factor of at least ten (10) percent to be subtracted from the calculated MAHL. The main purpose of a safety factor is to address data “uncertainties” that can affect the ability of the POTW to calculate accurate local limits such as variability and quality of the POTW’s data, potential for industrial user slug loadings (e.g., as a result of a chemical spill), and potential general service area growth. The MAHL minus a 10% safety factor is provided in Table 22.

Table 22 – MAHL Less Safety Factor

Dry Creek Water Reclamation Facility				
	Maximum Allowable Headworks Loading (lbs/Day)	Safety Factor %	Safety Factor (lbs/Day)	MAHL less Safety Factor (lbs/Day)
Arsenic	0.6554	10	0.06554	0.5899
Cadmium	0.0687	10	0.00687	0.0618
Chromium	18.1885	10	1.81885	16.3697
Copper	11.4660	10	1.1466	10.3194
Lead,	2.4697	10	0.24697	2.2227
Mercury	0.0063	10	0.00063	0.0057
Molybdenum	1.0882	10	0.10882	0.9794
Nickel	4.5769	10	0.45769	4.1192
Selenium	0.9242	10	0.09242	0.8318
Silver	1.4388	10	0.14388	1.2949
Zinc	29.5299	10	2.95299	26.5769
BOD	17200	10	1720	15480
TSS	19940	10	1994	17946
Crow Creek Water Reclamation Facility				
	Maximum Allowable Headworks Loading (lbs/Day)	Safety Factor %	Safety Factor (lbs/Day)	MAHL less Safety Factor (lbs/Day)
Arsenic	0.6294	10	0.0629	0.5664
Cadmium	0.0506	10	0.0051	0.0456
Chromium	14.0127	10	1.4013	12.6115
Copper	2.2397	10	0.2240	2.0158
Lead,	8.5839	10	0.8584	7.7256
Mercury	0.0277	10	0.0028	0.0249
Nickel	7.0720	10	0.7072	6.3648
Selenium	0.7212	10	0.0721	0.6491
Silver	1.1319	10	0.1132	1.0187
Zinc	25.7965	10	2.5797	23.2168
BOD	9340	10	934	8406
TSS	9340	10	934	8406

13.2 Loading From Uncontrolled Sources

As noted above, some sources of pollutant loadings to the POTW are considered uncontrolled. To determine the allocation of the MAHL available to the controllable permitted sources, the POTW must determine the pollutant loadings from non-controllable sources (commercial non-permitted facilities and residential domestic sources). Because the POTW does not control the loadings that these users discharge (except through the general and specific prohibitions in the city's sewer use ordinance), the POTW needs to subtract these loadings from its MAHLs before it can determine the MAIL (see the equation for calculating the MAIL above).

As provided in Section 3.5 and Attachment D, residential/commercial sampling results from January 1, 2020 through December 31, 2024 were used to determine the average pollutant concentrations for the pollutants of concern.

As provided in Section 5.3, the average residential/commercial contribution for the Dry Creek Facility is 4.798 MGD and the average residential/commercial contribution for the Crow Creek Facility is 2.5186 MGD.

The equation for calculating the Loading from domestic and commercial sources is as follows:

$$L_{UNC} = (C_{UNC})(Q_{UNC})(8.34)$$

L_{UNC} = Loading from uncontrolled sources (domestic or commercial) in lbs/Day

C_{UNC} = Pollutant concentration from residential/commercial sources, mg/L

Q_{UNC} = Residential/commercial flow rate, mgd

8.34 = Unit conversion factor.

The loading from residential/commercial sources is provided in Table 23.

Table 23 – Residential/Commercial Headworks Loading

Dry Creek Water Reclamation Facility				
		C_{UNC} = Average Residential/Commercial Pollutant Concentration (mg/L)	Q_{UNC} = Residential/Commercial Influent Flow Rate (MGD)	L_{UNC} = Loading (lbs/Day)
Arsenic	8.34	0.00294	4.798	0.1176
Cadmium	8.34	0.00050	4.798	0.0200
Chromium	8.34	0.00412	4.798	0.1649
Copper	8.34	0.06132	4.798	2.4537
Lead,	8.34	0.00122	4.798	0.0488
Mercury	8.34	0.00003	4.798	0.0012
Molybdenum	8.34	0.00388	4.798	0.1553
Nickel	8.34	0.00462	4.798	0.1849
Selenium	8.34	0.00097	4.798	0.0388
Silver	8.34	0.00073	4.798	0.0292
Zinc	8.34	0.11729	4.798	4.6934
BOD	8.34	322.64	4.798	12910.54
TSS	8.34	398.86	4.798	15960.51
Crow Creek Water Reclamation Facility				
		C_{UNC} = Average Residential/Commercial Pollutant Concentration (mg/L)	Q_{UNC} = Residential/Commercial Influent Flow Rate (MGD)	L_{UNC} = Loading (lbs/Day)
Arsenic	8.34	0.00191	2.5186	0.0401
Cadmium	8.34	0.00047	2.5186	0.0099
Chromium	8.34	0.00332	2.5186	0.0697
Copper	8.34	0.06954	2.5186	1.4606
Lead,	8.34	0.00625	2.5186	0.1312
Mercury	8.34	0.00017	2.5186	0.0036
Molybdenum	8.34	0.00127	2.5186	0.0267
Nickel	8.34	0.00426	2.5186	0.0896
Selenium	8.34	0.00125	2.5186	0.0313
Silver	8.34	0.00123	2.5186	0.0258
Zinc	8.34	0.13401	2.5186	2.8149
BOD	8.34	217.75	2.5186	4573.87
TSS	8.34	310.81	2.5186	6528.60

13.3 Determination of the Maximum Allowable Loadings (MAL)

Table 24 provides the MAL based on the information provided throughout this section and the following equation as provided in section 13.0

The equation for calculating the MAL for each pollutant is as follows:

$$\text{MAL} = (\text{MAHL Less Safety Factor} - \text{LUNC})$$

Table 24 – Maximum Allowable Loadings (MAL)

Dry Creek Water Reclamation Facility				
	Maximum Allowable Headworks Loading (lbs/Day)	Safety Factor (lbs/Day)	L_{UNC} Loading (lbs/Day)	MAL (lbs/Day)
Arsenic	0.6554	0.06554	0.1176	0.4722
Cadmium	0.0687	0.00687	0.0200	0.0418
Chromium	18.1885	1.81885	0.1649	16.2048
Copper	11.4660	1.1466	2.4537	7.8657
Lead,	2.4697	0.24697	0.0488	2.1739
Mercury	0.0063	0.00063	0.0012	0.0045
Molybdenum	1.0882	0.10882	0.1553	0.8241
Nickel	4.5769	0.45769	0.1849	3.9343
Selenium	0.9242	0.09242	0.0388	0.7930
Silver	1.4388	0.14388	0.0292	1.2657
Zinc	29.5299	2.95299	4.6934	21.8835
BOD	17200	1720	12910.54	2569.46
TSS	19940	1994	15960.51	1985.49
Crow Creek Water Reclamation Facility				
	Maximum Allowable Headworks Loading (lbs/Day)	Safety Factor (lbs/Day)	L_{UNC} Loading (lbs/Day)	MAL (lbs/Day)
Arsenic	0.6294	0.0629	0.0401	0.5263
Cadmium	0.0506	0.0051	0.0099	0.0356
Chromium	14.0127	1.4013	0.0697	12.5418
Copper	2.2397	0.2240	1.4606	0.5552
Lead,	8.5839	0.8584	0.1312	7.5944
Mercury	0.0277	0.0028	0.0036	0.0213
Nickel	7.0720	0.7072	0.0896	6.2753
Selenium	0.7212	0.0721	0.0313	0.6178
Silver	1.1319	0.1132	0.0258	0.9929
Zinc	25.7965	2.5797	2.8149	20.4020
BOD	9340	934	4573.87	3832.13
TSS	9340	934	6528.60	1877.40

13.4 Growth Allowance

Growth and Expansion Allowance in lbs/day, is the portion of the MAHL held in reserve to protect the POTW against expansion of current permitted industries.

Per EPA Guidance, A POTW that anticipates growth in the future can consider holding a portion of its MAHLs for this growth. This expansion/growth allowance is separate from the safety factor. By holding in reserve some of the MAHL, the POTW has a portion to allocate to accommodate growth or expansion of existing SIUs and may not need to revise its existing Industrial User permits or Sewer Use Ordinance due to the growth.

Based on Best Professional Judgement, the BOPU has applied a Growth and Expansion (GA) of 10% in the final calculation of the MAIL.

13.5 Determination of Final Maximum Allowable Industrial Loadings

Table 25 provides the MAIL based on the information provided throughout this section and the following equation as provided in section 13.0

$$\text{MAIL} = [(\text{MAHL Less Safety Factor} - L_{\text{UNC}}) - \text{GA}]$$

<u>Table 25 – Maximum Allowable Industrial Loadings (MAIL)</u>			
Dry Creek Water Reclamation Facility			
	MAL (lbs/Day)	Growth Allowance (10%) of MAL (lbs/Day)	Maximum Allowable Industrial Loadings (MAIL) (lbs/Day)
Arsenic	0.4722	0.0472	0.4250
Cadmium	0.0418	0.0042	0.0376
Chromium	16.2048	1.6205	14.5843
Copper	7.8657	0.7866	7.0791
Lead,	2.1739	0.2174	1.9565
Mercury	0.0045	0.0005	0.0040
Molybdenum	0.8241	0.0824	0.7417
Nickel	3.9343	0.3934	3.5409
Selenium	0.7930	0.0793	0.7137
Silver	1.2657	0.1266	1.1391
Zinc	21.8835	2.1884	19.6951
BOD	2569.46	256.95	2312.51
TSS	1985.49	198.55	1786.94
Crow Creek Water Reclamation Facility			
	MAL (lbs/Day)	Growth Allowance (10%) of MAL (lbs/Day)	Maximum Allowable Industrial Loadings (MAIL) (lbs/Day)
Arsenic	0.5263	0.0526	0.4737
Cadmium	0.0356	0.0036	0.0321
Chromium	12.5418	1.2542	11.2876
Copper	0.5552	0.0555	0.4996
Lead,	7.5944	0.7594	6.8349
Mercury	0.0213	0.0021	0.0192
Nickel	6.2753	0.6275	5.6478
Selenium	0.6178	0.0618	0.5560
Silver	0.9929	0.9929	0.8936
Zinc	20.4020	2.0402	18.3618
BOD	3832.13	383.21	3448.92
TSS	1877.40	187.74	1689.66

14.0 RESERVE LOADING

When a POTW allocates the EPA approved MAIL to individual industrial users, the POTW generally retains a portion of the MAIL in reserve so that new industrial users can be given an allocation out of the calculated MAIL. This also allows the POTW flexibility to release the reserve loading as necessary without changes to the EPA approved MAIL or changes to the Technically-Based Local Limits. Changes to the MAIL require EPA approval as a substantial program modification, and changes to numeric limitations require a public notice period. If the release of the reserve results in a change to the calculated uniform or mass based local limit, revisions to the POTW's legal authority must also occur. The allowable industrial loading held in reserve may also be allocated to industrial users requesting special isolated discharge events as long as the MAHL and MAIL are not exceeded. The BOPU is holding in reserve 20-percent of the calculated Maximum Allowable Industrial Loading to accommodate future Significant Industrial Users. Table 26 provides the loading held in reserve and loading allocated for the calculation on the uniform local limits.

Table 26 – Reserved Loadings

Dry Creek Water Reclamation Facility			
	Maximum Allowable Industrial Loadings (MAIL) (lbs/Day)	20% Loading Held in Reserve from the Calculated MAIL (lbs/Day)	MAIL Less Reserve Loading (lbs/Day)
Arsenic	0.4250	0.0850	0.3400
Cadmium	0.0376	0.0075	0.0301
Chromium	14.5843	2.9169	11.6675
Copper	7.0791	1.4158	5.6633
Lead,	1.9565	0.3913	1.5652
Mercury	0.0040	0.0008	0.0032
Molybdenum	0.7417	0.1483	0.5934
Nickel	3.5409	0.7082	2.8327
Selenium	0.7137	0.1427	0.5710
Silver	1.1391	0.2278	0.9113
Zinc	19.6951	3.9390	15.7561
BOD	2312.51	462.50	1850.01
TSS	1786.94	357.39	1429.55
Crow Creek Water Reclamation Facility			
	Maximum Allowable Industrial Loadings (MAIL) (lbs/Day)	20% Loading Held in Reserve from the Calculated MAIL (lbs/Day)	MAIL Less Reserve Loading (lbs/Day)
Arsenic	0.4737	0.0947	0.3790
Cadmium	0.0321	0.0064	0.0257
Chromium	11.2876	2.2575	9.0301
Copper	0.4996	0.0999	0.3997
Lead,	6.8349	1.3670	5.4679
Mercury	0.0192	0.0038	0.0153
Nickel	5.6478	1.1296	4.5182
Selenium	0.5560	0.1112	0.4448
Silver	0.8936	0.1787	0.7149
Zinc	18.3618	3.6724	14.6894
BOD	3448.92	689.78	2759.14
TSS	1689.66	337.93	1351.73

15.0 ALLOCATION OF MAILS AMONG CONTROLLED SOURCES

Per EPA guidance, “local limits can take many forms based on how MAILs are allocated to IUs. The designation and implementation of these MAILs, including the allocation of

loadings to IUs, are left to each POTW, as long as the implementation procedures do not allow the calculated MAHL to be exceeded and provide a reasonable method for making allocations to the IUs.”

“A POTW may select any allocation and implementation method that results in enforceable local limits to prevent pass through and interference and to comply with the prohibitions in the Federal regulations. The POTW should choose the allocation approach that best fits its own situation.”

“Ultimately, the POTW will want to allocate pollutant loadings in a fair and sensible way that does not favor any one industry or group of industries, considers the economic impacts, maintains compliance with the NPDES permit, and otherwise achieves the environmental goals of the program.”

Consistent with the city’s legal authority and current BOPU local limits allocation method, the Uniform Concentration Method based on each MAIL and total SIU flow will be implemented. The uniform limits yield one limit per pollutant expressed as a single concentration applicable to all controlled users. Separate local limits have been developed for both treatment facilities.

16.0 LIMIT DURATION

Per EPA guidance, “When applying its local limits, a POTW needs to determine the appropriate limit duration. The POTW may establish limits that are daily maximums, monthly averages, or instantaneous maximums. In general, a POTW should base the limit duration on the type of criteria – long-term or short-term – used to develop the local limit. However, most local limits will be implemented as daily maximums based upon two main factors: 1) the short-term nature of the event that the local limit is protecting against; and 2) the infrequency of IU sampling.”

The EPA recommends the use of a daily maximum.

Consistent with the city’s legal authority and current BOPU local limits duration, Local limits will be enforced upon SIUs as Daily Maximums.

17.0 CALCULATED UNIFORM LOCAL LIMITS

The uniform concentration limit calculation equation is as follows:

$$C_{\text{LIM}} = \text{MAIL less reserve} / (Q_{\text{CONT}} \times 8.34)$$

C_{LIM} = Uniform concentration limit, mg/L

MAIL = Maximum allowable industrial loading, lbs/day

Q_{CONT} = Maximum total flow rate from SIUs, MGD

8.34 = Unit conversion factor

As provided in Section 5.2 and Table 7, the total flow rate from SIUs based on daily maximum discharge is 383144 GPD for the Dry Creek facility and 444315 for the Crow Creek facility, thus Q_{CONT} = 0.383144 MGD and 0.444315 MGD respectively.

The MAIL less reserve for each pollutant is provided in Table 26.

The calculated Uniform Concentration based Local Limits are provided in Table 27.

Table 27 – Calculated Uniform Local Limits

Dry Creek Water Reclamation Facility				
	MAIL Less Reserve (lbs/Day)	Q_{CONT} = Maximum flow rate from SIUs (MGD)	Unit Conversion Factor	C_{LIM} = Uniform Concentration Limit (mg/L)
Arsenic (As)	0.3400	0.383	8.34	0.106
Cadmium (Cd)	0.0301	0.383	8.34	0.009
Chromium (Cr)	11.6675	0.383	8.34	3.653
Copper (Cu)	5.6633	0.383	8.34	1.773
Lead (Pb)	1.5652	0.383	8.34	0.490
Mercury (Hg)	0.0032	0.383	8.34	0.001
Molybdenum (Mo)	0.5934	0.383	8.34	0.186
Nickel (Ni)	2.8327	0.383	8.34	0.887
Selenium (Se)	0.5710	0.383	8.34	0.179
Silver (Ag)	0.9113	0.383	8.34	0.285
Zinc (Zn)	15.7561	0.383	8.34	4.933
BOD	1850.01	0.383	8.34	579
TSS	1429.55	0.383	8.34	448
Crow Creek Water Reclamation Facility				
	MAIL Less Reserve (lbs/Day)	Q_{CONT} = Maximum flow rate from SIUs (MGD)	Unit Conversion Factor	C_{LIM} = Uniform Concentration Limit (mg/L)
Arsenic (As)	0.3790	0.443	8.34	0.102
Cadmium (Cd)	0.0257	0.443	8.34	0.007
Chromium (Cr)	9.0301	0.443	8.34	2.437
Copper (Cu)	0.3997	0.443	8.34	0.108
Lead (Pb)	5.4679	0.443	8.34	1.476
Mercury (Hg)	0.0153	0.443	8.34	0.0041
Nickel (Ni)	4.5182	0.443	8.34	1.219
Selenium (Se)	0.4448	0.443	8.34	0.120
Silver (Ag)	0.7149	0.443	8.34	0.193
Zinc (Zn)	14.6894	0.443	8.34	3.964
BOD	2759.14	0.443	8.34	745
TSS	1351.73	0.443	8.34	365

18.0 LOCAL LIMITS ASSESSMENT

Per EPA guidance, after developing and allocating local limits, POTWs should determine whether their local limits pass a “common sense test.” Some of the questions a POTW should ask to determine if its limits pass the “common sense” test are:

- 1.) Are the limits technologically achievable? Are SIUs and other controlled dischargers likely to meet these limits with currently available forms of pretreatment and pollution prevention (e.g., process modifications)?
- 2.) Can the POTW and dischargers determine compliance with the local limits? Are the limits above sampling method detection levels?

18.1 Achievability of Calculated Local Limits by Treatment

A review of the calculated Local Limits versus SIU average and maximum pollutant concentrations, as provided in Attachment C, was completed. The calculated limits are achievable by all SIUs using current treatment processes.

18.2 Compliance of Calculated Local Limits versus Sampling Method Detection Levels

A review of the calculated Local Limits versus the analysis methods provided by the certified laboratories used locally by current SIUs shows that compliance can be determined versus the method and reporting limits.

18.3 Assessment of Calculated Local Limits versus Previously Calculated Limits Table

Table 28 provides a comparison of the numeric calculated Local Limits versus the numeric Local Limits established during the last study (March 4, 2021).

Table 28 – Calculated Local Limits Versus Previously Established Limits

Dry Creek Water Reclamation Facility						
	2021 Uniform Local Limits (mg/L)	2025 Calculated Uniform Local Limits (mg/L)	2021 Approved MAHL (lbs/Day)	2025 Calculated MAHL (lbs/Day)	2021 Approved MAIL (lbs/Day)	2025 Calculated MAIL (lbs/Day)
Arsenic	0.031	0.106	0.7151	0.6554	0.4348	0.4250
Cadmium	0.008	0.009	0.1342	0.0687	0.1172	0.0376
Chromium	2.43	3.653	37.8883	18.1885	34.0347	14.5843
Copper	0.70	1.773	13.6215	11.4660	9.7646	7.0791
Lead	0.187	0.490	2.9734	2.4697	2.6221	1.9565
Mercury	0.0096	0.001	0.1519	0.0063	0.1343	0.0040
Molybdenum	0.065	0.186	1.0922	1.0882	0.9110	0.7417
Nickel	0.46	0.887	7.4011	4.5769	6.4162	3.5409
Selenium	0.050	0.179	0.8815	0.9242	0.6962	0.7137
Silver	0.133	0.285	2.0939	1.4388	1.8665	1.1391
Zinc	1.67	4.933	29.8394	29.5299	23.4355	19.6951
BOD	N/A	579	17200	17200	9788	2313
TSS	N/A	448	19949	19940	10447	1787
Crow Creek Water Reclamation Facility						
	2021 Uniform Local Limits (mg/L)	2025 Calculated Uniform Local Limits (mg/L)	2021 Approved MAHL (lbs/Day)	2025 Calculated MAHL (lbs/Day)	2021 Approved MAIL (lbs/Day)	2025 Calculated MAIL (lbs/Day)
Arsenic	0.065	0.102	0.6321	0.6294	0.4625	0.4737
Cadmium	0.005	0.007	0.0426	0.0506	0.0333	0.0321
Chromium	2.40	2.437	19.0070	14.0127	17.0625	11.2876
Copper	0.24	0.108	3.4259	2.2397	1.7246	0.4996
Lead	0.162	1.476	1.4287	8.5839	1.1543	6.8349
Mercury	0.0023	0.0041	0.0265	0.0277	0.0163	0.0192
Nickel	0.69	1.219	5.6028	7.0720	4.9048	5.6478
Selenium	0.040	0.120	0.3776	0.7212	0.2876	0.5560
Silver	0.165	0.193	1.3232	1.1319	1.1763	0.8936
Zinc	3.04	3.964	26.7986	25.7965	21.6497	18.3618
BOD	N/A	745	9340	9340	4845	3449
TSS	N/A	365	9340	9340	3071	1690

Per EPA guidance, on occasion, a relaxation of local limits may be appropriate. “The “anti-backsliding” concept associated with NPDES permit limits does not apply to local

limits. Local limits apply to a particular IU and can be raised or lowered based on the periodic re-evaluation of the need for those limits.”

The Local Limits developed during this technically-based evaluation are based on the previous five-years of monitoring data and treatment facility and sludge management operations and have been developed to protect the treatment plants, sewage system, biosolids and receiving water from adverse impacts from industrial and commercial wastewater dischargers.

Since the 2021 Local Limits Study, there has been a substantial decrease in the SIU flows at both treatment facilities. Per the mathematical formula used to calculate the uniform local limit from the MAIL (Section 17), the reduction in total SIU flows may result in less stringent local limit for some pollutants. Concentration based Local Limit = MAIL (lbs/day) / Maximum SIU Flow (MGD) x 8.34

Per the Wyoming Department of Environmental Quality, Water Quality Rules and Regulations, Chapter 1, several of the metals for which Local Limitations have been developed have Water Quality Standards (Acute and Chronic) that are receiving stream hardness dependent see Section 8.0. The 2021 Local Limits Study developed limitations using a hardness of 250 mg/L as CaCO₃ for both treatment facilities. Per the Statement of Basis provided with each facilities’ current 2024 discharge permits, the measured hardness for the Dry Creek facility is 210.4 mg/L as CaCO₃ and the water hardness for the Crow Creek facility is 229.9 mg/L as CaCO₃. The lower hardness results in slightly lower numeric Water quality Standards than the previous local limits evaluation.

The volume and percent solids of sludge to disposal (Biosolids) has remained relatively consistent with the 2021 evaluation.

The 2021 Local Limits calculations applied a Safety factor of 10% when calculating the MAIL. This Local Limits evaluation applies a Safety Factor of 10%, a Growth & Expansion Factor of 10% to allow for potential increases in loadings from existing SIUs, and a Reserve Capacity of 20%, which is available for allocation to new SIUs.

18.4 Assessment of Calculated Local Limits versus Previously Calculated Limits Narrative – Dry Creek Water Reclamation Facility

Arsenic: The calculated MAHL and MAIL have remained relatively consistent with the 2021 evaluation. The concentration based Local Limit for Arsenic has been relaxed (increased) due to a substantial decrease in the SIU flows.

Cadmium: The calculated MAHL and MAIL have decreased from the 2021 evaluation due to the use of more accurate receiving water hardness data, which lowers the WQS

slightly. In addition, the removal efficiency used in this evaluation is based on an EPA literature value as there was not enough supporting influent and effluent sampling results above detection limits to support a mathematically developed removal efficiency. As a result, the removal efficiency used in this evaluation decreased. Even with the lower MAHL and MAIL, the concentration based Local Limit remained relatively consistent due to the substantial decrease in the SIU flows.

Chromium (total): The calculated MAHL and MAIL have substantially decreased from the 2021. The hardness dependent Water Quality Standards used in this evaluation are lower than the standards used in the 2021 evaluation (source unknown). In addition, the calculated removal efficiency has decreased from the 2021 evaluation. Even with the decreased MAHL and MAIL, the concentration based Local Limit has been relaxed due to substantial decrease in the SIU flows.

Copper: The calculated MAHL and MAIL have remained relatively consistent with the 2021 evaluation. The concentration based Local Limit for Copper has been relaxed due to a substantial decrease in the SIU flows.

Lead: The calculated MAHL and MAIL have remained relatively consistent with the 2021 evaluation. The concentration based Local Limit for Lead has been relaxed due to a substantial decrease in the SIU flows

Mercury: The calculated MAHL and MAIL have decreased from the 2021 evaluation. The removal efficiency used in this evaluation is based on an EPA literature value as there was not enough supporting effluent sampling results above the detection limits to support a mathematically developed removal efficiency. As a result, the removal efficiency used in this evaluation decreased resulting in the lower MAHL and MAIL. Even with the lower MAHL and MAIL, the concentration based Local Limit remained relatively consistent due to the substantial decrease in the SIU flows.

Molybdenum: The calculated MAHL and MAIL have remained relatively consistent with the 2021 evaluation. The concentration based Local Limit for Molybdenum has been relaxed due to a substantial decrease in the SIU flows.

Nickel: The calculated MAHL and MAIL have decreased from the 2021. The MAHL was based on the most stringent Allowable Headworks Loading (AHL) calculation. The AHL used for Nickel has been established based on Land Application of Biosolids criteria. There has been an increase in Nickel concentration in the Biosolids leaving the facility since the 2021 evaluation resulting in the decreased MAHL and MAIL. Even with the decreased MAHL and MAIL, the concentration based Local Limit has been relaxed due to substantial decrease in the SIU flows.

Selenium: The calculated MAHL and MAIL have remained relatively consistent with the 2021 evaluation. The concentration based Local Limit for Selenium has been relaxed due to a substantial decrease in the SIU flows.

Silver: The calculated MAHL and MAIL have decreased from the 2021 evaluation due to the use of more accurate receiving water hardness data, which lowers the WQS slightly. Even with the decreased MAHL and MAIL, the concentration based Local Limit for Silver has been relaxed due to substantial decrease in the SIU flows.

Zinc: The calculated MAHL and MAIL have remained relatively consistent with the 2021 evaluation. The concentration based Local Limit for Zinc has been relaxed due to a substantial decrease in the SIU flows.

18.5 Assessment of Calculated Local Limits versus Previously Calculated Limits Narrative – Crow Creek Water Reclamation Facility

Arsenic: The calculated MAHL and MAIL have remained relatively consistent with the 2021 evaluation. The concentration based Local Limit for Arsenic has been relaxed (increased) due to a substantial decrease in the SIU flows.

Cadmium: The calculated MAHL and MAIL have remained relatively consistent with the 2021 evaluation. The concentration based Local Limit for Cadmium has been relaxed (increased) due to a substantial decrease in the SIU flows.

Chromium (total): The calculated MAHL and MAIL have decreased from the 2021. The hardness dependent Water Quality Standards used in this evaluation are lower than the standards used in the 2021 evaluation (source unknown). In addition, the calculated removal efficiency has decreased from the 2021 evaluation. Even with the decreased MAHL and MAIL, the concentration based Local Limit has been relaxed due to substantial decrease in the SIU flows.

Copper: The calculated MAHL and MAIL have decreased substantially from the 2021 evaluation. The hardness dependent WQS for Copper has been lowered slightly due to the use of more accurate receiving water hardness data. The removal efficiency used in this evaluation is lower than the previous evaluation due to the use of the Decile method, which more accurately accounts for variabilities in removal efficiencies over time versus the previously used MRE method. Based on POTW sampling data, the influent loading of Copper has increased since the previous evaluation. Based on these factors, the concentration based Local limit for Copper has become more stringent (decreased) versus the previously developed limit.

Lead: The calculated MAHL and MAIL have increased substantially versus the 2021 evaluation. The increases are due to a substantial increase in the facilities calculated

removal efficiency percentage. The concentration based Local Limit for Lead has been relaxed substantially due to the increase in removal efficiency and substantial decrease in the SIU flows.

Mercury: The calculated MAHL and MAIL have remained relatively consistent with the 2021 evaluation. The concentration based Local Limit for Mercury has been relaxed (increased) due to a substantial decrease in the SIU flows.

Molybdenum: There are no Water Quality or Human Health Standards currently established for Molybdenum for which to base a technically-based local limit. There is Land Application of Biosolids Criteria established for Molybdenum per 40 CFR Part 503 but all sludge generated at the Crow Creek facility is transferred to the Dry Creek facility for treatment. A Local Limit has not been established for the Crow Creek facility.

Nickel: The calculated MAHL and MAIL have remained relatively consistent with the 2021 evaluation. The concentration based Local Limit for Nickel has been relaxed due to a substantial decrease in the SIU flows.

Selenium: The calculated MAHL and MAIL have increased versus the 2021 evaluation. The increases are due to an increase in the facilities calculated removal efficiency percentage. The concentration based Local Limit for Selenium has been relaxed substantially due to the increase in removal efficiency and substantial decrease in the SIU flows.

Silver: The calculated MAHL and MAIL have remained relatively consistent with the 2021 evaluation. The concentration based Local Limit for Silver has been relaxed due to a substantial decrease in the SIU flows.

Zinc: The calculated MAHL and MAIL have remained relatively consistent with the 2021 evaluation. The concentration based Local Limit for Zinc has been relaxed due to a substantial decrease in the SIU flows.

19.0 EPA REGION 8 LOCAL LIMITS DEVELOPMENT SPREADSHEET

EPA Region 8 has provided a spreadsheet in Microsoft Excel format to provide POTWs with established methods to calculate local limits. The BOPU completed the spreadsheet to verify the accuracy of calculations used throughout this study. The completed spreadsheets are included in Attachments E and F. The Link to the EPA Region 8 spreadsheet: [https://19january2021snapshot.epa.gov/npdes-permits/region-8-local-limits-spreadsheet .html](https://19january2021snapshot.epa.gov/npdes-permits/region-8-local-limits-spreadsheet.html)

ATTACHMENT A

Only pollutants with sampling results above the reporting limit are shown. The complete listing of influent/effluent sampling data has been provided in Attachment B of the Pollutants of Concern Evaluation Package

Sampling Data - Dry Creek Water Reclamation Facility

Parameter	INFLUENT				EFFLUENT			
	Number of Samples	Number of Samples >RL	Average mg/L	Maximum mg/L	Number of Samples	Number of Samples >RL	Average mg/L	Maximum mg/L
1,4-dichlorobenzene	20	14	0.00049	0.00083	21	0	0	0
4,4'-DDE	20	1	0.000004	0.000004	20	0	0	0
Aluminum	10	10	0.88510	2.18	33	2	0	0.04
Ammonia	540	540	28.11	57.5	529	528	0.52	6.7
Antimony	19	8	0.00165	0.003	43	1	0.001	0.001
Arsenic	20	17	0.0024	0.0043	42	38	0.0018	0.0124
bis(2-ethylhexyl)phthalate	20	2	0.00625	0.0072	21	0	0	0
BOD5	539	539	322.64	1821	532	532	4.14	35.6
Boron	10	10	0.25920	0.31	33	33	0.2661	0.37
BTEX	20	18	0.00148	0.00455	20	2	0.000545	0.00091
Cadmium	20	6	0.0003	0.0004	43	2	0.0004	0.0004
Calcium	10	10	60.62	82	33	33	53.40	74
Chloride	19	19	109.18	207	18	18	94.02	156
Chloroform	20	18	0.00212	0.0029	21	16	0.00035	0.00041
Chromium	20	12	0.0060	0.0092	43	1	0.0026	0.0026
COD	15	15	423.47	602	15	15	52.55	83.4
Copper	20	19	0.0968	0.315	20	9	0.0137	0.0812
Cyanide	19	2	0.06485	0.1	19	1	0.1	0.1
diethyl phthalate	20	2	0.00405	0.0056	20	0	0	0
di-n-octyl phthalate	20	1	0.011	0.011	20	0	0	0
Ethylbenzene	20	1	0.00025	0.00025	20	0	0	0
Iron	10	10	1.06500	1.99	10	10	0.0492	0.112
Lead	20	17	0.0049	0.013	20	1	0.0028	0.0028
m+p-Xylenes	18	2	0.00074	0.001	18	0	0	0
Magnesium	10	10	11.53	16	10	10	10.33	15
Mercury	20	17	0.0000868	0.00038	20	0	0	0
methylene chloride	20	10	0.00057	0.00082	20	0	0	0
Molybdenum	20	18	0.0034	0.0082	20	18	0.0019	0.0042
Nickel	20	14	0.0088	0.037	20	7	0.0039	0.01
Nitrate	519	49	0.6633	6.2	519	516	8.1538	28.8
Oil and Grease	20	19	37.84	158.1	35	16	1.12	2.5
o-Xylene	18	2	0.00048	0.00053	18	0	0	0
Pet Oil/Grease (TPH)	20	18	14.99667	46.9	20	0	0	0
phenol	20	14	0.03214	0.099	20	0	0	0
Phosphorus	18	18	6.8111	13.5	528	528	2.7867	8.7
Potassium	10	10	15.46	17	10	10	13.56	14.6

Selenium	20	16	0.0043	0.017	61	13	0.003	0.007
Silver	19	0	0.00000	0	19	0	0	0
Styrene	18	2	0.00064	0.00092	18	0	0	0
Sulfate	19	19	76.32632	145	18	18	72.46	170
Temperature	20	20	17.18	21.1	20	20	17.48	22.4
Tetrachloroethylene	20	2	0.00040	0.00054	20	0	0	0
Thallium	18	1	0.013	0.013	18	0	0	0
Toluene	20	18	0.00138	0.0038	20	2	0.000545	0.00091
Total Dissolved Solids (TDS)	542	542	520.95	986	542	542	424.70	868
Total Nitrogen	0	0	0.00000	0	58	58	9.63	19.65
Total Phenols	17	14	5.17943	71.7	16	5	0.038	0.066
Total Sulfide	20	18	2.45228	27.1	20	11	0.63363	4.57
Total Suspended Solids	541	541	398.86	560	542	542	5.05	85.9
Trichloroethylene	20	2	0.00031	0.00033	20	0	0	0
Xylene	20	3	0.00083	0.0016	20	0	0	0
Zinc	20	20	0.2004	0.532	20	19	0.0517	0.1617

Sampling Data - Crow Creek Water Reclamation Facility

Parameter	INFLUENT				EFFLUENT			
	Number of Samples	Number of Samples >RL	Average mg/L	Maximum mg/L	Number of Samples	Number of Samples >RL	Average mg/L	Maximum mg/L
1,4-dichlorobenzene	20	14	0.0004	0.0009	20	0	0	0
Aluminum	9	9	0.5856	1.08	31	31	0.06	0.10
Ammonia	540	540	24.31	50.0	540	538	0.47	17.7
Antimony	18	7	0.0053	0.0152	41	8	0.0019	0.003
Arsenic	20	15	0.0021	0.00400	41	31	0.0017	0.01240
Benzene	20	8	0.0004	0.0009	20	0	0	0
bis(2-ethylhexyl)phthalate	20	3	0.0027	0.0038	21	0	0	0
BOD	536	536	217.75	582.0	535	535	8.93	134
Boron	9	9	0.2533	0.40	31	31	0.2845	0.73
Bromochloromethane	19	1	0.0003	0.0003	18	0	0	0
Bromodichloromethane	20	3	0.0002	0.0002	20	0	0	0
BTEX	17	14	0.0017	0.0031	17	12	0.0011	0.0043
Cadmium	20	4	0.0005	0.0008	42	0	0	0
Calcium	9	9	63.22	94.0	31	31	56.16	86
Chloride	20	20	106.20	154.0	19	19	102.34	151
Chloroform	20	18	0.0031	0.0067	20	12	0.0012	0.0043
Chromium	20	7	0.0057	0.0100	42	1	0.0011	0.0011
cis-1,2-Dichloroethene	20	9	0.0003	0.0004	20	0	0	0
COD	17	17	374.2	667.0	17	17	102.8	386
Copper	20	20	0.0930	0.3150	42	41	0.0171	0.0320
Cyanide	20	4	0.1599	0.5010	20	2	0.273	0.446
diethyl phthalate	20	5	0.0041	0.0080	21	0	0	0
di-n-butyl phthalate	20	1	0.0032	0.0032	21	0	0	0
di-n-octyl phthalate	20	1	0.0025	0.0025	21	0	0	0

Iron	9	9	0.9011	1.190	31	31	0.1219	0.22
Lead	20	19	0.0158	0.180	42	1	0.0008	0.0008
m+p-Xylenes	18	2	0.0004	0.0004	18	0	0	0
Magnesium	9	9	12.1	19.0	31	31	11	18
Mercury, Total	20	18	0.000066	0.00050	20	6	0.0000073	0.000012
methylene chloride	20	9	0.0006	0.0009	19	0	0	0
Molybdenum	20	18	0.0045	0.0094	42	39	0.0029	0.0050
Nickel	20	11	0.0069	0.0144	42	6	0.0043	0.01
Nitrate	8	3	0.2667	0.4000	528	41	1.673	8.9
Oil and Grease	19	18	33.54	190.20	34	16	1.51	2.5
o-Xylene	18	1	0.0002	0.0002	18	0	0	0
Pet Oil/Grease (TPH)	19	14	15.02	83.50	19	0	0	0
phenol	20	13	0.0093	0.0180	21	1	0.003	0.003
Phosphorus	19	19	4.68	11.70	536	536	3.263	4.3
Potassium	9	9	12.89	14.00	31	31	12.29	14
Selenium	20	9	0.0061	0.0250	61	13	0.0027	0.004
Silver	19	6	0.0267	0.1440	40	0	0	0
Styrene	18	3	0.0003	0.0004	18	0	0	0
Sulfate	20	20	61.43	120.00	19	19	60.053	131
Temperature	20	20	16.87	25.50	20	20	16.86	22.2
Tetrachloroethylene	20	17	0.0014	0.0021	20	0	0	0
Toluene	20	15	0.0013	0.0024	20	14	0.00091	0.0037
Total Dissolved Solids (TDS)	541	541	501.1	952.0	541	541	335.65	825
Total Nitrogen	0	0	0.00000	0.00000	59	59	13.16	24.15
Total Phenols	18	14	0.0702	0.1570	17	3	0.04567	0.068
Total Sulfide	20	14	0.6726	3.2000	20	10	0.3754	1
Total Suspended Solids	541	541	310.8	1356.0	540	540	90.28	537
Trichloroethylene	20	18	0.0008	0.0016	19	0	0	0
Xylene	20	2	0.0005	0.0006	20	0	0	0
Zinc	19	19	0.1565	0.5320	42	40	0.0382	0.0882

Adjusted Sampling Data - Dry Creek Water Reclamation Facility

Technically-based Local Limitations are being established for the following pollutants. During the calculations of influent and effluent average concentrations used in maximum allowable loadings computations, the BOPU used a conservative approach of substituting $\frac{1}{2}$ the reporting limit for non-detections (see Section 7.0).

Parameter	INFLUENT				EFFLUENT			
	Number of Samples	Number of Samples >RL	Average mg/L	Average w/1/2 RL mg/L	Number of Samples	Number of Samples >RL	Average mg/L	Average w/1/2 RL mg/L
Arsenic	20	17	0.0024	0.0021	42	38	0.0018	0.0017
Cadmium	20	6	0.0003	0.0004	43	2	0.0004	0.0005
Chromium	20	12	0.0060	0.0046	43	1	0.0026	0.0025
Copper	20	19	0.0968	0.0921	20	9	0.0137	0.0075
Lead	20	17	0.0049	0.0042	20	1	0.0028	0.0006
Mercury	20	17	0.0000868	0.000074	20	0	0	0.0000025

Molybdenum	20	18	0.0034	0.0031	20	18	0.0019	0.0017
Nickel	20	14	0.0088	0.0069	20	7	0.0039	0.0030
Selenium	20	16	0.0043	0.0036	61	13	0.003	0.0010
Silver	19	0	0	0.0005	19	0	0	0.0005
Zinc	20	20	0.2004	0.2004	20	19	0.0517	0.0493
BOD	539	539	322.64	322.64	532	532	4.14	4.14
TSS	541	541	398.86	398.86	542	542	5.05	5.05

Adjusted Sampling Data - Crow Creek Water Reclamation Facility

Technically-based Local Limitations are being established for the following pollutants. During the calculations of influent and effluent average concentrations used in maximum allowable loadings computations, the BOPU used a conservative approach of substituting $\frac{1}{2}$ the reporting limit for non-detections (see Section 7.0).

Parameter	INFLUENT				EFFLUENT			
	Number of Samples	Number of Samples >RL	Average mg/L	Average w/1/2 RL mg/L	Number of Samples	Number of Samples >RL	Average mg/L	Average w/1/2 RL mg/L
Arsenic	20	15	0.0021	0.0017	41	31	0.0017	0.0014
Cadmium	20	4	0.0005	0.0005	42	0	0	0.0025
Chromium	20	7	0.0057	0.0036	42	1	0.0011	0.0025
Copper	20	20	0.0930	0.0930	42	41	0.0171	0.0168
Lead	20	19	0.0158	0.0151	42	1	0.0008	0.00002
Mercury	20	18	0.0000655	0.000059	20	6	0.000007	0.0000040
Molybdenum	20	18	0.0045	0.0041	42	39	0.0029	0.0029
Nickel	20	11	0.0069	0.0049	42	6	0.0043	0.0010
Selenium	20	9	0.0061	0.0030	61	13	0.0027	0.0010
Silver	19	6	0.0267	0.0088	40	0	0	0.0050
Zinc	19	19	0.1565	0.1565	42	40	0.0382	0.0364
BOD	536	536	217.75	217.75	535	535	8.9300	8.93
TSS	541	541	310.81	310.81	540	540	90.2800	90.28

ATTACHMENT B**Biosolids Sampling Results**

The following table includes the complied Biosolids sampling results (Dry Creek Facility) from January 1, 2022 through December 31, 2024

Parameter	Units	Number of Samples	Number of Samples >RL	Average	Maximum
Arsenic (As)	mg/Kg	57	57	6.3	8.0
Boron (B)	mg/Kg	57	57	28.4	36.0
Cadmium (Cd)	mg/Kg	57	57	1.9	2.7
Calcium (Ca)	mg/Kg	57	57	29171.9	36435.0
Chromium (Cr)	mg/Kg	57	57	44.9	71.0
Copper (Cu)	mg/Kg	57	57	728.1	962.0
Iron (Fe)	mg/Kg	57	57	10435.9	12800.0
Potassium (K)	mg/Kg	57	57	2738.7	3760.0
Phosphorus (P)	mg/Kg	57	57	20870.6	29500.0
Magnesium (Mg)	mg/Kg	57	57	5791.7	8230.0
Manganese (Mn)	mg/Kg	57	57	520.5	655.0
Mercury (Hg)	mg/Kg	57	57	0.7	1.8
Molybdenum (Mo)	mg/Kg	57	57	11.6	14.9
Nickel (Ni)	mg/Kg	57	57	28.5	40.0
Lead (Pb)	mg/Kg	57	57	25.6	39.0
Selenium (Se)	mg/Kg	57	57	18.1	53.0
Silver (Ag)	mg/Kg	57	57	2.5	6.0
Sulfur (S)	mg/Kg	57	57	1.1	1.6
Zinc (Zn)	mg/Kg	57	57	860.3	1130.0
Ammonia	mg/Kg	57	57	7256.2	19300.0
TKN	mg/Kg	57	57	37736.9	61400.0
TN	mg/Kg	57	57	38763.0	63500.0
Nitrite/Nitrate	mg/Kg	57	57	1022.0	4040.0
Organic Nitrogen	mg/Kg	57	57	30476.6	53900.0
Chloride	mg/Kg	57	57	374.5	575.0

ATTACHMENT C

Significant Industrial Users Sampling Results

SPF Store No #46 LAUST Facility ID #3337 WSP-SPF46-2022	Total Samples	Number of Sample Results > RL	Average of Samples Above RL mg/L	Maximum mg/L
2,4,-dimethylphenol	4	2	0.0105	0.011
Ammonia	2	2	0.605	1.11
Arsenic	3	2	0.012	0.022
Benzene	4	1	0.00077	0.00077
Boron	1	1	0.13	0.13
BTEX	4	2	0.005435	0.00957
Calcium	1	1	145	145
Chloride	4	4	147	174
Chloroform	1	1	0.00073	0.00073
Chromium	3	0	0	0
COD	2	2	58.3	75.3
Copper	3	1	0.005	0.005
Iron	1	1	0.28	0.28
Lead	3	2	0.0025	0.003
m+p-Xylenes	2	1	0.0005	0.0005
Magnesium	1	1	23	23
Mercury	4	0	0	0
Molybdenum	3	1	0.002	0.002
Nickel	3	0	0	0
Oil and Grease	4	0	0	0
o-Xylene	3	2	0.00375	0.0062
Pet Oil/Grease (TPH)	4	0	0	0
Phosphorus	2	2	0.3585	0.497
Potassium	1	1	3	3
Selenium	3	0	0	0
Silver	3	0	0	0
Sulfate	4	4	18.18	23.4
Temperature	4	4	19.4	22.3
Toluene	4	1	0.0021	0.0021
Total Dissolved Solids (TDS)	2	2	524.5	532
Total Phenols	4	2	0.0495	0.059
Total Sulfide	3	3	0.773	1.87
Total Suspended Solids (TSS)	1	0	0	0
Xylene	4	2	0.004	0.0067
Zinc	3	2	0.016	0.022
Flying J LAUST Facility ID #3476 WSP-Flying J-2022	Total Samples	Number of Sample Results > RL	Average of Samples Above RL mg/L	Maximum mg/L
Ammonia	3	3	10.76	23.3
Arsenic (total)	4	3	0.004	0.004
BOD5	1	1	6.2	6.2

Boron (total)	2	2	0.1	0.1
Bromodichloromethane	5	2	0.000365	0.00054
Cadmium (total)	4	0	0	0
Calcium	2	2	229.5	291
Chloride	4	4	284.375	464.5
Chlorodibromomethane	5	1	0.00023	0.00023
Chloroform	2	2	0.001315	0.0023
Chromium (total)	4	0	0	0
COD	3	3	196.8	253
Copper (total)	4	3	0.01367	0.018
Cyanide, (total)	5	2	0.0295	0.034
Iron (total)	2	2	2.23	3.94
Lead (total)	4	0	0	0
Magnesium (total)	2	2	55	67
Mercury, Total	5	1	0.008	0.008
Molybdenum	4	1	0.003	0.003
Nickel (total)	4	3	0.01267	0.017
Oil and Grease	5	2	6.75	7.5
Pet Oil/Grease (TPH)	5	0	0	0
Phosphorus (total)	3	3	0.811	0.964
Potassium, (total)	2	2	5.5	6
Selenium (total)	4	0	0	0
Silver, (total)	4	0	0	0
Sulfate	5	5	17.26	25.2
Temperature ©	5	5	18.28	22.2
Total Dissolved Solids (TDS)	3	2	870.5	917
Total Phenols	5	2	0.038	0.046
Total Sulfide	5	4	1.846	6.8
Total Suspended Solids (TSS)	2	2	8	9
Zinc (total)	4	3	0.03133	0.054
Walmart Distribution Center #7077	Total Samples	Number of Sample Results > RL	Average of Samples Above RL mg/L	Maximum mg/L
Ammonia	5	3	0.1	0.1
Antimony	4	4	0.003375	0.004
Arsenic	5	3	0.012233333	0.0346
BOD	5	2	4.25	5.7
Cadmium	5	0	0	0
Calcium	2	2	32.5	43
Chloride	5	5	63.26	82.1
Chloroform	4	2	0.000395	0.00049
Chromium	5	0	0	0
COD	5	5	50.64	62.7
Copper	5	5	0.01536	0.047
Endosulfan-I	5	1	0.00029	0.00029
Lead	5	0	0	0
Magnesium	2	2	7	8
Mercury	5	5	0.00000864	0.000012

Molybdenum	5	4	0.00255	0.004
Nickel	5	2	0.0035	0.0053
Nitrate	2	2	39	40
Oil and Grease	5	0	0	0
Pet Oil/Grease (TPH)	5	0	0	0
Phosphorus	5	5	5.066	6.87
Potassium	2	2	19.5	22
Selenium	5	0	0	0
Silver	5	0	0	0
Sulfate	5	5	134.6	174
Temperature	5	5	24.7	29.6
Total Dissolved Solids (TDS)	5	5	549.2	640
Total Phenols	5	1	0.066	0.066
Total Sulfide	5	3	0.494	1.14
Total Suspended Solids (TSS)	5	2	1.75	2.2
Zinc	5	4	0.029475	0.0591
Union Pacific Railroad UPRR-2022	Total Samples	Number of Sample Results > RL	Average of Samples Above RL mg/L	Maximum mg/L
Aluminum	5	3	1.7567	4.46
Ammonia	8	6	0.3135	0.68
Antimony	7	2	0.0015	0.002
Arsenic	9	4	0.0065	0.016
BOD	8	7	7.84	13.6
Boron	5	5	0.916	1.2
Cadmium	9	0	0	0
Calcium	5	5	95.2	156
Chloride	8	8	118.25	300
Chromium	9	0	0	0
COD	6	6	70.22	107
Copper	9	5	0.0082	0.017
Endosulfan-II	9	1	0.037	0.037
endrin aldehyde	9	1	0.011	0.011
Iron	5	3	0.9267	1.43
Lead	9	2	0.0015	0.002
Magnesium	5	5	18	26
Mercury	9	0	0	0
Molybdenum	9	7	0.0528	0.239
Nickel	9	3	0.0122	0.0137
Nitrate	7	1	20.1	20.1
Oil and Grease	8	0	0	0
Pet Oil/Grease (TPH)	8	0	0	0
Phosphorus	8	7	0.584	2.6
Potassium	5	5	8.6	14
Selenium	9	2	0.0015	0.002
Silver	7	0	0	0
Sulfate	8	8	78.89	160
Temperature	8	8	17.81	25

Tetrachloroethylene	9	3	0.00044	0.00061
Total Dissolved Solids (TDS)	8	8	643.25	992
Total Phenols	8	4	0.0466	0.0534
Total Sulfide	8	7	3.0281	14.7
Total Suspended Solids (TSS)	8	5	11.66	38
Zinc	9	6	0.1282	0.37
Black Hills Energy - Cheyenne Prairie Generating Station BHE-2022	Total Samples	Number of Sample Results > RL	Average of Samples Above RL mg/L	Maximum mg/L
Aluminum	9	5	3.658	13.8
Ammonia	20	20	1.5724	18
Antimony	17	16	0.0037	0.009
Arsenic	19	18	0.0213	0.05
Beryllium	17	1	0.002	0.002
BOD	21	11	11.09	39.2
Boron	9	9	0.2956	0.35
Bromodichloromethane	19	6	0.0019	0.0081
Bromoform	19	1	0.0007	0.00074
BTEX	19	3	0.0003	0.00059
Cadmium	19	3	0.0008	0.0011
Calcium	9	9	280.67	487
Chloride	20	20	91.87	123
Chlorodibromomethane	19	6	0.0014	0.0053
Chloroform	17	17	0.0023	0.013
Chromium	19	18	0.0382	0.356
COD	19	19	65.15	130
Copper	19	10	0.0871	0.598
Cyanide	18	1	0.021	0.021
Iron	9	9	4.2011	28.9
Lead	19	3	0.0093	0.022
Magnesium	9	9	50.33	68
Mercury	19	0	0	0
Molybdenum	19	18	0.0141	0.023
Nickel	19	9	0.0145	0.071
Nitrate	17	6	11.7	15.4
Oil and Grease	21	2	12	14
Pet Oil/Grease (TPH)	20	1	5	5
Phosphorus	20	18	0.4813	1.8
Potassium	9	9	22.44	29
Selenium	18	18	0.0087	0.018
Silver	17	2	0.0048	0.0057
Sulfate	20	18	1158.28	3840
Temperature	20	20	25.045	31.6
Toluene	19	3	0.00034	0.00059
Total Dissolved Solids (TDS)	21	16	992.94	1367
Total Phenols	20	4	0.063	0.147
Total Sulfide	18	9	0.1257	0.439

Total Suspended Solids (TSS)	21	16	56.26	354
Zinc	19	13	0.1880	1.71
Cheyenne Renewable Diesel Company CRDC-2022-Mod	Total Samples	Number of Sample Results > RL	Average of Samples Above RL mg/L	Maximum mg/L
Aluminum (total)	5	4	1.0425	3.89
Ammonia	3	3	0.681	1.52
Arsenic (total)	11	3	0.002	0.002
Benzene	11	4	0.02142	0.04
BOD	3	3	14.95	25.5
Boron (total)	5	1	0.47	0.47
Bromodichloromethane	11	1	0.0002	0.0002
BTEX	10	4	0.67943	1.5184
Cadmium (total)	11	0	0	0
Calcium	5	5	31.8	68
Chloride	3	3	91.2	103
Chlorodibromomethane	11	1	0.0003	0.00027
Chloroform	12	10	0.0079	0.0276
Chromium (total)	11	2	0.0115	0.014
COD	4	4	85.9	135
Copper (total)	12	9	0.0622	0.21
endrin aldehyde	3	1	0.000009	0.000009
Ethylbenzene	12	3	0.01147	0.018
gamma-BHC (Lindane)	3	1	0.000008	0.0000075
Iron (total)	5	5	0.718	1.77
Lead (total)	11	3	0.0107	0.029
m+p-Xylenes	12	4	0.0698	0.099
Magnesium (total)	5	5	5.2	11
Mercury	3	3	0.0000152	0.000019
Molybdenum	11	2	0.001	0.001
Nickel (total)	11	3	0.011	0.018
Oil and Grease	12	8	41.3	188.4
o-Xylene	10	5	0.0261	0.05
Pet Oil/Grease (TPH)	12	2	6.2	6.7
Phosphorus (total)	3	3	0.876	1.07
Potassium, (total)	5	5	12	27
Selenium (total)	11	0	0	0
Silver, (total)	9	0	0	0
Sulfate	3	3	28.93	31.6
Temperature	15	15	17.21	25.9
Toluene	12	4	0.0313	0.057
Total Dissolved Solids (TDS)	3	3	369.67	380
Total Sulfide	3	1	0.385	0.385
Total Suspended Solids (TSS)	3	3	48.17	63
Xylene (total)	12	6	0.4506	1.49
Zinc (total)	14	13	0.139	1.3

ATTACHMENT D

Residential/Commercial Sampling Results

The following tables provide residential/commercial sampling results. Only pollutants with sampling results above the reporting limit are shown

Dry Creek Water Reclamation Facility

	Total Samples	Number of Sample Results > RL	Average of Samples Above RL mg/L	Maximum mg/L
1,4-Dichlorobenzene	20	14	0.0007	0.0011
Aluminum	7	7	0.3529	0.53
Ammonia	20	20	30.41	46.0
Antimony	14	3	0.0075	0.009
Arsenic	17	15	0.0033	0.0152
Benzene	20	3	0.0023	0.0056
bis(2-ethylhexyl)phthalate	20	1	0.0043	0.0043
BOD	19	19	176.52	306
Boron (total)	7	6	0.2167	0.27
Bromochloromethane	18	1	0.3100	0.3100
Bromodichloromethane	20	1	0.2100	0.2100
BTEX	16	16	0.0025	0.0190
Cadmium	17	0	0	0
Calcium	7	7	59.86	98
Chloride	19	19	102.09	219
Chloroform	20	18	0.0032	0.0048
Chromium	17	7	0.0064	0.009
COD	18	18	433.56	681.5
Copper	17	17	0.0613	0.125
diethyl phthalate	20	2	0.0047	0.0061
di-n-octyl phthalate	20	1	0.0074	0.0074
Ethylbenzene	20	2	0.0086	0.016
Iron	7	7	0.5300	0.75
Lead	17	12	0.0015	0.0035
m+p-Xylenes	18	4	0.0193	0.0710
Magnesium	7	7	12.14	19
Mercury	20	17	0.00004	0.000240
methylene chloride	20	10	0.0007	0.0010
Molybdenum	17	15	0.0043	0.0164
Nickel	17	10	0.0061	0.0077
Nitrate	8	2	0.1000	0.1
Oil and Grease	20	20	23.19	85.3
o-Xylene	18	4	0.0102	0.0380
Pet Oil/Grease (TPH)	20	19	10.19	21.0
Phenol	20	15	0.03207	0.0590
Phosphorus	18	18	4.38	7.01
Potassium	7	7	13.43	16

Selenium	18	11	0.0013	0.002
Silver	15	1	0.0040	0.004
Styrene	18	1	0.0006	0.0006
Sulfate	19	19	64.60	113
Temperature	19	19	17.41	22.4
Toluene	20	18	0.00631	0.06
Total Dissolved Solids (TDS)	20	20	490.20000	593
Total Phenols	18	15	0.08253	0.174
Total Sulfide	19	17	1.22506	4.27
Total Suspended Solids (TSS)	20	20	160.42000	438
Trichloroethylene	20	1	0.00098	0.00098
Xylene	20	4	0.02950	0.109
Zinc	17	17	0.11729	0.1956

Adjusted Sampling Data - Dry Creek Water Reclamation Facility

Technically-based Local Limitations are being established for the pollutants listed in the table below. During the calculations of the residential/commercial contribution used in maximum allowable loadings computations, the BOPU used a conservative approach of substituting $\frac{1}{2}$ the reporting limit for non-detections (see Section 7.0).

	Number of Samples	Number of Samples >RL	Average of Samples >RL mg/L	Adjusted Average w/1/2 RL mg/L
Arsenic	17	15	0.0033	0.00294
Cadmium	17	0	0	0.00050
Chromium	17	7	0.0064	0.00412
Copper	17	17	0.0613	0.06132
Lead	17	12	0.0015	0.00122
Mercury	20	17	0.00004	0.00003
Molybdenum	17	15	0.0043	0.00388
Nickel	17	10	0.0061	0.00462
Selenium	18	11	0.0013	0.00097
Silver	15	1	0.0040	0.00073
Zinc	17	17	0.11729	0.11729

Crow Creek Water Reclamation Facility

	Total Samples	Number of Sample Results > RL	Average of Samples Above RL mg/L	Maximum mg/L
1,4-Dichlorobenzene	11	9	0.0007	0.0013
4,4'-DDE	11	1	0.000003	0.000003
4,4'-DDT	11	1	0.00001	0.00001
Aluminum	4	4	0.7775	2.15
Ammonia	11	11	32.38	46

Antimony	10	2	0.0012	0.0012
Arsenic	11	8	0.0024	0.0097
bis(2-ethylhexyl)phthalate	11	4	0.0068	0.0082
BOD	10	10	195.18	350
Boron	4	4	0.260	0.3
Bromochloromethane	10	2	0.0003	0.0003
Bromodichloromethane	11	4	0.0004	0.0007
BTEX	11	9	0.0024	0.0043
Cadmium	11	1	0.0002	0.0002
Calcium	4	4	50.75	57
chlordan	11	1	0.0001	0.0001
Chloride	9	9	94.34	170
Chloroform	11	10	0.0051	0.0076
Chromium	11	4	0.00475	0.0150
COD	8	8	426.94	636.0
Copper (total)	11	11	0.0695	0.1411
diethyl phthalate	11	3	0.0050	0.0079
Iron	4	4	1.60	3.75
Lead	11	10	0.0068	0.022
m+p-Xylenes	10	2	0.0004	0.0004
Magnesium (total)	4	4	11.25000	13
Mercury	11	10	0.0002	0.00081
methylene chloride	11	6	0.0007	0.0008
Molybdenum	11	9	0.0014	0.0028
Nickel	11	7	0.0053	0.0078
Nitrate	6	2	0.65	0.7
Oil and Grease	11	11	24.40	39.6
o-Xylene	10	1	0.0002	0.0002
Pet Oil/Grease (TPH)	11	10	10.84	19.1
Phenol	11	10	0.0333	0.0610
Phosphorus	9	9	5.21	9.65
Potassium	4	4	15.25	17
Selenium	10	7	0.0016	0.003
Silver	10	2	0.0042	0.0073
Styrene	10	2	0.0004	0.0005
Sulfate	9	9	59.79	95.3
Temperature	11	11	19.09	23.2
Tetrachloroethylene	11	5	0.0004	0.0010
Toluene	11	10	0.0023	0.0043
Total Dissolved Solids (TDS)	11	11	458.45	582
Total Phenols	10	8	0.1361	0.3750
Total Sulfide	9	8	1.45	5.89
Total Suspended Solids (TSS)	11	11	143.27	322
Zinc	11	11	0.1340	0.290

Adjusted Sampling Data - Crow Creek Water Reclamation Facility				
Technically-based Local Limitations are being established for the pollutants listed in the table below. During the calculations of the residential/commercial contribution used in maximum allowable loadings computations, the BOPU used a conservative approach of substituting $\frac{1}{2}$ the reporting limit for non-detections (see Section 7.0).				
	Number of Samples	Number of Samples >RL	Average of Samples >RL mg/L	Adjusted Average w/1/2 RL mg/L
Arsenic	11	8	0.0024	0.00191
Cadmium	11	1	0.0002	0.00047
Chromium	11	4	0.00475	0.00332
Copper	11	11	0.06954	0.06954
Lead	11	10	0.0068	0.00625
Mercury	11	10	0.0002	0.00017
Molybdenum	11	9	0.0014	0.00127
Nickel	11	7	0.0053	0.00426
Selenium	10	7	0.0016	0.00125
Silver	10	2	0.0042	0.00123
Zinc	11	11	0.13401	0.13401

ATTACHMENT E

EPA Local Limits Spreadsheet - Dry Creek Water Reclamation Facility

TABLE 1 - GENERAL DATA ENTRY		
POTW NAME:	Cheyenne BOPU-Dry Creek	
POTW INFLUENT FLOW FOR LOCAL LIMITS STUDY (MGD):	5.908	Highest Influent Flow 2020-2024
POTW AVERAGE INFLUENT FLOW (MGD)	4.989	Average Influent Flow 2020-2024
COMBINED DOMESTIC + COMMERCIAL FLOW (MGD):	4.798	Daily Average (MGD)
MAXIMUM SIU FLOW (MGD):	0.383	Maximum SIU Flow (MGD)
AVERAGE SIU FLOW (MGD):	0.1911	Average SIU Flow (MGD)
POTW AVERAGE EFFLUENT FLOW (MGD)	4.722	Average Effluent Flow 2020-2024
SPECIFIC GRAVITY OF SLUDGE TO DISPOSAL (kg/l)	1	N/A when using DMT in sludge flow calculation
SLUDGE FLOW TO DISPOSAL (MGD):	0.00110	Calculated using DMT to Disposal
% SOLIDS TO DISPOSAL (%)	77.9	Dry Sludge % Solids from Data
BIOSOLIDS TABLE (1,3 OR "O" THER) BASED ON DISPOSAL OPTION:	3	40 CFR Part 503
ARE YOU USING TABLE 2 FOR BIOSOLIDS (Y/N)?:	n	
SITE AREA (ACRES):	100	
SITE LIFE (YEARS):	100	
CHRONIC RECEIVING WATER FLOW (MGD):	0	Per Permit Statement of Basis
ACUTE RECEIVING WATER FLOW (MGD):	0	Per Permit Statement of Basis
HARDNESS FOR METALS CALCULATIONS (MG/L):	210.4	Per Permit Statement of Basis
IS YOUR RECEIVING WATER A DRINKING WATER SUPPLY (Y/N)?:	N	Per Surface Water Designation
APPLICABLE STANDARDS (ACUTE, CHRONIC, BOTH):	B	A,C OR B

TABLE 2: DAILY/WEEKLY APPLICABLE STANDARDS	WYPDES DAILY/WEEKLY AVERAGE MAXIMUM MG/L	AQUATIC LIFE ACUTE WQS MG/L	POTW DESIGN CAPACITY LBS/DAY	FINAL APPLICABLE DAILY/WEEKLY STANDARD	
POLLUTANT					POLLUTANT
ARSENIC		0.3400		0.3400	ARSENIC
CADMIUM		0.0045		0.0045	CADMIUM
CHROMIUM		1.7645		1.7645	CHROMIUM
COPPER		0.0282		0.0282	COPPER
LEAD		0.2105		0.2105	LEAD
MERCURY		0.0016		0.0016	MERCURY
MOLYBDENUM					MOLYBDENUM
NICKEL		0.8803		0.8803	NICKEL
SELENIUM		0.0200		0.0200	SELENIUM
SILVER		0.0073		0.0073	SILVER
ZINC		0.2250		0.2250	ZINC
BOD 5			17200	17200	BOD 5
TSS			19940	19940	TSS

Aquatic Life Acute Water Quality Standards are Hardness Dependent for Cadmium, Chromium, Copper, Lead, Mercury, Nickel, Silver, and Zinc. The WQS for these pollutants are calculated using the Hardness-Dependent Formulas provided in Wyoming Water Quality Rules and Regulations - Chapter 1. WQS expressed as Dissolved have been converted to Total Recoverable based on the EPA guidance - The Metals Translator: Guidance For Calculating a Total Recoverable Permit Limit From a Dissolved Criterion -1996.

TABLE 3:					
MONTHLY APPLICABLE STANDARDS	WYPDES MONTHLY AVERAGE MAXIMUM MG/L	AQUATIC LIFE CHRONIC WQS MG/L	HUMAN HEALTH WQS MG/L	FINAL CHRONIC CRITERIA MG/L	
POLLUTANT					POLLUTANT
ARSENIC		0.1500	0.01	0.0100	ARSENIC
CADMIUM	0.00046	0.00046		0.0005	CADMIUM
CHROMIUM		0.1554		0.1554	CHROMIUM
COPPER	0.0176	0.0176		0.0176	COPPER
LEAD		0.0082		0.0082	LEAD
MERCURY		0.00077	0.000051	0.000051	MERCURY
MOLYBDENUM					MOLYBDENUM
NICKEL		0.0979	4.6	0.0979	NICKEL
SELENIUM	0.005	0.0050	4.2	0.0050	SELENIUM
SILVER					SILVER
ZINC		0.2250	26	0.2250	ZINC
BOD 5					BOD 5
TSS					TSS

Aquatic Life Chronic Water Quality Standards are Hardness Dependent for Cadmium, Chromium, Copper, Lead, Nickel, and Zinc. The WQS for these pollutants are calculated using the Hardness-Dependent Formulas provided in Wyoming Water Quality Rules and Regulations - Chapter 1. WQS expressed as Dissolved have been converted to Total Recoverable based on the EPA guidance - The Metals Translator: Guidance For Calculating a Total Recoverable Permit Limit From a Dissolved Criterion -1996.

TABLE 4: INFLUENT AND EFFLUENT DATA	COMMENTS AND NOTES	RL HANDLING	AVERAGE INFLUENT CONC. MG/L	AVERAGE POTW INFLUENT FLOW MGD	POTW INFLUENT LBS/DAY	COMMENTS AND NOTES	RL HANDLING	ADJ. AVERAGE EFFLUENT CONC. MG/L	AVERAGE POTW EFFLUENT FLOW MGD	POTW EFFLUENT LBS/DAY
POLLUTANT										
ARSENIC	n=20, 3 <RL	1/2 RL	0.0021	4.989	0.0892	n=42, 4 <RL	1/2 RL	0.0017	4.722	0.0662
CADMIUM	n=20, 14 <RL	1/2 RL	0.0004	4.989	0.0186	n=43, 41 <RL	1/2 RL	0.0005	4.722	0.0194
CHROMIUM	n=20, 8 <RL	1/2 RL	0.0046	4.989	0.1904	n=43, 42 <RL	1/2 RL	0.0025	4.722	0.0985
COPPER	n=20, 1 <RL	1/2 RL	0.0921	4.989	3.8309	n=20, 11 <RL	1/2 RL	0.0075	4.722	0.2971
LEAD	n=20, 3 <RL	1/2 RL	0.0042	4.989	0.1766	n=20, 19 <RL	1/2 RL	0.0006	4.722	0.0242
MERCURY	n=20, 3 <RL	1/2 RL	0.000074	4.989	0.0031	n=20, 20 <RL	1/2 RL	0.000003	4.722	0.0001
MOLYBDENUM	n=20, 2 <RL	1/2 RL	0.0031	4.989	0.1290	n=20, 2 <RL	1/2 RL	0.0017	4.722	0.0681
NICKEL	n=20, 6 <RL	1/2 RL	0.0069	4.989	0.2861	n=20, 13 <RL	1/2 RL	0.0030	4.722	0.1178
SELENIUM	n=20, 4 <RL	1/2 RL	0.0036	4.989	0.1477	n=61, 48 <RL	1/2 RL	0.0010	4.722	0.0407
SILVER	n=20, 20 <RL	1/2 RL	0.0005	4.989	0.0208	n=19, 19 <RL	1/2 RL	0.0005	4.722	0.0197
ZINC	n=20, 0 <RL	N/A	0.2004	4.989	8.3402	n=20, 1 <RL	N/A	0.0493	4.722	1.9431
BOD 5	n=539	N/A	322.64	4.989	13424.49	n=532	N/A	4.14	4.722	163.04
TSS	n=541	N/A	398.86	4.989	16595.87	n=542	N/A	5.05	4.722	198.88

TABLE 5: RESIDENTIAL + COMMERCIAL CONTRIBUTION	COMMENTS AND NOTES	RL HANDLING	RESIDENTIAL+ COMMERCIAL DISCHARGE TO POTW MG/L	RESIDENTIAL+ COMMERCIAL DISCHARGE TO POTW MGD	FINAL CALCULATED RESIDENTIAL + COMMERCIAL CONTRIBUTION LBS/DAY	
POLLUTANT						POLLUTANT
ARSENIC	n=17, 2 <RL	1/2 RL	0.00294	4.798	0.1176	ARSENIC
CADMIUM	n=17, 17 <RL	1/2 RL	0.00050	4.798	0.0200	CADMIUM
CHROMIUM	n=17, 10 <RL	1/2 RL	0.00412	4.798	0.1649	CHROMIUM
COPPER	n=17, 0 <RL	N/A	0.06132	4.798	2.4537	COPPER
LEAD	n=17, 5 <RL	1/2 RL	0.00122	4.798	0.0488	LEAD
MERCURY	n=20, 3 <RL	1/2 RL	0.00003	4.798	0.0012	MERCURY
MOLYBDENUM	n=17, 2 <RL	1/2 RL	0.00388	4.798	0.1553	MOLYBDENUM
NICKEL	n=17, 7 <RL	1/2 RL	0.00462	4.798	0.1849	NICKEL
SELENIUM	n=18, 9 <RL	1/2 RL	0.00097	4.798	0.0388	SELENIUM
SILVER	n=15, 14 <RL	1/2 RL	0.00073	4.798	0.0292	SILVER
ZINC	n=17, 0 <RL	N/A	0.11729	4.798	4.6934	ZINC
BOD 5	n=539	N/A	322.64	4.798	12910.54	BOD 5
TSS	n=541	N/A	398.86	4.798	15960.51	TSS

TABLE 6: BIOSOLIDS	MAXIMUM POTW BIOSOLIDS TO DISPOSAL MG/KG DRY WT.	TABLE 1 MAXIMUM LAND APP SLUDGE CRITERIA MG/KG	TABLE 3 "CLEAN" LAND APP SLUDGE CRITERIA MG/KG	FINAL SLUDGE CRITERIA MG/KG	
POLLUTANT					POLLUTANT
ARSENIC	8.0		41	41	ARSENIC
CADMIUM	2.7		39	39	CADMIUM
CHROMIUM	71.0				CHROMIUM
COPPER	962.0		1500	1500	COPPER
LEAD	39.0		300	300	LEAD
MERCURY	1.8		17	17	MERCURY
MOLYBDENUM (1)	14.9	75		75	MOLYBDENUM
NICKEL	40.0		420	420	NICKEL
SELENIUM	53.0		100	100	SELENIUM
SILVER	6.0				SILVER
ZINC	1130.0		2800	2800	ZINC
BOD 5					BOD 5
TSS					TSS

(1) Molybdenum Standard Based on Table 1 Maximum Ceiling Concentration Per 40 CFR Part 503.

TABLE 7: REMOVAL EFFICIENCY CALCULATIONS	ADRE ADRE METHOD REMOVAL EFFICIENCY %	MRE MRE METHOD REMOVAL EFFICIENCY %	DECILE DECILE METHOD REMOVAL EFFICIENCY %	LIT LITERATURE REMOVAL EFFICIENCY %	SOURCE OF LITERATURE REMOVAL EFFICIENCY DATA	ENTER THE NAME OF THE REMOVAL EFFICIENCY TO BE USED: ADRE, MRE, DECILE, OR LIT	FINAL POTW REMOVAL %
POLLUTANT							
ARSENIC	26.97	2.82	44.83	45		DECILE	44.83
CADMIUM				67	EPA	LIT	67.00
CHROMIUM		57.90		82		MRE	57.90
COPPER	78.66	91.71	93.75	86		DECILE	93.75
LEAD		87.05		61		MRE	87.05
MERCURY				60	EPA	LIT	60.00
MOLYBDENUM	34.24	44.92	49.39			DECILE	49.39
NICKEL	60.10	65.76		42		MRE	65.76
SELENIUM	81.20	77.54		50		MRE	77.54
SILVER				75	EPA	LIT	75.00
ZINC	68.46	74.22	67.95	79.0		DECILE	67.95
BOD 5		98.7				MRE	98.70
TSS		98.7				MRE	98.70

TABLE 8:						
DAILY MAHL CALCULATIONS	DAILY/7 DAY NPDES LOADING LBS/DAY	ACUTE LOADING LBS/DAY	POTW LOADING LBS/DAY	MOST STRINGENT CRITERIA LBS/DAY	NAME OF MAHL FOR DAILY MAX LIMITS	
POLLUTANT						POLLUTANT
ARSENIC		30.3656		30.366	WQ-ACUTE	ARSENIC
CADMIUM		0.6719		0.672	WQ-ACUTE	CADMIUM
CHROMIUM		206.5227		206.523	WQ-ACUTE	CHROMIUM
COPPER		22.2319		22.232	WQ-ACUTE	COPPER
LEAD		80.0942		80.094	WQ-ACUTE	LEAD
MERCURY		0.1971		0.197	WQ-ACUTE	MERCURY
MOLYBDENUM						MOLYBDENUM
NICKEL		126.6835		126.684	WQ-ACUTE	NICKEL
SELENIUM		4.3869		4.387	WQ-ACUTE	SELENIUM
SILVER		1.4388		1.439	WQ-ACUTE	SILVER
ZINC		34.5908		34.591	WQ-ACUTE	ZINC
BOD 5			17200	17200	POTW DESIGN	BOD 5
TSS			19940	19940	POTW DESIGN	TSS

TABLE 9:							
MONTHLY MAHL CALCULATIONS	MONTHLY NPDES LOADING LBS/DAY	CHRONIC TOXICITY LOADING LBS/DAY	LOADING FOR HUMAN HEALTH LBS/DAY	TABLE 1 OR 3 SLUDGE LOADING LBS/DAY	MOST STRINGENT CRITERIA LBS/DAY	NAME OF MAHL FOR MONTHLY LIMITS	
POLLUTANT							POLLUTANT
ARSENIC		13.3966	0.8931	0.6554	0.6554	Biosolids	ARSENIC
CADMIUM	0.0687	0.0687		0.4171	0.0687	NPDES Monthly	CADMIUM
CHROMIUM		18.1885			18.1885	WQ-CHRONIC	CHROMIUM
COPPER	13.8752	13.8752		11.4660	11.4660	Biosolids	COPPER
LEAD		3.12006		2.4697	2.4697	Biosolids	LEAD
MERCURY		0.0948	0.0063	0.2030	0.0063	HH	MERCURY
MOLYBDENUM				1.0882	1.0882	Biosolids	MOLYBDENUM
NICKEL		14.0887	661.9837	4.5769	4.5769	Biosolids	NICKEL
SELENIUM	1.0967	1.0967	921.2409	0.9242	0.9242	Biosolids	SELENIUM
SILVER						No Criteria	SILVER
ZINC		34.5908	3997.1629	29.5299	29.5299	Biosolids	ZINC
BOD 5							BOD 5
TSS							TSS

TABLE 10:							
DAILY LOCAL LIMITS		FINAL		SAFETY	MAHL	MAHL MINUS	MAXIMUM
POLLUTANT	MOST	MAHL FOR	NAME	FACTOR %	WITH SAFETY -	RES + COM	ALLOWABLE
	STRINGENT	DAILY	FOR MAHL	FOR DAILY	FACTOR	LOADING	LOAD
	MAHL	LOCAL LIMITS		MAX LIMITS	LBS/DAY	LBS/DAY	MAL
							LBS/DAY
ARSENIC	0.6554	0.6554	Biosolids	10	0.5899	0.4722	0.4722
CADMIUM	0.0687	0.0687	NPDES Monthly	10	0.0618	0.0418	0.0418
CHROMIUM	18.1885	18.1885	WQ-CHRONIC	10	16.3697	16.2048	16.2048
COPPER	11.4660	11.4660	Biosolids	10	10.3194	7.8657	7.8657
LEAD	2.4697	2.4697	Biosolids	10	2.2227	2.1739	2.1739
MERCURY	0.0063	0.0063	HH	10	0.0057	0.0045	0.0045
MOLYBDENUM	1.0882	1.0882	Biosolids	10	0.9794	0.8241	0.8241
NICKEL	4.5769	4.5769	Biosolids	10	4.1192	3.9343	3.9343
SELENIUM	0.9242	0.9242	Biosolids	10	0.8318	0.7930	0.7930
SILVER	1.4388	1.4388	WQ-ACUTE	10	1.2949	1.2657	1.2657
ZINC	29.5299	29.5299	Biosolids	10	26.5769	21.8835	21.8835
BOD 5	17200	17200	POTW DESIGN	10	15480	2569.46	2569.46
TSS	19940	19940	POTW DESIGN	10	17946	1985.49	1985.49

TABLE 10 (Continued)

DAILY LOCAL LIMITS

MAXIMUM ALLOWABLE LOAD MAL LBS/DAY	SIU EXPANSION / GROWTH FACTOR FUTURE LOADING %	MAL WITH EXPANSION / GROWTH FUTURE LOADING LBS/DAY	MAIL CALCULATED ALLOCATION FOR SIUS LBS/DAY	RESERVE CAPACITY 20% OF MAIL LBS/DAY	CALCULATED MAIL LESS RESERVE CAPACITY LBS/DAY	CALCULATED UNIFORM LOCAL LIMITS FOR SIUS MG/L	POLLUTANT
0.4722	10	0.4250	0.4250	0.0850	0.3400	0.106	ARSENIC
0.0418	10	0.0376	0.0376	0.0075	0.0301	0.009	CADMIUM
16.2048	10	14.5843	14.5843	2.9169	11.6675	3.653	CHROMIUM
7.8657	10	7.0791	7.0791	1.4158	5.6633	1.773	COPPER
2.1739	10	1.9565	1.9565	0.3913	1.5652	0.490	LEAD
0.0045	10	0.0040	0.0040	0.0008	0.0032	0.0010	MERCURY
0.8241	10	0.7417	0.7417	0.1483	0.5934	0.186	MOLYBDENUM
3.9343	10	3.5409	3.5409	0.7082	2.8327	0.887	NICKEL
0.7930	10	0.7137	0.7137	0.1427	0.5710	0.179	SELENIUM
1.2657	10	1.1391	1.1391	0.2278	0.9113	0.285	SILVER
21.8835	10	19.6951	19.6951	3.9390	15.7561	4.933	ZINC
2569.46	10	2312.51	2312.51	462.50	1850.01	579	BOD 5
1985.49	10	1786.94	1786.94	357.39	1429.55	448	TSS

TABLE 11: Daily LOCAL LIMIT SUMMARY POLLUTANT	Dry Creek Water Reclamation Facility			
	MAIL FOR SIUs LBS/DAY	UNIFORM CONCENTRATION FOR SIUs (MG/L)	PROPOSED DAILY MAXIMUM LIMIT (MG/L)	POLLUTANT
ARSENIC	0.4250	0.1064	0.106	ARSENIC
CADMIUM	0.0376	0.0094	0.009	CADMIUM
CHROMIUM	14.5843	3.6527	3.653	CHROMIUM
COPPER	7.0791	1.7730	1.773	COPPER
LEAD	1.9565	0.4900	0.490	LEAD
MERCURY	0.0040	0.0010	0.001	MERCURY
MOLYBDENUM	0.7417	0.1858	0.186	MOLYBDENUM
NICKEL	3.5409	0.8868	0.89	NICKEL
SELENIUM	0.7137	0.1787	0.179	SELENIUM
SILVER	1.1391	0.2853	0.285	SILVER
ZINC	19.6951	4.9327	4.933	ZINC
BOD 5	2312.51	579	579	BOD 5
TSS	1786.94	448	448	TSS

ATTACHMENT F

EPA Local Limits Spreadsheet - Crow Creek Water Reclamation Facility

TABLE 1 - GENERAL DATA ENTRY		
POTW NAME:	Cheyenne BOPU-Crow Creek	
POTW INFLUENT FLOW FOR LOCAL LIMITS STUDY (MGD):	4.088	Highest Influent Flow 2020-2024
POTW AVERAGE INFLUENT FLOW (MGD)	2.756	Average Influent Flow 2020-2024
COMBINED DOMESTIC + COMMERCIAL FLOW (MGD):	2.5186	Daily Average (MGD)
MAXIMUM SIU FLOW (MGD):	0.444315	Maximum SIU Flow (MGD)
AVERAGE SIU FLOW (MGD):	0.237419	Average SIU Flow (MGD)
POTW AVERAGE EFFLUENT FLOW (MGD)	2.062	Average Effluent Flow 2020-2024
SPECIFIC GRAVITY OF SLUDGE TO DISPOSAL (kg/l)	0	N/A when using DMT in sludge flow calculation
SLUDGE FLOW TO DISPOSAL (MGD):	0.00000	Sludge to Dry Creek Facility
% SOLIDS TO DISPOSAL (%)	0.0	N/A
BIOSOLIDS TABLE (1,3 OR "OTHER") BASED ON DISPOSAL OPTION:	3	40 CFR Part 503
ARE YOU USING TABLE 2 FOR BIOSOLIDS (Y/N)?:	n	
SITE AREA (ACRES):	100	
SITE LIFE (YEARS):	100	
CHRONIC RECEIVING WATER FLOW (MGD):	0	Per Permit Statement of Basis
ACUTE RECEIVING WATER FLOW (MGD):	0	Per Permit Statement of Basis
HARDNESS FOR METALS CALCULATIONS (MG/L):	229.9	Per Permit Statement of Basis
IS YOUR RECEIVING WATER A DRINKING WATER SUPPLY (Y/N)?:	N	Per Surface Water Designation
APPLICABLE STANDARDS (ACUTE, CHRONIC, BOTH):	B	A,C OR B

TABLE 2: DAILY/WEEKLY APPLICABLE STANDARDS		WYPDES DAILY/WEEKLY AVERAGE MAXIMUM MG/L	AQUATIC LIFE ACUTE WQS MG/L	POTW DESIGN CAPACITY LBS/DAY	FINAL APPLICABLE DAILY/WEEKLY STANDARD	
POLLUTANT						POLLUTANT
	ARSENIC		0.3400		0.3400	ARSENIC
	CADMIUM		0.0049		0.0049	CADMIUM
	CHROMIUM		1.8973		1.8973	CHROMIUM
	COPPER		0.0306		0.0306	COPPER
	LEAD		0.2098		0.2098	LEAD
	MERCURY		0.0016		0.0016	MERCURY
	MOLYBDENUM					MOLYBDENUM
	NICKEL		0.9488		0.9488	NICKEL
	SELENIUM		0.0200		0.0200	SELENIUM
	SILVER		0.0083		0.0083	SILVER
	ZINC		0.2425		0.2425	ZINC
	BOD 5			9340	9340	
	TSS			9340	9340	

Aquatic Life Acute Water Quality Standards are Hardness Dependent for Cadmium, Chromium, Copper, Lead, Mercury, Nickel, Silver, and Zinc. The WQS for these pollutants are calculated using the Hardness-Dependent Formulas provided in Wyoming Water Quality Rules and Regulations - Chapter 1. WQS expressed as Dissolved have been converted to Total Recoverable based on the EPA guidance - The Metals Translator: Guidance For Calculating a Total Recoverable Permit Limit From a Dissolved Criterion -1996.

TABLE 3: MONTHLY APPLICABLE STANDARDS	WYPDES MONTHLY AVERAGE MAXIMUM MG/L	AQUATIC LIFE CHRONIC WQS MG/L	HUMAN HEALTH WQS MG/L	FINAL CHRONIC CRITERIA MG/L	
POLLUTANT					POLLUTANT
ARSENIC		0.1500	0.01	0.0100	ARSENIC
CADMIUM	0.00049	0.00049		0.0005	CADMIUM
CHROMIUM		0.1671		0.1671	CHROMIUM
COPPER	0.0189	0.0189		0.0189	COPPER
LEAD		0.0082		0.0082	LEAD
MERCURY		0.00077	0.000051	0.000051	MERCURY
MOLYBDENUM					MOLYBDENUM
NICKEL		0.1055	4.6	0.1055	NICKEL
SELENIUM	0.005	0.0050	4.2	0.0050	SELENIUM
SILVER					SILVER
ZINC		0.2425	26	0.2425	ZINC
BOD 5					
TSS					

Aquatic Life Chronic Water Quality Standards are Hardness Dependent for Cadmium, Chromium, Copper, Lead, Nickel, and Zinc. The WQS for these pollutants are calculated using the Hardness-Dependent Formulas provided in Wyoming Water Quality Rules and Regulations - Chapter 1. WQS expressed as Dissolved have been converted to Total Recoverable based on the EPA guidance - The Metals Translator: Guidance For Calculating a Total Recoverable Permit Limit From a Dissolved Criterion -1996.

TABLE 4:										
INFLUENT AND EFFLUENT DATA	COMMENTS AND NOTES	RL HANDLING	AVERAGE INFLUENT CONC. MG/L	AVERAGE POTW INFLUENT FLOW MGD	POTW INFLUENT LBS/DAY	COMMENTS AND NOTES	RL HANDLING	ADJ. AVERAGE EFFLUENT CONC. MG/L	AVERAGE POTW EFFLUENT FLOW MGD	POTW EFFLUENT LBS/DAY
POLLUTANT										
ARSENIC	n=20, 5 <RL	1/2 RL	0.0017	2.756	0.0393	n=41, 10 <RL	1/2 RL	0.0014	2.062	0.0240
CADMIUM	n=20, 16 <RL	1/2 RL	0.0005	2.756	0.0115	n=42, 42 <RL	1/2 RL	0.0025	2.062	0.0430
CHROMIUM	n=20, 13 <RL	1/2 RL	0.0036	2.756	0.0829	n=42, 41 <RL	1/2 RL	0.0025	2.062	0.0424
COPPER	n=20, 0 <RL	N/A	0.0930	2.756	2.1366	n=42, 1 <RL	1/2 RL	0.0168	2.062	0.2881
LEAD	n=20, 1 <RL	1/2 RL	0.0151	2.756	0.3464	n=42, 41 <RL	1/2 RL	0.0000	2.062	0.0004
MERCURY	n=20, 2 <RL	1/2 RL	0.000059	2.756	0.0014	n=20, 14 <RL	1/2 RL	0.000004	2.062	0.0001
MOLYBDENUM	n=20, 2 <RL	1/2 RL	0.0041	2.756	0.0932	n=42, 3 <RL	1/2 RL	0.0029	2.062	0.0494
NICKEL	n=20, 9 <RL	1/2 RL	0.0049	2.756	0.1126	n=42, 36 <RL	1/2 RL	0.0010	2.062	0.0180
SELENIUM	n=20, 11 <RL	1/2 RL	0.0030	2.756	0.0695	n=61, 48 <RL	1/2 RL	0.0010	2.062	0.0166
SILVER	n=19, 13 <RL	1/2 RL	0.0088	2.756	0.2023	n=40, 40 <RL	1/2 RL	0.0050	2.062	0.0860
ZINC	n=20, 0 <RL	N/A	0.1565	2.756	3.5979	n=42, 2 <RL	1/2 RL	0.0364	2.062	0.6260
BOD 5	n=536	N/A	217.75	2.756	5004.99	n=535	N/A	8.93	2.062	153.58
TSS	n=541	N/A	310.81	2.756	7143.98	n=540	N/A	90.28	2.062	1552.70

TABLE 5: RESIDENTIAL + COMMERCIAL CONTRIBUTION	COMMENTS AND NOTES	RL HANDLING	RESIDENTIAL + COMMERCIAL DISCHARGE TO POTW MG/L	RESIDENTIAL + COMMERCIAL DISCHARGE TO POTW MGD	FINAL CALCULATED RESIDENTIAL + COMMERCIAL CONTRIBUTION LBS/DAY	
POLLUTANT						POLLUTANT
ARSENIC	n=11, 3 <RL	1/2 RL	0.00191	2.519	0.0401	ARSENIC
CADMIUM	n=11, 10 <RL	1/2 RL	0.00047	2.519	0.0099	CADMIUM
CHROMIUM	n=11, 7 <RL	1/2 RL	0.00332	2.519	0.0697	CHROMIUM
COPPER	n=11, 0 <RL	N/A	0.06954	2.519	1.4606	COPPER
LEAD	n=11, 1 <RL	1/2 RL	0.00625	2.519	0.1312	LEAD
MERCURY	n=11, 1 <RL	1/2 RL	0.00017	2.519	0.0036	MERCURY
MOLYBDENUM	n=11, 2 <RL	1/2 RL	0.00127	2.519	0.0267	MOLYBDENUM
NICKEL	n=11, 4 <RL	1/2 RL	0.00426	2.519	0.0896	NICKEL
SELENIUM	n=10, 3 <RL	1/2 RL	0.00125	2.519	0.0313	SELENIUM
SILVER	n=10, 8 <RL	1/2 RL	0.00123	2.519	0.0258	SILVER
ZINC	n=11, 0 <RL	N/A	0.13401	2.519	2.8149	ZINC
BOD 5	n=10, 0 <RL	N/A	217.75	2.519	4573.87	BOD 5
TSS	n=11, 0 <RL	N/A	310.81	2.519	6528.60	TSS

TABLE 6: BIOSOLIDS	MAXIMUM POTW BIOSOLIDS TO DISPOSAL MG/KG DRY WT.	TABLE 1 MAXIMUM LAND APP SLUDGE CRITERIA MG/KG	TABLE 3 "CLEAN" LAND APP SLUDGE CRITERIA MG/KG	FINAL SLUDGE CRITERIA MG/KG	
POLLUTANT					POLLUTANT
ARSENIC	8.0		41	41	ARSENIC
CADMIUM	2.7		39	39	CADMIUM
CHROMIUM	71.0				CHROMIUM
COPPER	962.0		1500	1500	COPPER
LEAD	1.8		300	300	LEAD
MERCURY	14.9		17	17	MERCURY
MOLYBDENUM (1)	40.0	75		75	MOLYBDENUM
NICKEL	39.0		420	420	NICKEL
SELENIUM	53.0		100	100	SELENIUM
SILVER	6.0				SILVER
ZINC	1130.0		2800	2800	ZINC
BOD 5					BOD 5
TSS					TSS

(1) Molybdenum Standard Based on Table 1 Maximum Ceiling Concentration Per 40 CFR Part 503.

TABLE 7: REMOVAL EFFICIENCY CALCULATIONS	ADRE ADRE METHOD REMOVAL EFFICIENCY	MRE METHOD REMOVAL EFFICIENCY	DECILE DECILE METHOD REMOVAL EFFICIENCY	LIT LITERATURE REMOVAL EFFICIENCY	SOURCE OF LITERATURE REMOVAL EFFICIENCY DATA	NAME OF THE REMOVAL EFFICIENCY TO BE USED: ADRE, MRE DECILE, OR LIT	FINAL POTW REMOVAL
POLLUTANT	%	%	%	%			%
ARSENIC	38.06	21.66	45.83	45		DECILE	45.83
CADMIUM				67	EPA	LIT	67.00
CHROMIUM		59.34		82		MRE	59.34
COPPER	75.55	83.08	71.23	86		DECILE	71.23
LEAD		96.74		61		MRE	96.74
MERCURY	84.30	93.72		60		MRE	93.72
MOLYBDENUM	29.24	41.82	33.33			DECILE	33.33
NICKEL	40.80	49.14		42		MRE	49.14
SELENIUM	69.56	76.36		50		MRE	76.36
SILVER				75		LIT	75.00
ZINC	71.48	76.56	67.95	79.0		DECILE	67.95
BOD 5		95.9				MRE	95.90
TSS		71.0				MRE	71.00

TABLE 8:						
DAILY MAHL CALCULATIONS	DAILY/7 DAY NPDES LOADING LBS/DAY	ACUTE LOADING LBS/DAY	POTW LOADING LBS/DAY	MOST STRINGENT CRITERIA LBS/DAY	NAME OF MAHL FOR DAILY MAX LIMITS	
POLLUTANT						POLLUTANT
ARSENIC		21.3992		21.3992	WQ-ACUTE	ARSENIC
CADMIUM		0.5062		0.5062	WQ-ACUTE	CADMIUM
CHROMIUM		159.1044		159.1044	WQ-ACUTE	CHROMIUM
COPPER		3.6263		3.6263	WQ-ACUTE	COPPER
LEAD		219.6234		219.6234	WQ-ACUTE	LEAD
MERCURY		0.8691		0.8691	WQ-ACUTE	MERCURY
MOLYBDENUM					No Criteria	MOLYBDENUM
NICKEL		63.6015		63.6015	WQ-ACUTE	NICKEL
SELENIUM		2.8849		2.8849	WQ-ACUTE	SELENIUM
SILVER		1.1319		1.1319	WQ-ACUTE	SILVER
ZINC		25.7965		25.7965	WQ-ACUTE	ZINC
BOD 5			9340	9340	POTW DESIGN	BOD 5
TSS			9340	9340	POTW DESIGN	TSS

TABLE 9:							
MONTHLY MAHL CALCULATIONS	MONTHLY NPDES LOADING LBS/DAY	CHRONIC TOXICITY LOADING LBS/DAY	LOADING FOR HUMAN HEALTH LBS/DAY	TABLE 1 OR 3 SLUDGE LOADING LBS/DAY	MOST STRINGENT CRITERIA LBS/DAY	NAME OF MAHL FOR MONTHLY LIMITS	
POLLUTANT							POLLUTANT
ARSENIC		9.4408	0.6294		0.629	HH	ARSENIC
CADMIUM	0.0506	0.0506			0.051	NPDES Monthly	CADMIUM
CHROMIUM		14.0127			14.013	WQ-CHRONIC	CHROMIUM
COPPER	2.2397	2.2397			2.240	NPDES Monthly	COPPER
LEAD		8.5839			8.584	WQ-CHRONIC	LEAD
MERCURY		0.4183	0.0277		0.028	HH	MERCURY
MOLYBDENUM						No Criteria	MOLYBDENUM
NICKEL		7.0720	308.3546		7.072	WQ-CHRONIC	NICKEL
SELENIUM	0.7212	0.7212	605.8227		0.721	NPDES Monthly	SELENIUM
SILVER						No Criteria	SILVER
ZINC		25.7965	2765.8094		25.796	WQ-CHRONIC	ZINC
BOD 5							BOD 5
TSS							TSS

Sludge Flow to Dry Creek Facility

TABLE 10:							
DAILY LOCAL LIMITS	MOST STRINGENT	FINAL MAHL FOR DAILY LOCAL LIMITS	NAME FOR MAHL	SAFETY FACTOR % FOR DAILY MAX LIMITS	MAHL WITH SAFETY - FACTOR LBS/DAY	MAHL MINUS RES + COM LOADING LBS/DAY	MAXIMUM ALLOWABLE LOAD MAL LBS/DAY
POLLUTANT	MAHL						
ARSENIC	0.6294	0.6294	HH	10	0.5664	0.5263	0.5263
CADMIUM	0.0506	0.0506	NPDES Monthly	10	0.0456	0.0356	0.0356
CHROMIUM	14.0127	14.0127	WQ-CHRONIC	10	12.6115	12.5418	12.5418
COPPER	2.2397	2.2397	NPDES Monthly	10	2.0158	0.5552	0.5552
LEAD	8.5839	8.5839	WQ-CHRONIC	10	7.7256	7.5944	7.5944
MERCURY	0.0277	0.0277	HH	10	0.0249	0.0213	0.0213
MOLYBDENUM			No Criteria				
NICKEL	7.0720	7.0720	WQ-CHRONIC	10	6.3648	6.2753	6.2753
SELENIUM	0.7212	0.7212	NPDES Monthly	10	0.6491	0.6178	0.6178
SILVER	1.1319	1.1319	WQ-ACUTE	10	1.0187	0.9929	0.9929
ZINC	25.7965	25.7965	WQ-ACUTE	10	23.2168	20.4020	20.4020
BOD 5	9340	9340	POTW DESIGN	10	8406	3832.13	3832.13
TSS	9340	9340	POTW DESIGN	10	8406	1877.40	1877.40

TABLE 10 (Continued)

DAILY LOCAL LIMITS

MAXIMUM ALLOWABLE LOAD MAL LBS/DAY	SIU EXPANSION / GROWTH FACTOR FUTURE LOADING %	MAL WITH EXPANSION / GROWTH FUTURE LOADING LBS/DAY	MAIL ALLOCATION FOR SIUS LBS/DAY	RESERVE CAPACITY 20% OF MAIL LBS/DAY	CALCULATED MAIL LESS RESERVE CAPACITY LBS/DAY	CALCULATED UNIFORM LOCAL LIMITS FOR SIUS MG/L	POLLUTANT
0.5263	10	0.4737	0.4737	0.0947	0.3790	0.102	ARSENIC
0.0356	10	0.0321	0.0321	0.0064	0.0257	0.007	CADMIUM
12.5418	10	11.2876	11.2876	2.2575	9.0301	2.437	CHROMIUM
0.5552	10	0.4996	0.4996	0.0999	0.3997	0.108	COPPER
7.5944	10	6.8349	6.8349	1.3670	5.4679	1.476	LEAD
0.0213	10	0.0192	0.0192	0.0038	0.0153	0.0041	MERCURY
							MOLYBDENUM
6.2753	10	5.6478	5.6478	1.1296	4.5182	1.219	NICKEL
0.6178	10	0.5560	0.5560	0.1112	0.4448	0.120	SELENIUM
0.9929	10	0.8936	0.8936	0.1787	0.7149	0.193	SILVER
20.4020	10	18.3618	18.3618	3.6724	14.6894	3.964	ZINC
3832.13	10	3448.92	3448.92	689.78	2759.14	745	BOD 5
1877.40	10	1689.66	1689.66	337.93	1351.73	365	TSS

TABLE 11: DAILY LOCAL LIMIT SUMMARY	MAIL FOR SIUS LBS/DAY	UNIFORM CONCENTRATION FOR SIUS (MG/L)	PROPOSED DAILY MAXIMUM LIMIT (MG/L)	POLLUTANT
POLLUTANT				
ARSENIC	0.4737	0.1023	0.102	ARSENIC
CADMIUM	0.0321	0.0069	0.007	CADMIUM
CHROMIUM	11.2876	2.4369	2.437	CHROMIUM
COPPER	0.4996	0.1079	0.108	COPPER
LEAD	6.8349	1.4756	1.476	LEAD
MERCURY	0.0192	0.0041	0.004	MERCURY
NICKEL	5.6478	1.2193	1.22	NICKEL
SELENIUM	0.5560	0.120	0.120	SELENIUM
SILVER	0.8936	0.193	0.193	SILVER
ZINC	18.3618	3.9641	3.964	ZINC
BOD 5	3448.92	745	745	BOD 5
TSS	1689.66	365	365	TSS

ATTACHMENT G**Revised Legal Authority Language****Current language from the City of Cheyenne Municipal Code Title 13,
Chapter 13.20, Article 1 (13.20.070)**

- A. The BOPU is authorized to establish local limits pursuant to 40 CFR Section 403.5(c).
- B. It shall be unlawful for any significant industrial user or other designated and permitted industrial user to discharge or cause to be discharged any waste or wastewater to the Crow Creek Water Reclamation Facility or the Dry Creek Water Reclamation Facility that exceeds the following daily maximum limit or pollutant allocations as specified in the individual wastewater discharge permit established in Section 13.20.210 of this chapter and issued by the BOPU.

Dry Creek Water Reclamation Facility		
Pollutant	Daily Maximum Discharge Limits mg/L	Total Maximum Allowable Industrial Load (MAIL) ^(*) lbs/day
Arsenic	0.031	0.4348
Cadmium	0.008	0.1172
Chromium	2.43	34.0347
Copper	0.70	9.7646
Lead	0.187	2.6221
Mercury	0.0096	0.1343
Molybdenum	0.065	0.9110
Nickel	0.46	6.4162
Selenium	0.050	0.6962
Silver	0.133	1.8665
Zinc	1.67	23.4355
Biological Oxygen Demand (BOD ₅), lbs/day	9788	9788
Total Suspended Solids (TSS), lbs/day	10447	10447

Crow Creek Water Reclamation Facility		
Pollutant	Daily Maximum Discharge Limits mg/L	Total Maximum Allowable Industrial Load (MAIL) ^(*) lbs/day
Arsenic	0.065	0.4625
Cadmium	0.005	0.0333
Chromium	2.40	17.0625
Copper	0.24	1.7246
Lead	0.162	1.1543
Mercury	0.0023	0.0163
Nickel	0.69	4.9048
Selenium	0.040	0.2876
Silver	0.165	1.1763
Zinc	3.04	21.6497
Biological Oxygen Demand (BOD ₅), lbs/day	4845	4845
Total Suspended Solids (TSS), lbs/day	3071	3071

(*) The BOPU may allocate the MAIL for a pollutant on a case-by-case basis for the Crow Creek and Dry Creek facilities to significant industrial users or other permitted industrial user discharges. The total allocations included in individual wastewater discharge permits shall not exceed the lbs/day values listed. The BOPU will base its allocation decisions upon available loading and industrial user needs. Allocations are at the sole discretion of the BOPU.

- C. Benzene and BTEX. The following limits shall apply to wastewaters that are discharged from:
1. Groundwater cleanup of petroleum or gasoline underground storage tanks or other remediation wastewaters containing these pollutants;
 2. Discharges where one or more of these pollutants are present; or
 3. Where these pollutants are appropriate surrogates.

It shall be unlawful for any industrial user to discharge or cause to be discharged any waste or wastewater to the POTW that exceeds the following limits:

Pollutant ⁽¹⁾⁽³⁾	Daily Maximum Limit (mg/L)
Benzene	0.050
BTEX ⁽²⁾	0.750

- (1) All pollutants shown in the table are total
- (2) BTEX shall be measured as the sum of Benzene, Ethylbenzene, Toluene and Xylenes
- (3) These limits are based upon installation of air stripping technology as described in the EPA document: "Model NPDES Permit for Discharges Resulting from the Cleanup of Gasoline Released from Underground Storage Tanks. June 1989."

D. It shall be unlawful for any industrial user to discharge or cause to be discharged any waste or wastewater to the Crow Creek Water Reclamation Facility or the Dry Creek Water Reclamation Facility that exceeds the following limits.

Pollutant	Daily Maximum Limit (mg/L)
Total Petroleum Hydrocarbons	50
Total Fats, Oils and Grease	100

E. The BOPU may establish more stringent pollutant limits, additional site-specific pollutant limits, BMPs, or additional pretreatment requirements, by ordinance or in individual wastewater discharge permits when, in the judgment of the BOPU, such limitations are necessary to implement the provisions of this chapter.

**The following language is being adopted into the City of Cheyenne
Municipal Code 13.20.070:**

- A. The BOPU is authorized to establish local limits pursuant to 40 CFR Section 403.5(c).
- B. It shall be unlawful for any significant industrial user or other designated and permitted industrial user to discharge or cause to be discharged any waste or wastewater to the Crow Creek Water Reclamation Facility or the Dry Creek Water Reclamation Facility that exceeds the following daily maximum limit or pollutant allocations as specified in the individual wastewater discharge permit established in Section 13.20.210 of this chapter and issued by the BOPU.

Dry Creek Water Reclamation Facility		
Pollutant	Daily Maximum Discharge Limits mg/L	Total Maximum Allowable Industrial Load (MAIL) ^(*) lbs/day
Arsenic	0.106	0.4250
Cadmium	0.009	0.0376
Chromium	3.653	14.5843
Copper	1.773	7.0791
Lead	0.490	1.9565
Mercury	0.001	0.0040
Molybdenum	0.186	0.7417
Nickel	0.89	3.5409
Selenium	0.179	0.7137
Silver	0.285	1.1391
Zinc	4.933	19.6951
Biological Oxygen Demand (BOD ₅)	579	2312.51
Total Suspended Solids (TSS)	448	1786.94

Crow Creek Water Reclamation Facility		
Pollutant	Daily Maximum Discharge Limits mg/L	Total Maximum Allowable Industrial Load (MAIL) ^(*) lbs/day
Arsenic	0.102	0.4737
Cadmium	0.007	0.0321
Chromium	2.437	11.2876
Copper	0.108	0.4996
Lead	1.476	6.8349
Mercury	0.004	0.0192
Nickel	1.22	5.6478
Selenium	0.120	0.5560
Silver	0.193	0.8936
Zinc	3.964	18.3618
Biological Oxygen Demand (BOD ₅)	745	3448.92
Total Suspended Solids (TSS)	365	1689.66

(*) The BOPU may allocate the MAIL for a pollutant on a case-by-case basis for the Crow Creek and Dry Creek facilities to significant industrial users or other permitted industrial user discharges. The total allocations included in individual wastewater discharge permits shall not exceed the lbs/day values listed. The BOPU will base its allocation decisions upon available loading and industrial user needs. Allocations are at the sole discretion of the BOPU.

- C. Benzene and BTEX. The following limits shall apply to wastewaters that are discharged from:
4. Groundwater cleanup of petroleum or gasoline underground storage tanks or other remediation wastewaters containing these pollutants;
 5. Discharges where one or more of these pollutants are present; or
 6. Where these pollutants are appropriate surrogates.

It shall be unlawful for any industrial user to discharge or cause to be discharged any waste or wastewater to the POTW that exceeds the following limits:

Pollutant ⁽¹⁾⁽³⁾	Daily Maximum Limit (mg/L)
Benzene	0.050
BTEX ⁽²⁾	0.750

- (4) All pollutants shown in the table are total
- (5) BTEX shall be measured as the sum of Benzene, Ethylbenzene, Toluene and Xylenes
- (6) These limits are based upon installation of air stripping technology as described in the EPA document: "Model NPDES Permit for Discharges Resulting from the Cleanup of Gasoline Released from Underground Storage Tanks. June 1989."

D. It shall be unlawful for any industrial user to discharge or cause to be discharged any waste or wastewater to the Crow Creek Water Reclamation Facility or the Dry Creek Water Reclamation Facility that exceeds the following limits.

Pollutant	Daily Maximum Limit (mg/L)
Total Petroleum Hydrocarbons	50
Total Fats, Oils and Grease	100

- E. The BOPU may establish more stringent pollutant limits, additional site-specific pollutant limits, BMPs, or additional pretreatment requirements, by ordinance or in individual wastewater discharge permits when, in the judgment of the BOPU, such limitations are necessary to implement the provisions of this chapter.